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## 1. Introduction

This architecture models the engineering-model-go repository as a model-driven toolchain. It captures authored functional intent, export surfaces, and verification-oriented traceability. The model is intended to guide implementation work in this repository using stable IDs and support paths.

### 1.1. Scope and Assumptions

- Functional Groups and Functional Units are the stable authored design backbone.
- Runtime elements and code elements are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- Requirements and reference indexes are provided for traceability, not as primary narrative flow.

## 2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

### 3. Document Health Snapshot

| View                                                | Authored Status | Authored Status Explanation                                                                                       | Units in Scope | Direct Evidence Coverage | Gap Count | Heuristic Confidence | Heuristic Basis                                                                                                  |
|-----------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|----------------|--------------------------|-----------|----------------------|------------------------------------------------------------------------------------------------------------------|
| Architecture Intent View (VIEW-ARCHITECTURE-INTENT) | in-review       | Initial repository self-model with stable top-level domains; implementation details will be expanded iteratively. | 15             | 15                       | 0         | high                 | Defaults authored units to covered; use authoredStatus as the primary publication signal.                        |
| Security View (VIEW-SECURITY)                       | in-review       | Security scope starts from MCP boundary and export-validation supply chain paths.                                 | 10             | 10                       | 7         | high                 | Counts a unit as covered when explicit authored attack-vector mapping exists.                                    |
| Traceability View (VIEW-TRACEABILITY)               | in-review       | Traceability scope is seeded and will be tightened as requirements are added.                                     | 8              | 8                        | 8         | high                 | Counts a unit as covered when direct inferred runtime or code evidence exists.                                   |
| State Lifecycle View (VIEW-STATE-LIFECYCLE)         | draft           | Captures model validity and artifact freshness lifecycle behavior.                                                | 0              | 0                        | 0         | n/a                  | Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.                                  |
| Interaction Flow View (VIEW-INTERACTION-FLOW)       | draft           | Core implementation workflow is captured to guide model-driven repository changes.                                | 3              | 3                        | 0         | high                 | Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal. |

### 4. Terms and Definitions

| Term                    | Definition                                                                         |
|-------------------------|------------------------------------------------------------------------------------|
| actor (TERM-ACTOR)      | A person or operational role that interacts with functional units.                 |
| ai agent (ACT-AI-AGENT) | Consumes MCP outputs to plan and execute model-guided edits. Aliases: coding agent |

| Term                                                                            | Definition                                                                                                                                        |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| allocation trace (FU-ALLOCATION-TRACE)                                          | Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems. Aliases:allocation                            |
| architecture author (ACT-ARCHITECTURE-AUTHOR)                                   | Maintains model structure, mappings, and architecture intent. Aliases:model author                                                                |
| architecture decision (TERM-DECISION)                                           | Authored architecture decision record with status, context, decision text, and consequences.                                                      |
| architecture decision records are authored (FEAT-ARCHITECTURE-DECISION-RECORDS) | Feature indicating the architecture model includes first-class ADR entries. Aliases:adrs authored                                                 |
| architecture decision records document (DATA-ARCHITECTURE-DECISION-RECORDS)     | Generated document containing authored architecture decision records. Aliases:adr document, decisions document                                    |
| architecture model (DATA-ARCHITECTURE-MODEL)                                    | Authored architecture source describing functional intent and mappings. Aliases:architecture yml                                                  |
| artifact generated event is raised (EVT-ARTIFACT-GENERATED)                     | Event indicating one or more generated outputs were produced successfully. Aliases:artifact generated                                             |
| artifact generation (FG-ARTIFACT-GENERATION)                                    | Generation of AsciiDoc and exchange artifacts from model inputs. Aliases:generation                                                               |
| AsciiDoc diagram (TERM-ASCIIDOC-DIAGRAM)                                        | Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.                                 |
| AsciiDoc document (TERM-ASCIIDOC-DOCUMENT)                                      | Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions. |
| asciidoc generator (FU-ASCIIDOC-GENERATOR)                                      | Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts. Aliases:engdoc                                              |
| AsciiDoc section (TERM-ASCIIDOC-SECTION)                                        | Structured generated document section or row model used to render authored and inferred architecture content.                                     |
| attack vector (TERM-ATTACK-VECTOR)                                              | A technical misuse or attack path that targets functional, referenced, or runtime elements.                                                       |
| authored mapping (TERM-AUTHORED-MAPPING)                                        | An explicit relationship declared in architecture inputs between authored or referenced elements.                                                 |
| catalog (TERM-CATALOG)                                                          | Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.                                          |
| catalog entry (TERM-CATALOG-ENTRY)                                              | A stable catalog term with a definition and aliases for linting, linking, and generated references.                                               |

| Term                                                                      | Definition                                                                                                                                         |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| catalog references resolve successfully (COND-CATALOG-REFERENCES-RESOLVE) | Condition requiring all authored references to match known catalog entries. Aliases:catalog resolved                                               |
| ci pipeline (ACT-CI-PIPELINE)                                             | Runs automated validation and export checks for pull requests. Aliases:ci                                                                          |
| ci validation mode is active (MODE-CI-VALIDATION)                         | Automated pipeline mode focused on deterministic checks and artifacts. Aliases:ci mode                                                             |
| CLI command (TERM-CLI-COMMAND)                                            | Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.                       |
| cli orchestration (FU-CLI-ORCHESTRATION)                                  | Command entrypoints that orchestrate loading, validation, generation, and export flows. Aliases:cmd layer                                          |
| code element (TERM-CODE-ELEMENT)                                          | An inferred code structure or ownership element discovered from source trees and build metadata.                                                   |
| codemap inference (FU-CODEMAP-INFERENCE)                                  | Infers runtime, code, and verification evidence from repository sources and test markers. Aliases:inference                                        |
| compliance engineer (ACT-COMPLIANCE-ENGINEER)                             | Reviews controls, risks, and traceability outputs for compliance needs. Aliases:assurance                                                          |
| compliance mapping (TERM-COMPLIANCE-MAPPING)                              | Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.  |
| control (TERM-CONTROL)                                                    | An authored security, policy, or operational control used to constrain behavior and reduce risk.                                                   |
| control verification (TERM-CONTROL-VERIFICATION)                          | Authored control verification record with method, status, threat/risk scope, findings, and evidence.                                               |
| data object (TERM-DATA-OBJECT)                                            | An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.                                          |
| deployment target (TERM-DEPLOYMENT-TARGET)                                | An authored deployment destination such as environment, cluster, namespace, or registry scope.                                                     |
| design document (TERM-DESIGN)                                             | Authored design narrative organized by model entities and architecture views.                                                                      |
| design view (TERM-DESIGN-VIEW)                                            | View-scoped design narrative attached to an authored functional group or functional unit.                                                          |
| engineering model go (SYS-ENGINEERING-MODEL-GO)                           | Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts. Aliases:engineering model toolchain, engmodel |
| event (TERM-EVENT)                                                        | An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.                                                       |

| Term                                                                        | Definition                                                                                                                                  |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| evidence (TERM-EVIDENCE)                                                    | A file, artifact, or observation used to support control, risk, threat, or verification claims.                                             |
| external schema dependencies are available (COND-EXTERNAL-SCHEMA-AVAILABLE) | Condition requiring external schema artifacts to be present for validation.<br>Aliases:schema available                                     |
| flow (TERM-FLOW)                                                            | An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.                                   |
| flow step (TERM-FLOW-STEP)                                                  | A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.                         |
| functional group (TERM-FUNCTIONAL-GROUP)                                    | A major authored capability area that groups related functional units.                                                                      |
| functional unit (TERM-FUNCTIONAL-UNIT)                                      | An authored working unit inside a functional group that owns specific behavior.                                                             |
| gemara export is enabled (FEAT-GEMARA-EXPORT)                               | Feature enabling OpenSSF Gemara catalog and evaluation log generation.<br>Aliases:gemara generation                                         |
| gemara exporter (FU-GEMARA-EXPORTER)                                        | Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK. Aliases:enggemara |
| generated artifacts are fresh (STATE-ARTIFACTS-FRESH)                       | State where generated outputs reflect the latest authored model inputs. Aliases:up to date                                                  |
| generation run is requested (EVT-GENERATION-RUN-REQUESTED)                  | Event indicating documentation or export generation execution starts.<br>Aliases:generation requested                                       |
| generation workflow (TERM-GENERATION-WORKFLOW)                              | Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.                                     |
| go toolchain (REF-GO-TOOLCHAIN)                                             | Go compiler, test runner, and module tooling used by the repository. Aliases:go                                                             |
| hardware interfaces are authored (FEAT-HARDWARE-INTERFACES)                 | Feature enabling hardware items and hardware/software interface modeling.<br>Aliases:hw interfaces                                          |
| implementation engineer (ACT-IMPLEMENTATION-ENGINEER)                       | Implements code changes guided by model requirements and support paths.<br>Aliases:engineer                                                 |
| inference hint (TERM-INFERENCE-HINT)                                        | Authored source and ownership configuration used to discover runtime, code, and verification evidence.                                      |
| inferred mapping (TERM-INFERRED-MAPPING)                                    | A discovered relationship that links inferred runtime/code elements upward to authored design.                                              |
| interface (TERM-INTERFACE)                                                  | An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.                                |

| Term                                                      | Definition                                                                                                                         |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| lint run (TERM-LINT-RUN)                                  | Requirement lint configuration that controls parsing and quality checks for requirement text.                                      |
| LOBSTER activity trace (TERM-LOBSTER-ACTIVITY-TRACE)      | Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.                           |
| lobster exporter (FU-LOBSTER-EXPORTER)                    | Generates LOBSTER traceability inputs and reports from model and verification links. Aliases:englobster                            |
| lobster toolchain (REF-LOBSTER-TOOLCHAIN)                 | LOBSTER core and TRLC integration tools used for traceability reporting. Aliases:lobster                                           |
| lobster trace report (DATA-LOBSTER-TRACE-REPORT)          | Traceability report output generated from TRLC and test links. Aliases:lobster report                                              |
| local development mode is active (MODE-LOCAL-DEVELOPMENT) | Local developer mode for iterative model and code updates. Aliases:local mode                                                      |
| malformed model input (AV-MALFORMED-MODEL-INPUT)          | Invalid or ambiguous model content that causes incorrect graph construction. Aliases:invalid model                                 |
| mcp integration (FG-MCP-INTEGRATION)                      | MCP server interfaces that expose model-aware tooling for AI agents. Aliases:mcp                                                   |
| mcp server (FU-MCP-SERVER)                                | Serves model-backed MCP tools with validated inputs and structured responses. Aliases:engmcp                                       |
| MCP server (TERM-MCP-SERVER)                              | Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.                       |
| mcp server is enabled (FEAT-MCP-SERVER)                   | Feature enabling MCP runtime APIs for model-aware tooling. Aliases:mcp enabled                                                     |
| MCP stdio transport (TERM-MCP-STDIO-TRANSPORT)            | Content-Length framed standard input/output transport used by the MCP command-line server.                                         |
| MCP tool (TERM-MCP-TOOL)                                  | Named MCP operation with a constrained input schema and structured response payload.                                               |
| mcp tool call is received (EVT-MCP-TOOL-CALL-RECEIVED)    | Event indicating the MCP server received a tool invocation request. Aliases:tool call                                              |
| MCP tool response (TERM-MCP-TOOL-RESPONSE)                | Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error. |
| mcp tool result (DATA-MCP-TOOL-RESULT)                    | Structured MCP tool response payload carrying model-aware query results. Aliases:tool result                                       |

| Term                                                     | Definition                                                                                                                              |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| model (TERM-MODEL)                                       | The authored architecture model root that composes functional structure, relationships, inference hints, and views.                     |
| model authoring (FG-MODEL-AUTHORING)                     | Authoring and loading of architecture, catalog, requirements, and design inputs.<br>Aliases:authoring                                   |
| model bundle (TERM-BUNDLE)                               | Loaded architecture, catalog, and companion documents resolved as one working model context.                                            |
| model is invalid (STATE-MODEL-INVALID)                   | State where validation diagnostics include one or more errors. Aliases:invalid                                                          |
| model is valid (STATE-MODEL-VALID)                       | State where authored model passes validation checks. Aliases:valid                                                                      |
| model loader (FU-MODEL-LOADER)                           | Loads and normalizes architecture, catalog, requirements, and design documents.<br>Aliases:bundle loader                                |
| model updated event is received (EVT-MODEL-UPDATED)      | Event indicating architecture or requirement model source changed. Aliases:model change                                                 |
| Open OTM document (TERM-OPEN-OTM-DOCUMENT)               | Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.                     |
| open otm schema (REF-OPEN-OTM-SCHEMA)                    | Open Threat Model schema used to validate OTM output artifacts. Aliases:otm schema                                                      |
| OSCAL assessment results (TERM-OSCAL-ASSESSMENT-RESULTS) | Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.                          |
| oscal exporter (FU-OSCAL-EXPORTER)                       | Exports the OSCAL system security plan, assessment results, and plan of action and milestones. Aliases:engoscal                         |
| OSCAL POA&M (TERM-OSCAL-POAM)                            | Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.                     |
| OSCAL SSP (TERM-OSCAL-SSP)                               | Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON. |
| oscal ssp artifact (DATA-OSCAL-SSP)                      | Generated OSCAL system security plan JSON output. Aliases:ssp json                                                                      |
| path traversal in mcp calls (AV-PATH-TRAVERSAL-IN-MCP)   | Host file access attempts through untrusted path arguments in MCP tools. Aliases:path traversal                                         |
| POA&M item (TERM-POAM-ITEM)                              | Plan of action and milestones item linked to a modeled risk and supporting artifacts.                                                   |
| reference index (TERM-REFERENCE-INDEX)                   | Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.                     |



| Term                                                            | Definition                                                                                                                                                          |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| referenced element (TERM-REFERENCED-ELEMENT)                    | An architecture-relevant external, platform, or third-party dependency represented by role.                                                                         |
| repository index (TERM-REPO-INDEX)                              | Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.                                                  |
| requirement (TERM-REQUIREMENT)                                  | A structured requirement statement that defines expected system behavior and maps to functional ownership.                                                          |
| requirement trace is complete (COND-REQUIREMENT-TRACE-COMPLETE) | Condition requiring requirements to map to implementation and verification evidence. Aliases:trace complete                                                         |
| risk (TERM-RISK)                                                | Authored risk record with likelihood, impact, response, scope, controls, and evidence.                                                                              |
| runtime element (TERM-RUNTIME-ELEMENT)                          | An inferred runtime realization element discovered from infrastructure and deployment sources.                                                                      |
| schema supply chain tamper (AV-SCHEMA-SUPPLY-CHAIN-TAMPER)      | Tampering or drift in external schemas used for export validation. Aliases:schema tamper                                                                            |
| security context (TERM-SECURITY-CONTEXT)                        | Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.            |
| source block (TERM-SOURCE-BLOCK)                                | Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.                                                     |
| state (TERM-STATE)                                              | An authored lifecycle state used to model transition behavior, guards, and event-driven progression.                                                                |
| Structurizr DSL (TERM-STRUCTURIZR-DSL)                          | Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.                                     |
| structurizr dsl artifact (DATA-STRUCTURIZR-DSL)                 | Generated Structurizr workspace DSL document. Aliases:dsl                                                                                                           |
| structurizr export is enabled (FEAT-STRUCTURIZR-EXPORT)         | Feature enabling Structurizr DSL generation and validation. Aliases:dsl export                                                                                      |
| structurizr exporter (FU-STRUCTURIZR-EXPORTER)                  | Exports Structurizr DSL and supports validation/static-site workflows. Aliases:engstruct                                                                            |
| structurizr validator (REF-STRUCTURIZR-VALIDATOR)               | Structurizr DSL validation runtime used to verify generated DSL. Aliases:structurizr                                                                                |
| system composition (FU-SYSTEM-COMPOSITION)                      | Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model. Aliases:composition |

| Term                                                      | Definition                                                                                                                                                 |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| system composition is enabled (FEAT-SYSTEM-COMPOSITION)   | Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems. Aliases:system of systems                     |
| threat assumption (TERM-THREAT-ASSUMPTION)                | Authored security assumption that records scope, rationale, owner, status, and evidence.                                                                   |
| Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)      | Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.                                                     |
| threat dragon json artifact (DATA-THREAT-DRAGON-JSON)     | Generated threat model artifact for OWASP Threat Dragon. Aliases:threat dragon output                                                                      |
| threat dragon schemas (REF-THREAT-DRAGON-SCHEMAS)         | External JSON schemas used to validate Threat Dragon exports. Aliases:threat dragon                                                                        |
| threat export is enabled (FEAT-THREAT-EXPORT)             | Feature enabling Threat Dragon and Open OTM exports. Aliases:threat modeling export                                                                        |
| threat exporter (FU-THREAT-EXPORTER)                      | Exports threat model views to Threat Dragon and Open OTM representations. Aliases:engdragon                                                                |
| threat mitigation (TERM-THREAT-MITIGATION)                | Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.                                           |
| threat model export (TERM-THREAT-MODEL-EXPORT)            | Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema. |
| threat out-of-scope decision (TERM-THREAT-OUT-OF-SCOPE)   | Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.                                              |
| threat scenario (TERM-THREAT-SCENARIO)                    | Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.                                          |
| trace marker (TERM-TRACE-MARKER)                          | Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.                                  |
| traceability and compliance (FG-TRACEABILITY-COMPLIANCE)  | Requirement trace exports and compliance-oriented output generation. Aliases:compliance                                                                    |
| traceability export is enabled (FEAT-TRACEABILITY-EXPORT) | Feature enabling TRLC and LOBSTER generation flows. Aliases:trace export                                                                                   |
| traceability gap drift (AV-TRACEABILITY-GAP-DRIFT)        | Requirement links or tests drifting from implementation over time. Aliases:trace drift                                                                     |
| trlc exporter (FU-TRL-EXPORTER)                           | Exports requirements and model structures to TRLC artifacts. Aliases:engtrlc                                                                               |
| trlc package (DATA-TRL-PACKAGE)                           | Generated TRLC requirements package and model metadata. Aliases:trlc output                                                                                |

| Term                                                      | Definition                                                                                                                           |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| TRLC package (TERM-TRLC-PACKAGE)                          | Generated TRLC model and requirements package used for formal requirement traceability processing.                                   |
| trlc toolchain (REF-TRLC-TOOLCHAIN)                       | TRLC parser and runtime used for requirements language validation. Aliases:trlc                                                      |
| trust boundary (TERM-TRUST-BOUNDARY)                      | An authored trust separation zone that marks policy, identity, or security control boundaries.                                       |
| upward linking (TERM-UPWARD-LINKING)                      | Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. |
| validation (TERM-VALIDATION)                              | Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.                 |
| validation and analysis (FG-VALIDATION-ANALYSIS)          | Validation, inference, and evidence analysis over authored and inferred model content. Aliases:analysis                              |
| validation diagnostic (TERM-VALIDATION-DIAGNOSTIC)        | Structured validation finding with code, severity, message, and optional source path.                                                |
| validation engine (FU-VALIDATION-ENGINE)                  | Validates IDs, references, and model consistency across authored entities and mappings. Aliases:validator                            |
| validation failed event is raised (EVT-VALIDATION-FAILED) | Event indicating model or export validation failed. Aliases:validation failure                                                       |
| verification check (TERM-VERIFICATION-CHECK)              | An inferred or authored verification artifact that validates requirement behavior with test evidence.                                |
| view (TERM-VIEW)                                          | Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.                             |
| view projection (FU-VIEW-PROJECTION)                      | Builds view-specific projections for architecture communication and traceability outputs. Aliases:projection                         |

## 5. Architecture Decision Records

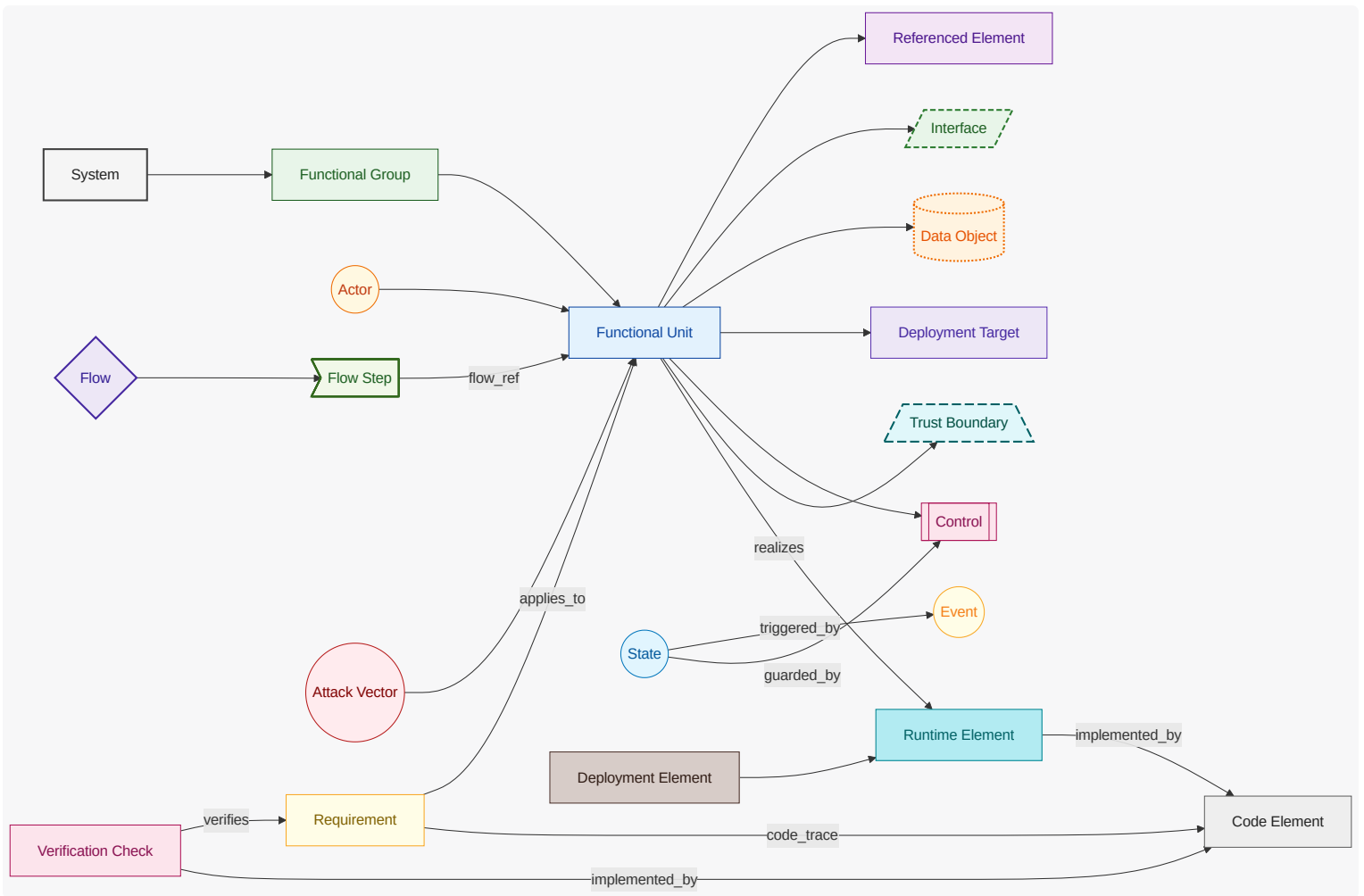
Architecture decision records are generated separately and linked here as a compact index.

| ID                          | Title                                                                                 | Status   | Date       |
|-----------------------------|---------------------------------------------------------------------------------------|----------|------------|
| <a href="#">ADR-EMG-001</a> | Generate architecture decision records separately from the main architecture document | accepted | 2026-04-29 |
| <a href="#">ADR-EMG-002</a> | Keep architecture decision records model-authored and separate from code links        | accepted | 2026-04-29 |

| ID                 | Title                                                                                             | Status   | Date       |
|--------------------|---------------------------------------------------------------------------------------------------|----------|------------|
| <u>ADR-EMG-003</u> | Render Gemara as a first-class output using the official go-gemara SDK types                      | accepted | 2026-06-27 |
| <u>ADR-EMG-004</u> | Source the Gemara evaluation log from authored control verifications, with pass and fail outcomes | accepted | 2026-06-27 |
| <u>ADR-EMG-005</u> | Keep hand-written OSCAL SSP/AR/POA&M; add Gemara-to-OSCAL only as an additive bridge              | accepted | 2026-06-27 |
| <u>ADR-EMG-006</u> | Synthesize Gemara structure that the model does not author; with explicit fallbacks               | accepted | 2026-06-27 |
| <u>ADR-EMG-007</u> | Attach trace and model-link markers to package-level var and const declarations                   | accepted | 2026-06-27 |
| <u>ADR-EMG-008</u> | Model multi-repository work as a recursive system-of-systems                                      | accepted | 2026-06-27 |
| <u>ADR-EMG-009</u> | Subsystem references are downward-only with parent-owned allocation                               | accepted | 2026-06-27 |
| <u>ADR-EMG-010</u> | Resolve subsystem references from local subdirectories first                                      | accepted | 2026-06-27 |
| <u>ADR-EMG-011</u> | Requirements carry no tiers; delegation is the link                                               | accepted | 2026-06-27 |
| <u>ADR-EMG-012</u> | Trace links are existence-checked, scoped to the owning model, and exported as a matrix           | accepted | 2026-06-27 |
| <u>ADR-EMG-013</u> | CI enforces the generation gates and artifact freshness                                           | accepted | 2026-06-28 |

## 5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



## 6. Architecture Intent View

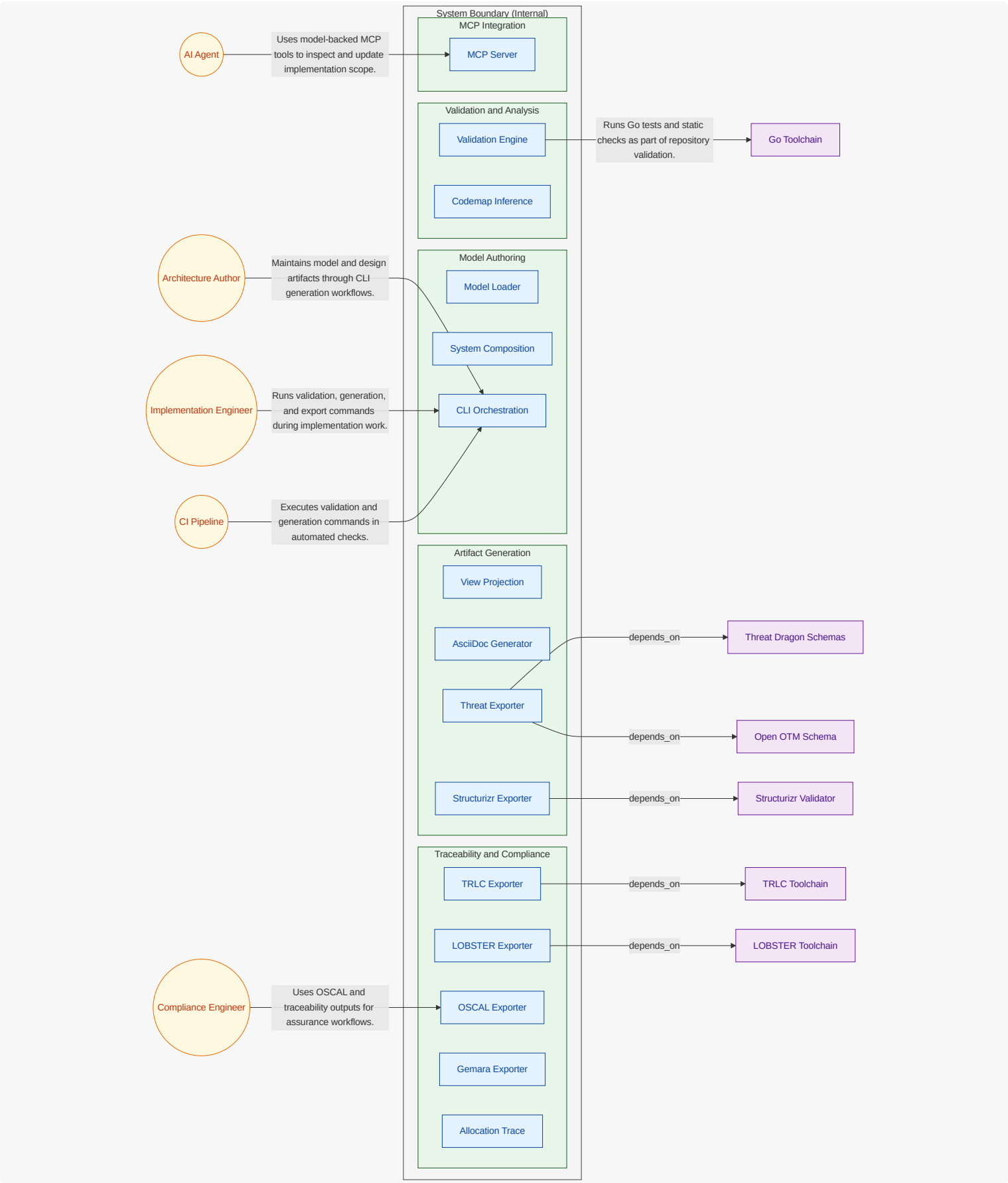
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

### 6.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

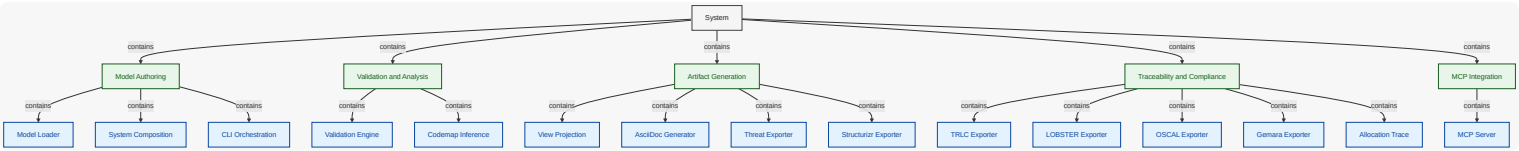
### 6.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the **System Boundary (Internal)** container, while actors and referenced dependencies are shown outside the boundary.



6.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



6.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

|                    |                   |                      |                             |                 |
|--------------------|-------------------|----------------------|-----------------------------|-----------------|
|                    |                   | Allocation Trace     |                             |                 |
|                    |                   | AsciDoc Generator    | Gemara Exporter             |                 |
|                    |                   | Structurizr Exporter | LOBSTER Exporter            |                 |
| CLI Orchestration  |                   | Threat Exporter      | OSCAL Exporter              |                 |
| Model Loader       | Codemap Inference | View Projection      | TRLC Exporter               | MCP Server      |
| System Composition | Validation Engine |                      |                             |                 |
| Model Authoring    |                   | Artifact Generation  | Traceability and Compliance | MCP Integration |

6.5. Functional Groups and Units (View Scoped)

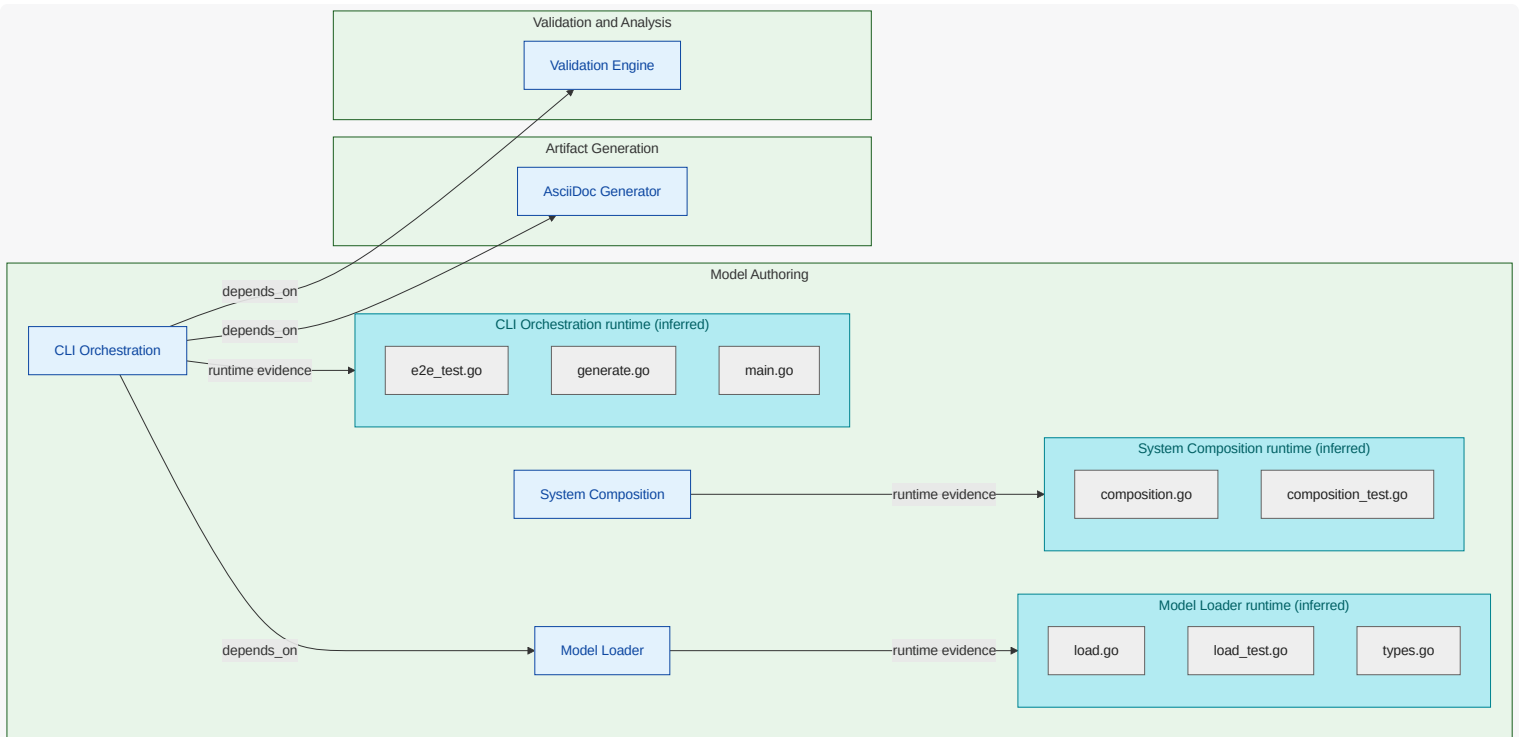
6.5.1. Model Authoring (FG-MODEL-AUTHORING)

Inputs and model loading responsibilities.

Maintains authored architecture, <<REF\_IDX-ENGMODEL\_TERM\_REQUIREMENT,requirements>>, design, and <<REF\_IDX-ENGMODEL\_TERM\_CATALOG,catalog>> handling.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



CLI Orchestration (FU-CLI-ORCHESTRATION)

Command entrypoints orchestrating model load, validation, generation, and exports.

Owned Responsibility



functional responsibility for cli orchestration ; includes decision and orchestration flow to AsciiDoc Generator and decision and orchestration flow to Model Loader

Key Collaborators

AsciiDoc Generator , Model Loader , Validation Engine

Linked Requirements

REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-009 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015

Model Loader (FU-MODEL-LOADER)

Loads and normalizes architecture, catalog, requirements, and design documents .

Owned Responsibility

functional responsibility for model loader

Key Collaborators

none

Linked Requirements

REQ-EMG-001 , REQ-EMG-016 , REQ-EMG-027

System Composition (FU-SYSTEM-COMPOSITION)

Resolves downward subsystem references from local subdirectories or external git repositories cloned into the .engmod/subsystems cache, and composes a federated system-of-systems model .

Owned Responsibility

functional responsibility for system composition

Key Collaborators

none

Linked Requirements

REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-027

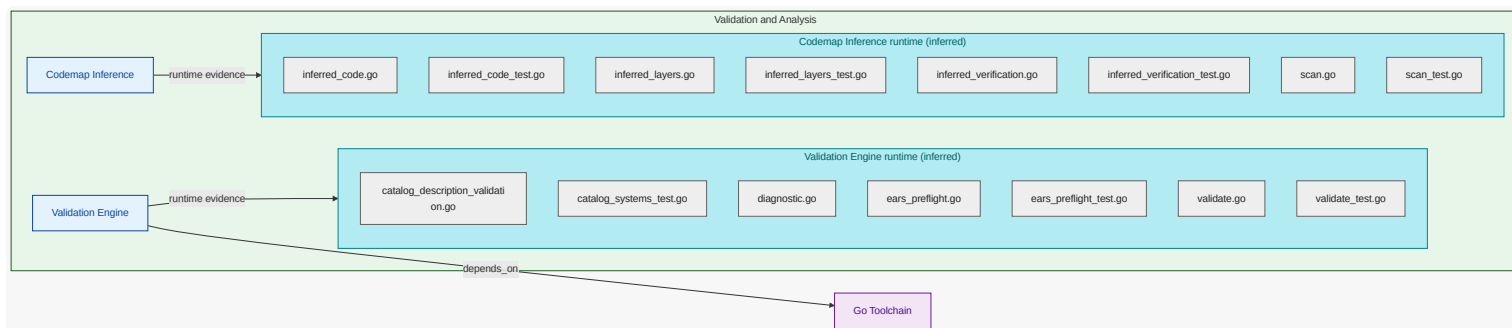
6.5.2. Validation and Analysis (FG-VALIDATION-ANALYSIS)

Validation and inference responsibilities.

Performs consistency checks and derives runtime, code, and verification <<REF\_IDX-ENGMODEL\_TERM\_EVIDENCE,evidence>>.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



### Codemap Inference (FU-CODEMAP-INFERENCE)

Infers code/runtime ownership and verification links from source and tests.

#### Owned Responsibility

functional responsibility for `codemap inference`

#### Key Collaborators

none

#### Linked Requirements

REQ-EMG-010, REQ-EMG-012, REQ-EMG-028, REQ-EMG-030

### Validation Engine (FU-VALIDATION-ENGINE)

Validates authored entities, IDs, references, and mapping consistency.

#### Owned Responsibility

functional responsibility for `validation engine`; includes decision and orchestration `flow` to `Go Toolchain`

#### Key Collaborators

`Go Toolchain`

#### Linked Requirements

REQ-EMG-001, REQ-EMG-009, REQ-EMG-011, REQ-EMG-018, REQ-EMG-021, REQ-EMG-022, REQ-EMG-023, REQ-EMG-026, REQ-EMG-028, REQ-EMG-029

### 6.5.3. Artifact Generation (FG-ARTIFACT-GENERATION)

Publication and exchange `artifact generation`.

Produces AsciiDoc publication docs and external modeling outputs.

### Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Owned Responsibility

functional responsibility for view projection

Key Collaborators

none

Linked Requirements

REQ-EMG-003, REQ-EMG-024

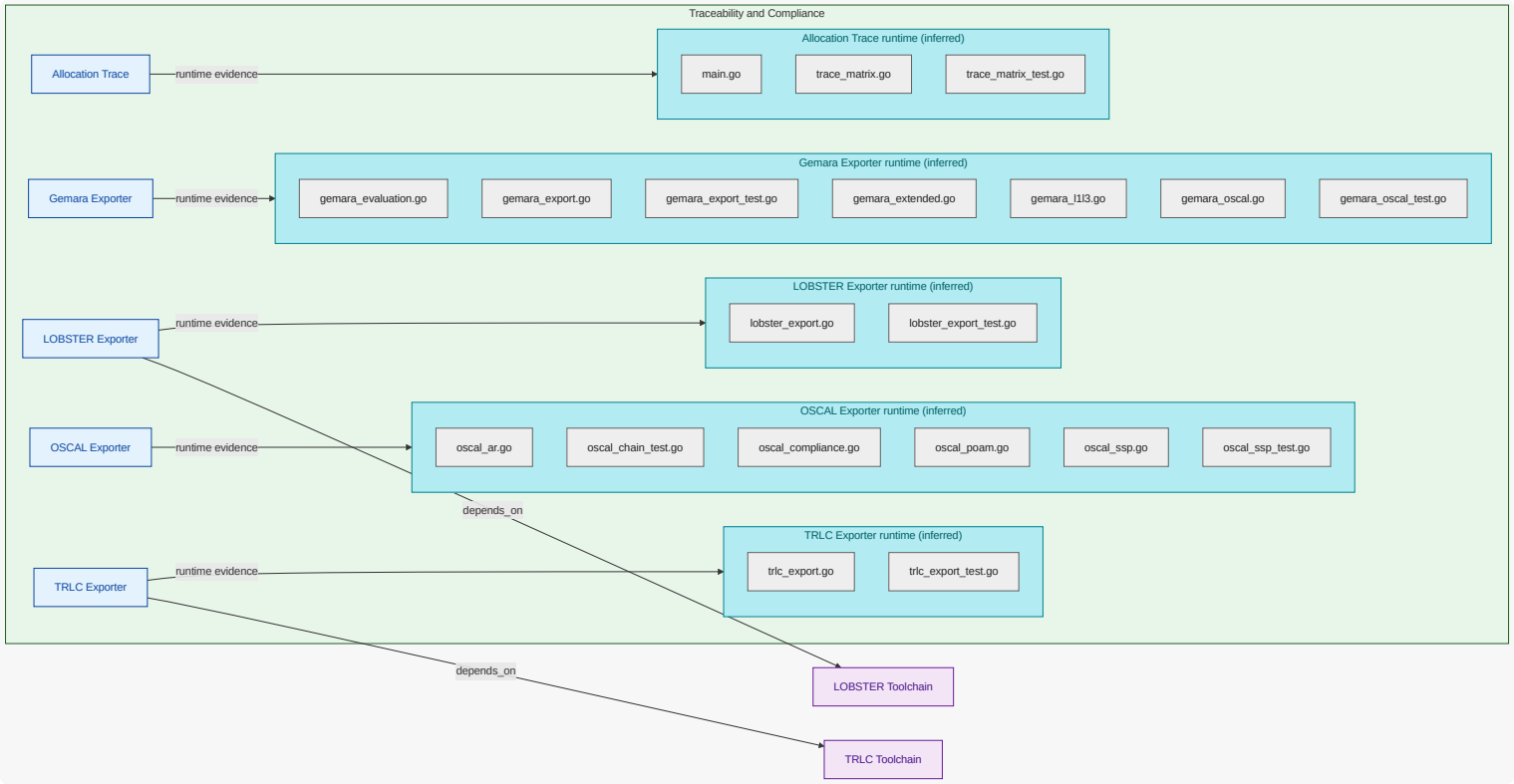
6.5.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)

Trace and compliance export responsibilities.

Produces <<REF\_IDX-CATALOG\_TRLC,TRLC>>, <<REF\_IDX-CATALOG\_LOBSTER,LOBSTER>>, and OSCAL outputs from modeled content.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Allocation Trace (FU-ALLOCATION-TRACE)

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems, without modifying subsystem models.

Owned Responsibility

functional responsibility for allocation trace

Key Collaborators

none

#### Linked Requirements

REQ-EMG-012 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030

#### Gemara Exporter (FU-GEMARA-EXPORTER)

Renders OpenSSF Gemara L1-L7 catalogs and logs (vector, principle, guidance, control , capability, threat, risk , evaluation, enforcement, audit) with an optional OSCAL catalog and assessment-results bridge, validated by the Gemara SDK.

#### Owned Responsibility

functional responsibility for gemara exporter

#### Key Collaborators

none

#### Linked Requirements

REQ-EMG-015

#### LOBSTER Exporter (FU-LOBSTER-EXPORTER)

Generates LOBSTER traceability inputs and reports.

#### Owned Responsibility

functional responsibility for lobster exporter ; includes decision and orchestration flow to LOBSTER Toolchain

#### Key Collaborators

LOBSTER Toolchain

#### Linked Requirements

REQ-EMG-006

#### OSCAL Exporter (FU-OSCAL-EXPORTER)

Exports OSCAL System Security Plan (SSP), Assessment Results (AR), and Plan of Action and Milestones (POA&M).

#### Owned Responsibility

functional responsibility for oscal exporter

#### Key Collaborators

none

#### Linked Requirements

REQ-EMG-001 , REQ-EMG-013

#### TRLC Exporter (FU-TRLC-EXPORTER)

Emits TRLC model and requirement artifacts.

#### Owned Responsibility

functional responsibility for trlc exporter ; includes decision and orchestration flow to TRLC Toolchain

#### Key Collaborators

TRLC Toolchain

Linked Requirements

REQ-EMG-006

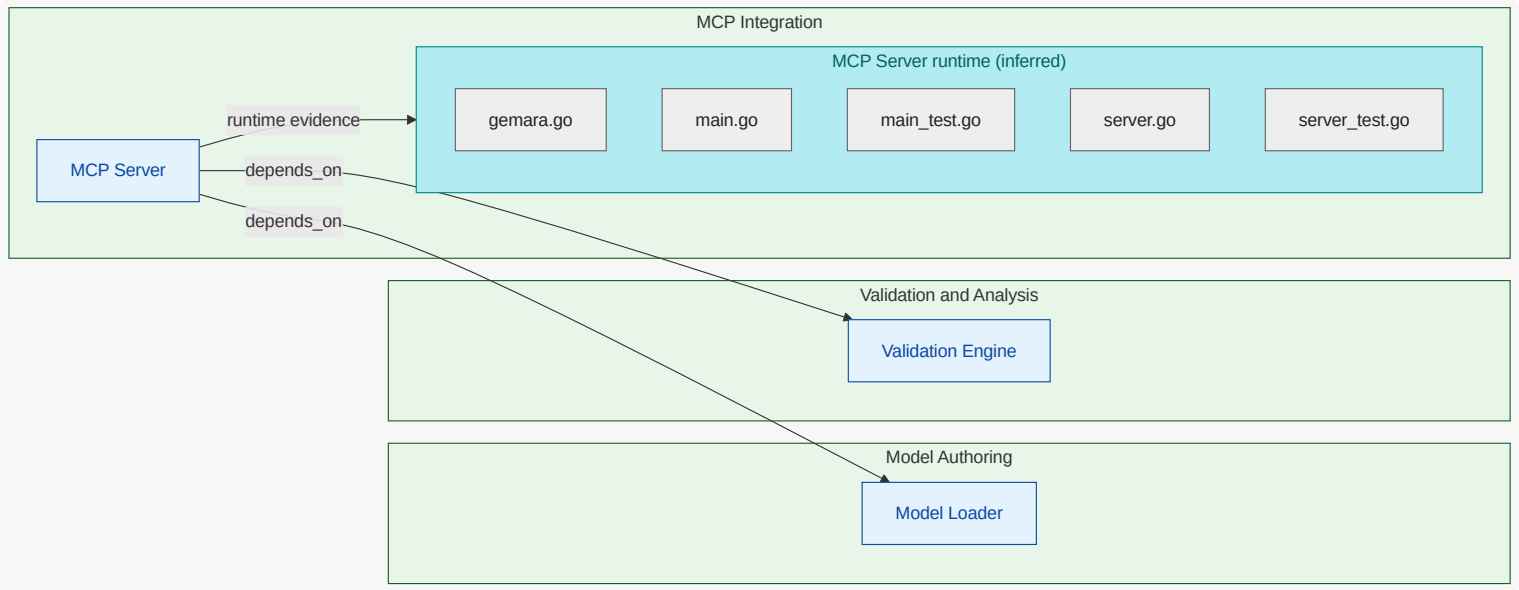
6.5.5. MCP Integration (FG-MCP-INTEGRATION)

AI-agent integration and runtime API responsibilities.

Exposes model-aware tools over MCP with strict input and error contracts.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



MCP Server (FU-MCP-SERVER)

Serves model-backed tools over MCP with path safety and structured errors.

Owned Responsibility

functional responsibility for mcp server ; includes decision and orchestration flow to Model Loader and decision and orchestration flow to Validation Engine

Key Collaborators

Model Loader , Validation Engine

Linked Requirements

REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-015

## 7. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence .

### 7.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

### 7.2. Coverage Gaps

Coverage summary: 3/10 units have direct evidence coverage in this view .

- AsciiDoc Generator has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- CLI Orchestration has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- LOBSTER Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Structurizr Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- TRLC Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Threat Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Validation Engine has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

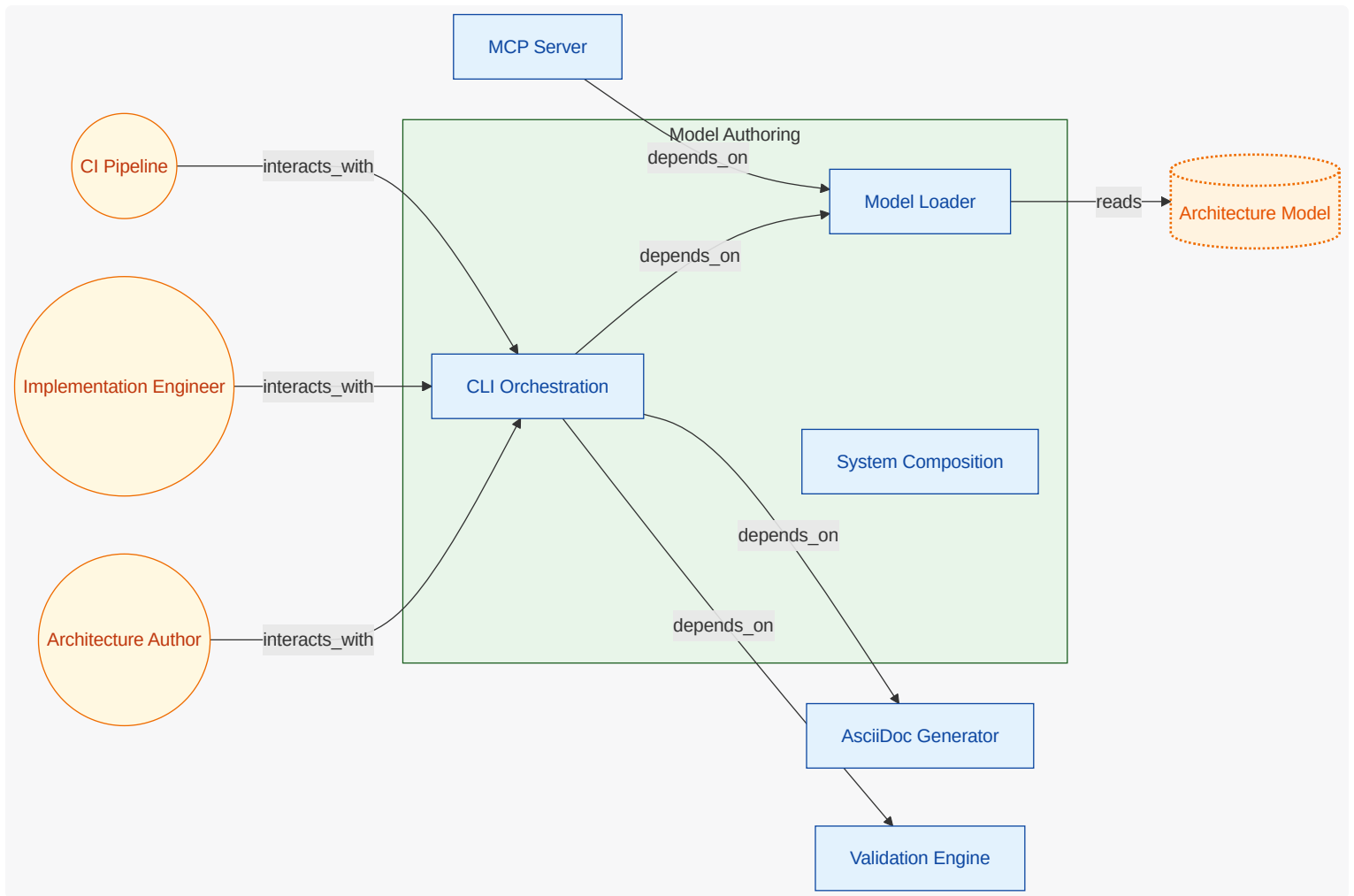
### 7.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/ events ) for unresolved observability evidence .
- Map additional security-relevant dependencies and exposures into authored mappings for traceable coverage.

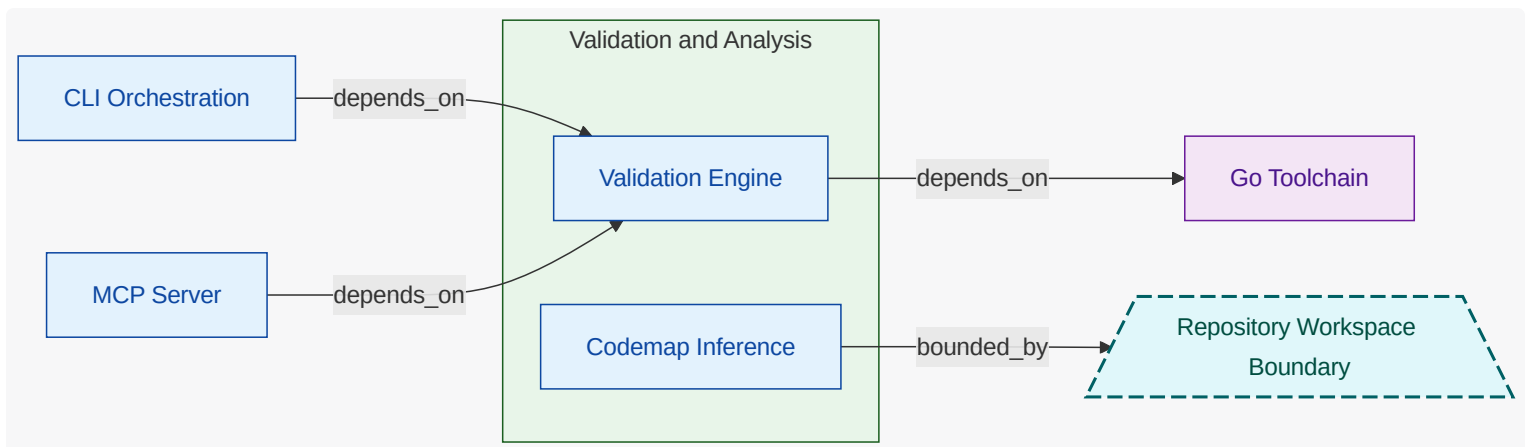
### 7.4. Threat Model Context Diagram

What these diagrams show: OWASP-style context views split by functional group. Each group diagram keeps owned functional units and owned security-relevant entities inside the functional-group container; connected actors, referenced elements, external functional units, and other external entities are rendered outside it.

#### 7.4.1. Model Authoring (FG-MODEL-AUTHORING)

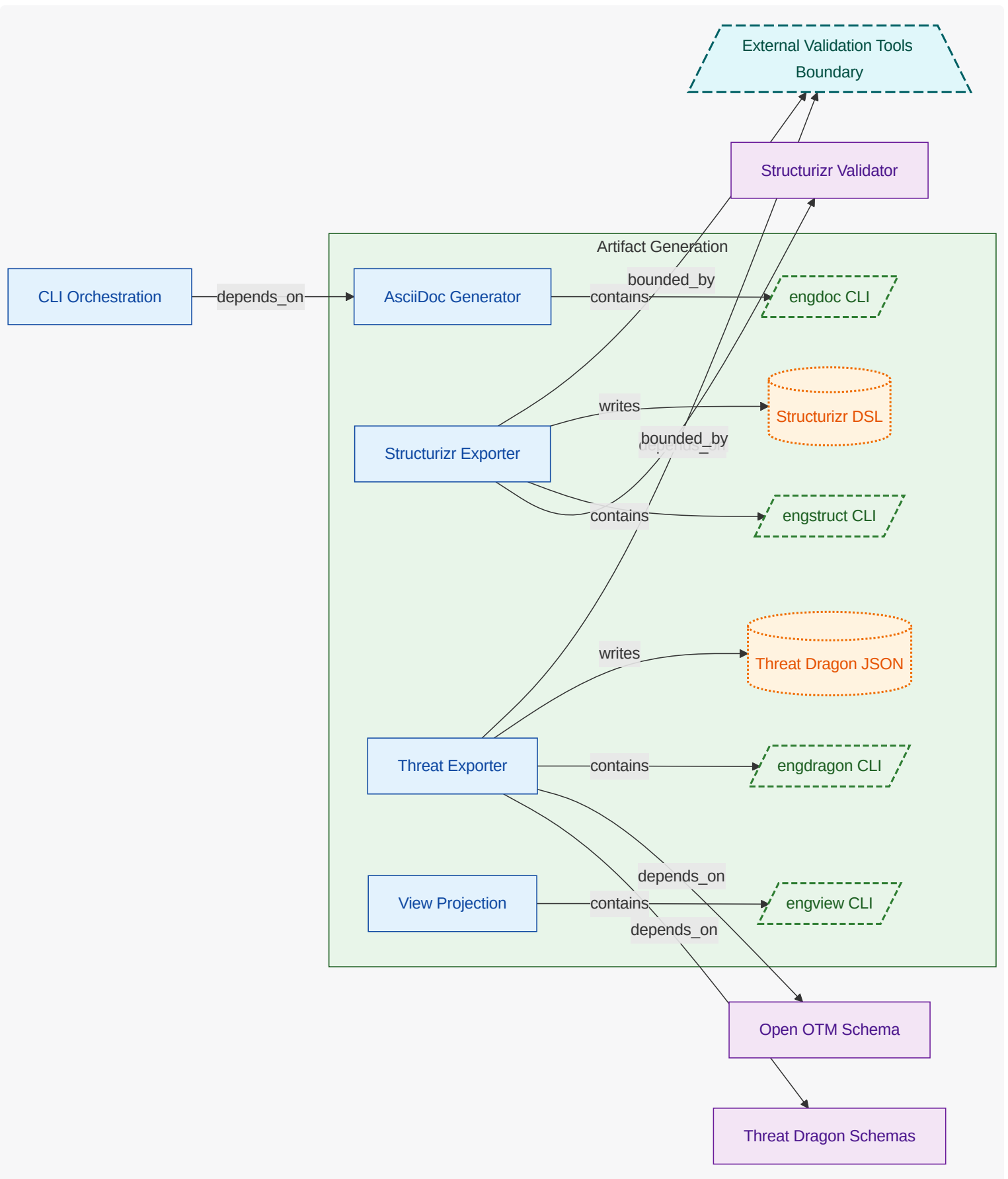


#### 7.4.2. Validation and Analysis (FG-VALIDATION-ANALYSIS)

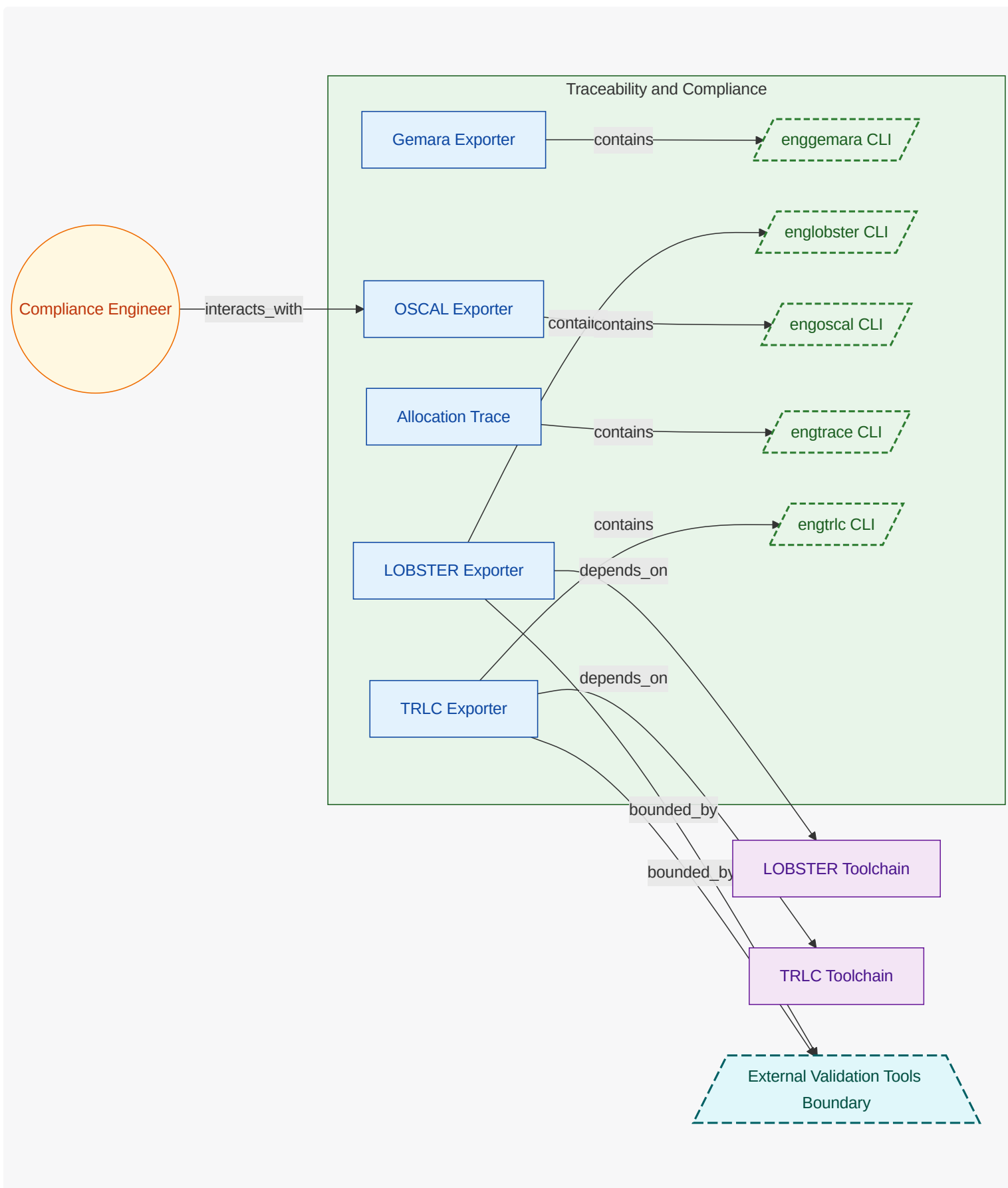


#### 7.4.3. Artifact Generation (FG-ARTIFACT-GENERATION)

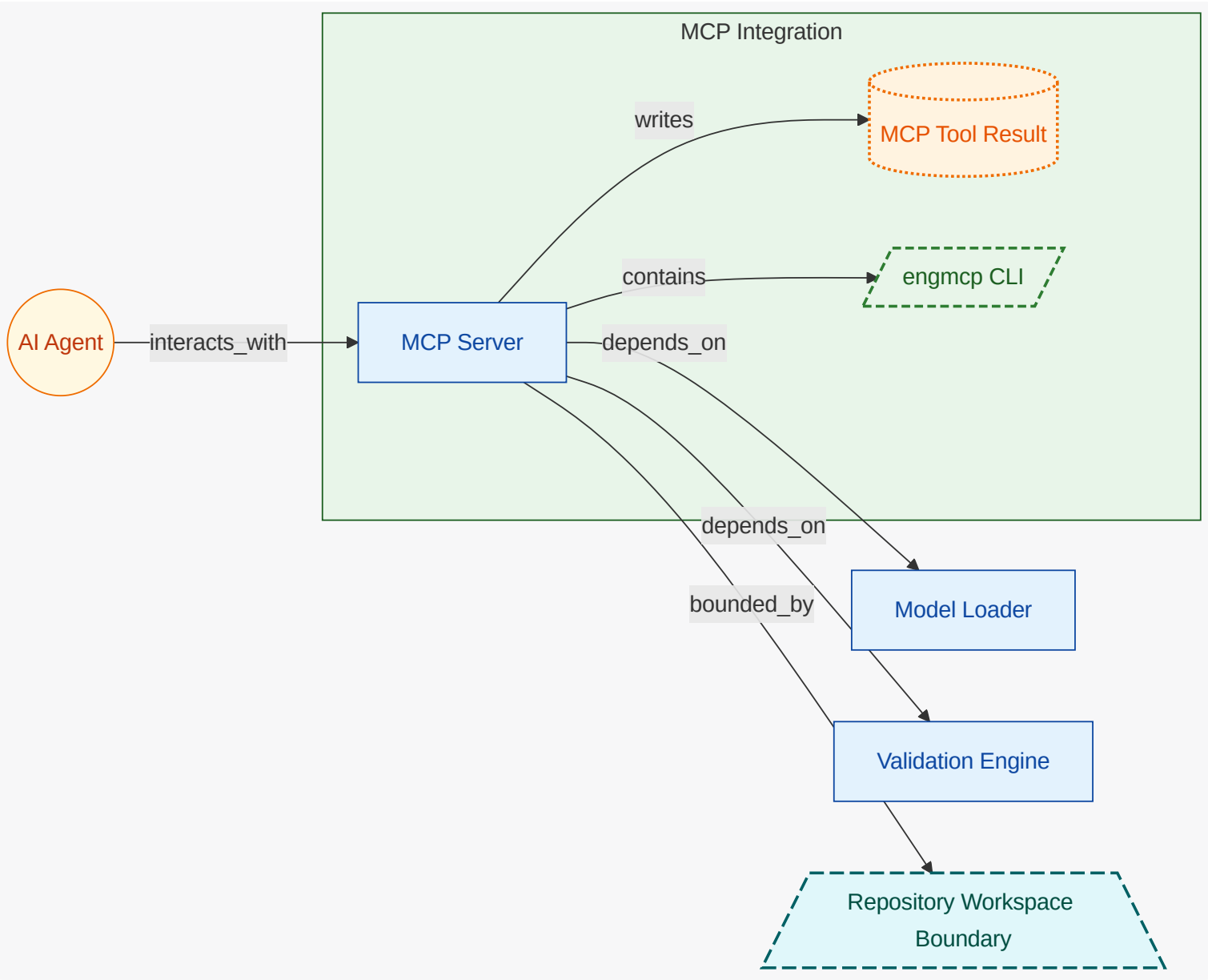




#### 7.4.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)



#### 7.4.5. MCP Integration (FG-MCP-INTEGRATION)



7.5. Threat Scenario Register

7.5.1. Requirement traceability drifts from implementation (TS-TRACEABILITY-DRIFT)

Code and tests change without corresponding requirement trace updates.

| Field         | Value                                                |
|---------------|------------------------------------------------------|
| Status        | mitigating                                           |
| Owner         | Compliance Engineer                                  |
| Attack Vector | Traceability Gap Drift                               |
| Scope         | Codemap Inference , LOBSTER Exporter , TRLC Exporter |
| Flows         | none                                                 |

| Field             | Value                                                      |
|-------------------|------------------------------------------------------------|
| Likelihood        | medium                                                     |
| Impact            | medium                                                     |
| Severity          | medium                                                     |
| Risk              | none                                                       |
| Controls          | Requirement Traceability Coverage                          |
| Mitigations       | none                                                       |
| Verifications     | none                                                       |
| Operational Notes | Track closure through mitigation and verification linkage. |

Control Verification Status

VER-TRACEABILITY-COVERAGE

| Field       | Value                                                            |
|-------------|------------------------------------------------------------------|
| Control     | Requirement Traceability Coverage                                |
| Method      | analysis                                                         |
| Status      | partial                                                          |
| Owner       | Compliance Engineer                                              |
| Last Tested | unknown                                                          |
| Risks       | none                                                             |
| Findings    | Coverage depends on TRLC-LINKS completeness in source and tests. |

7.5.2. MCP path traversal accesses non-repo files (TS-MCP-PATH-TRAVERSAL)

Untrusted MCP input attempts to escape repo root for sensitive file reads.

| Field         | Value                                   |
|---------------|-----------------------------------------|
| Status        | mitigating                              |
| Owner         | Architecture Author                     |
| Attack Vector | Path Traversal in MCP Calls             |
| Scope         | MCP Server                              |
| Flows         | Model Change to Verified Artifacts Flow |

| Field             | Value                                                      |
|-------------------|------------------------------------------------------------|
| Likelihood        | medium                                                     |
| Impact            | high                                                       |
| Severity          | high                                                       |
| Risk              | none                                                       |
| Controls          | MCP Path Boundary Enforcement , Strict MCP Input Schemas   |
| Mitigations       | none                                                       |
| Verifications     | none                                                       |
| Operational Notes | Track closure through mitigation and verification linkage. |

Control Verification Status

VER-MCP-PATH-BOUNDARY

| Field       | Value                                                            |
|-------------|------------------------------------------------------------------|
| Control     | MCP Path Boundary Enforcement                                    |
| Method      | test                                                             |
| Status      | pass                                                             |
| Owner       | Architecture Author                                              |
| Last Tested | unknown                                                          |
| Risks       | none                                                             |
| Findings    | Path traversal rejected and error payload returned in MCP tests. |

7.6. Threat Mitigation Coverage

| ID                   | Threat Scenario       | Control                           | Status      | Effectiveness | Owner               | Verification Links    | Notes |
|----------------------|-----------------------|-----------------------------------|-------------|---------------|---------------------|-----------------------|-------|
| TM-MCP-PATH-BOUNDARY | TS-MCP-PATH-TRAVERSAL | MCP Path Boundary Enforcement     | implemented | high          | Architecture Author | VER-MCP-PATH-BOUNDARY |       |
| TM-TRACE-COVERAGE    | TS-TRACEABILITY-DRIFT | Requirement Traceability Coverage | implemented | medium        | Compliance Engineer | none                  |       |

7.7. Security Flow Semantics

| ID                                      | Title                                   | Kind    | Direction | Frequency | Source                  | Destination       | Protocol |
|-----------------------------------------|-----------------------------------------|---------|-----------|-----------|-------------------------|-------------------|----------|
| FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS | Model Change to Verified Artifacts Flow | control | internal  | on-demand | Implementation Engineer | CLI Orchestration | cli      |

| ID                                      | Auth              | Encryption | Integrity   | Boundary Crossing | Trust Boundary | Threats | Data                                                                        |
|-----------------------------------------|-------------------|------------|-------------|-------------------|----------------|---------|-----------------------------------------------------------------------------|
| FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS | repository-access | none       | git-history | false             | none           | none    | Architecture Model , MCP Tool Result , Structurizr DSL , Threat Dragon JSON |

7.8. Security Logging and Observability

| Signal                               | Layer | Owner                | Evidence                                                                |
|--------------------------------------|-------|----------------------|-------------------------------------------------------------------------|
| application security telemetry hooks | code  | FU-ALLOCATION-TRACE  | code cmd/engtrace/main.go                                               |
| application security telemetry hooks | code  | FU-ALLOCATION-TRACE  | code trace_matrix.go                                                    |
| application security telemetry hooks | code  | FU-ALLOCATION-TRACE  | code trace_matrix_test.go                                               |
| application security telemetry hooks | code  | FU-RISK-SCORING      | code examples/payments-engineering-sample/src/support/audit_envelope.ts |
| application security telemetry hooks | code  | FU-VALIDATION-ENGINE | code catalog_description_validation.go                                  |
| application security telemetry hooks | code  | FU-VALIDATION-ENGINE | code catalog_systems_test.go                                            |

| Signal                               | Layer | Owner           | Evidence                                                                                     |
|--------------------------------------|-------|-----------------|----------------------------------------------------------------------------------------------|
| application security telemetry hooks | code  | unresolved      | code examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts  |
| application security telemetry hooks | code  | unresolved      | code examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts       |
| application security telemetry hooks | code  | unresolved      | code examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts |
| fraud and abuse detection code paths | code  | FU-RISK-SCORING | code examples/payments-engineering-sample/src/risk_scorer.ts                                 |
| fraud and abuse detection code paths | code  | unresolved      | code examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs               |

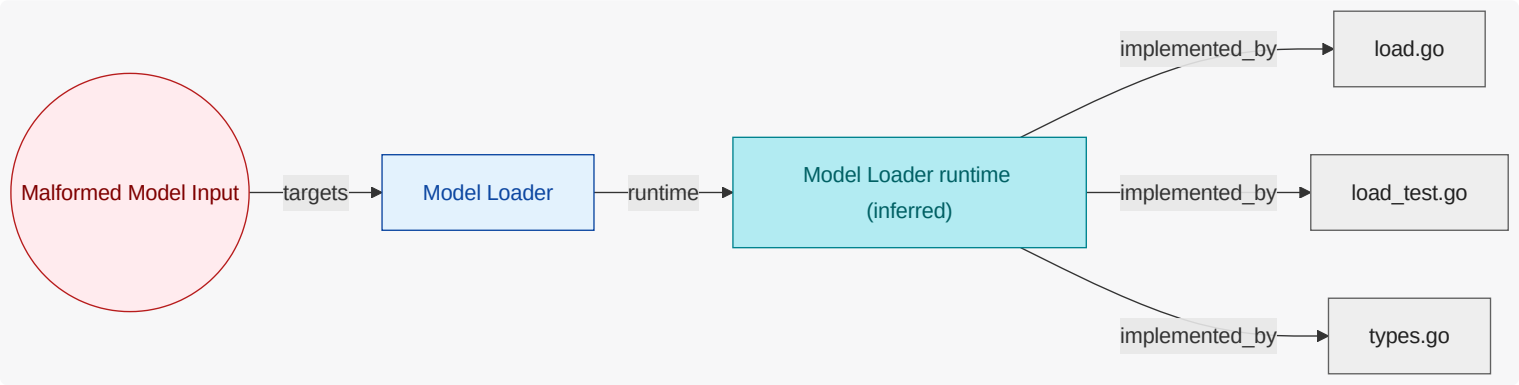
## 7.9. Attack Vector Chapters

### 7.9.1. Malformed Model Input (AV-MALFORMED-MODEL-INPUT)

Invalid or ambiguous model content causing incorrect parsing or graph interpretation.

Mitigated by: none

Trust boundaries: none



### Model Loader (FU-MODEL-LOADER)

Rejects invalid parsing states and unresolved core model references.

#### Attack Surface

none

#### Security Evidence Present

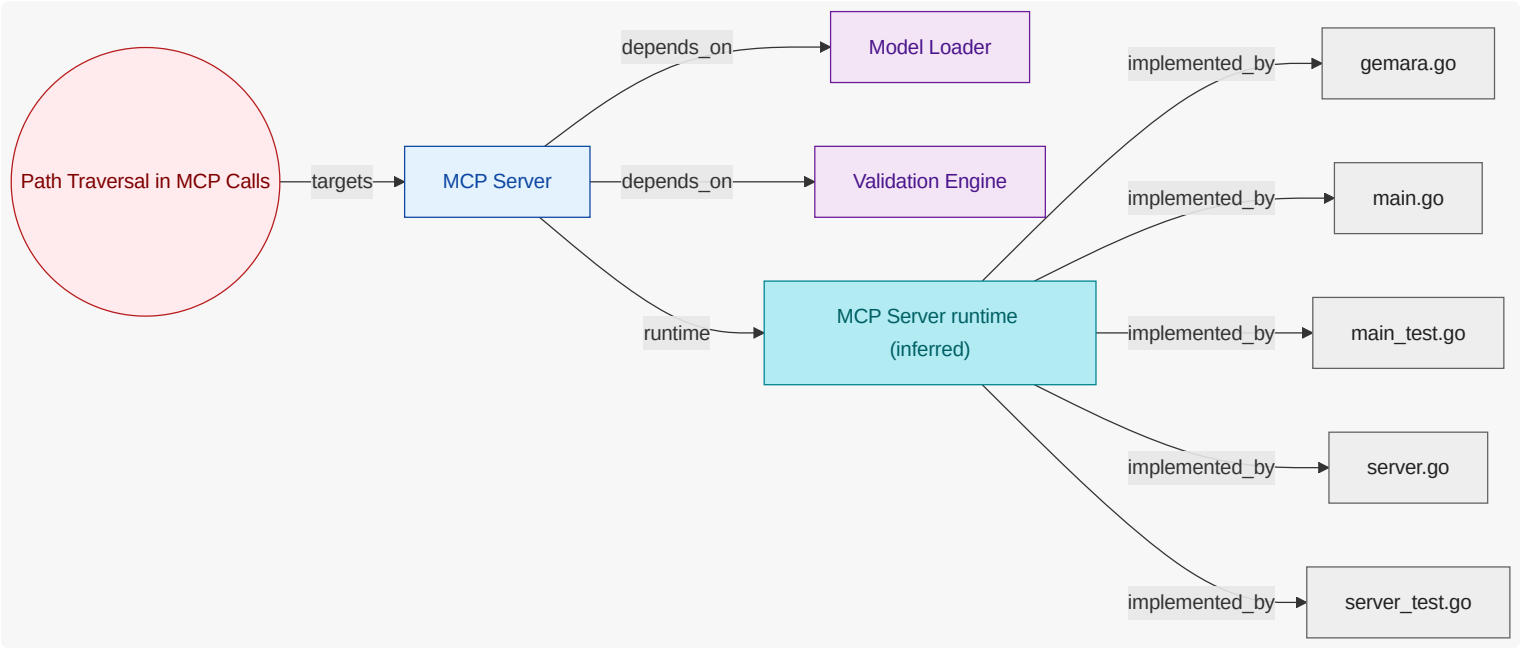
code modules: model

### 7.9.2. Path Traversal in MCP Calls (AV-PATH-TRAVERSAL-IN-MCP)

Attempts to read files outside repository root via MCP path arguments.

Mitigated by: MCP Path Boundary Enforcement, Strict MCP Input Schemas

Trust boundaries: Repository Workspace Boundary



MCP Server (FU-MCP-SERVER)

Enforces strict schemas, panic containment, and repo-root path constraints.

Attack Surface

Model Loader , Validation Engine

Security Evidence Present

code modules: cmd/ engmcp , mcp

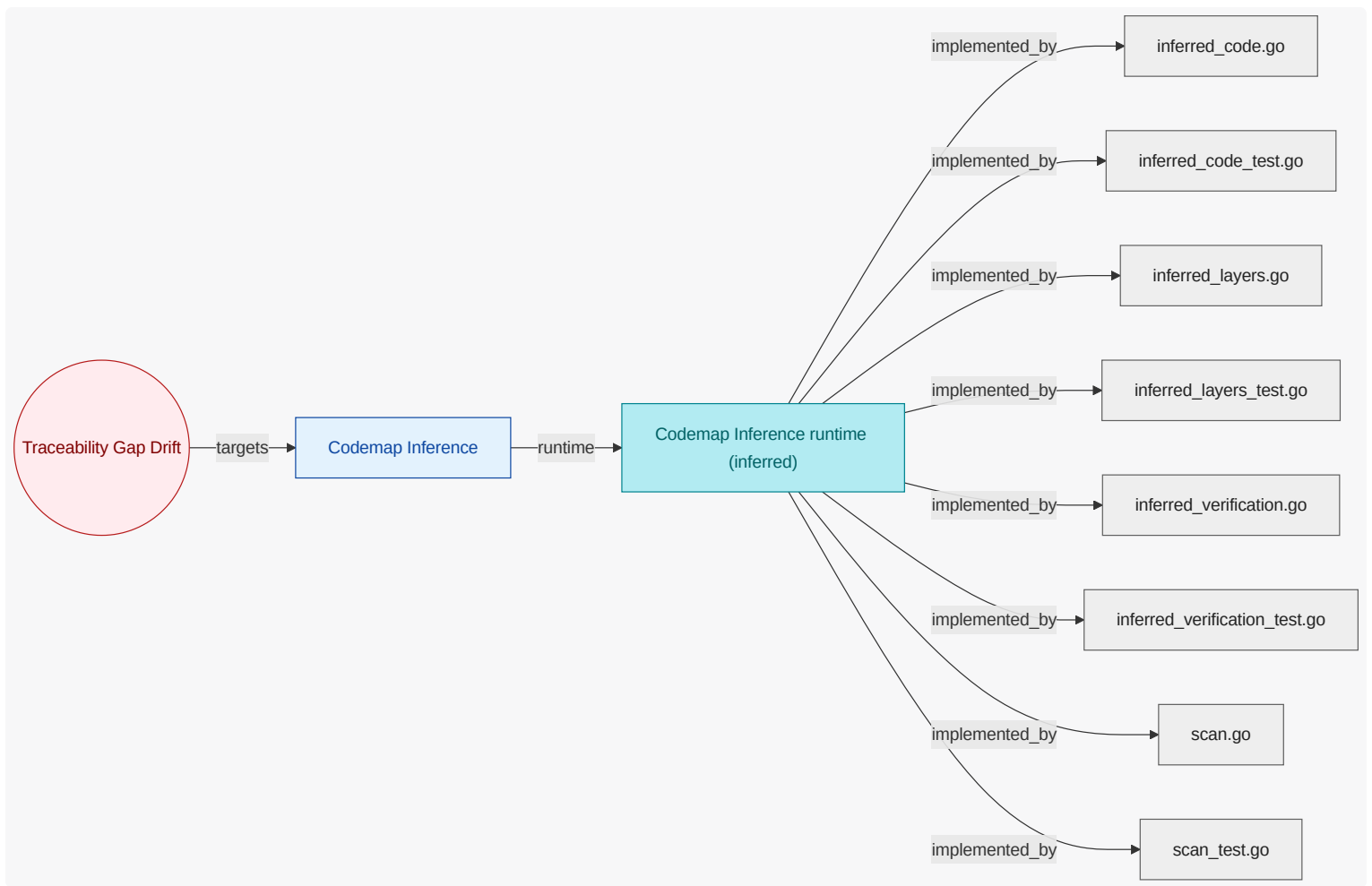
7.9.3. Traceability Gap Drift (AV-TRACEABILITY-GAP-DRIFT)

Requirement and verification links diverge from implementation over time.

Mitigated by: Requirement Traceability Coverage

Trust boundaries: Repository Workspace Boundary





Codemap Inference (FU-CODEMAP-INFERENCE)

Limits inference scope to trusted repo paths and evidence formats.

Attack Surface

none

Security Evidence Present

code modules: codemap, inferred\_code, inferred\_code\_test, inferred\_layers, inferred\_layers\_test, inferred\_verification, inferred\_verification\_test

## 8. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

### 8.1. What This View Answers

- Which requirements map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

### 8.2. Coverage Gaps

Coverage summary: 8/8 units have direct evidence coverage in this view .

- AsciiDoc Generator has no direct inferred runtime evidence yet.
- CLI Orchestration has no direct inferred runtime evidence yet.
- Codemap Inference has no direct inferred runtime evidence yet.
- LOBSTER Exporter has no direct inferred runtime evidence yet.
- MCP Server has no direct inferred runtime evidence yet.
- Model Loader has no direct inferred runtime evidence yet.
- TRLC Exporter has no direct inferred runtime evidence yet.
- Validation Engine has no direct inferred runtime evidence yet.

### 8.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where requirement scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence .

## 9. State Lifecycle View

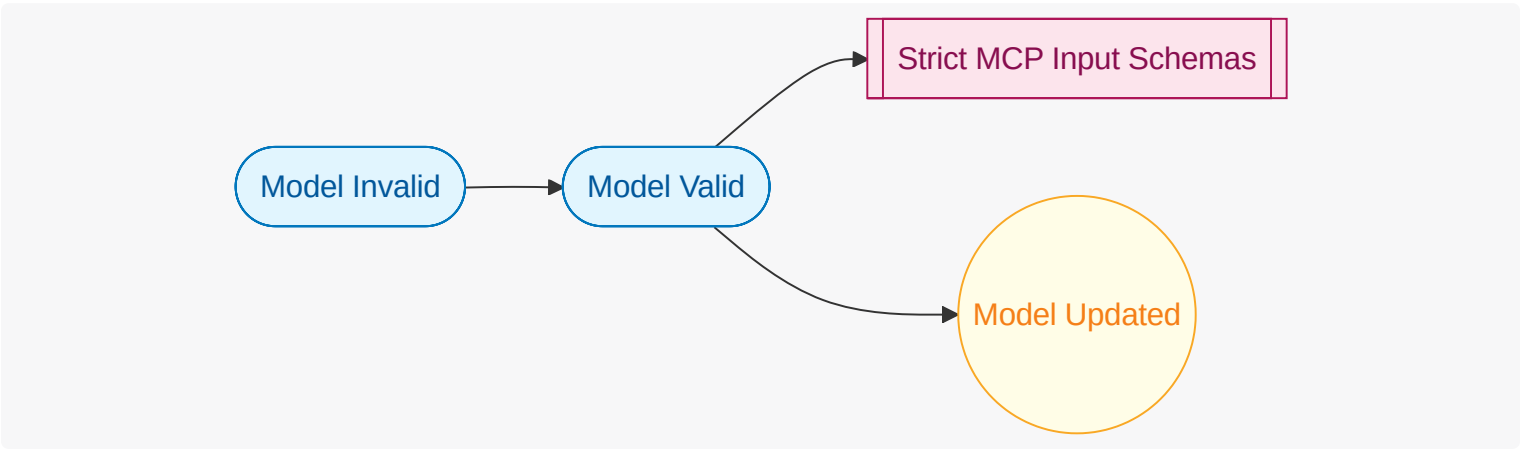
Show workflow state transitions, ownership of transitions, and key exceptional paths.

### 9.1. What This View Answers

- What states exist and which events trigger transitions?
- Which unit owns each state transition?
- Where do automatic and manual flows diverge?

### 9.2. State Lifecycle Diagram

What this diagram shows: authored lifecycle states, triggering events, guard controls, trust boundaries, and lifecycle transitions for this view.



### 9.3. Coverage Gaps

Coverage summary: No functional units are in scope for this view.

- No major coverage gaps detected from current authored and inferred inputs.

### 9.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

### 9.5. Functional Groups and Units (View Scoped)

## 10. Interaction Flow View

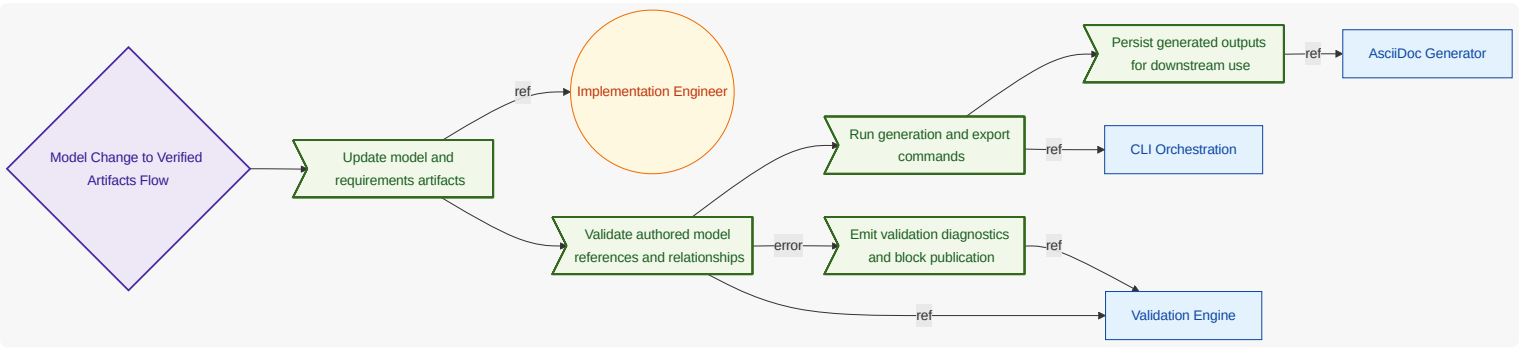
Show user-intent-to-outcome causal flow, including ordered steps, data movement, and error/async branches.

### 10.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

### 10.2. Interaction Flow Diagram

What this diagram shows: authored interaction flow steps, step ordering, async markers, and referenced actors/interfaces/units for this view.



### 10.3. Coverage Gaps

Coverage summary: 3/3 units have direct evidence coverage in this view.

- No major coverage gaps detected from current authored and inferred inputs.

### 10.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

### 10.5. Functional Groups and Units (View Scoped)

### 10.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

#### Functional Unit Columns 1-7

| Requirement | FU-ALLOCATION-TRACE | FU-ASCIIDOC-GENERATOR | FU-CLI-ORCHESTRATION | FU-CODEMAP-INFERENCE | FU-GEMARA-EXPORTER | FU-LOBSTER-EXPORTER | FU-MCP-SERVER |
|-------------|---------------------|-----------------------|----------------------|----------------------|--------------------|---------------------|---------------|
| REQ-EMG-001 |                     | X                     | X                    |                      |                    |                     |               |

| Requirement | FU-<br>ALLOCATION-<br>TRACE | FU-<br>ASCIIDOC-<br>GENERATOR | FU-CLI-<br>ORCHESTRATION | FU-<br>CODEMAP-<br>INFERENCE | FU-<br>GEMARA-<br>EXPORTER | FU-<br>LOBSTER-<br>EXPORTER | FU-<br>MCP-<br>SERVER |
|-------------|-----------------------------|-------------------------------|--------------------------|------------------------------|----------------------------|-----------------------------|-----------------------|
| REQ-EMG-003 |                             | X                             | X                        |                              |                            |                             |                       |
| REQ-EMG-004 |                             |                               | X                        |                              |                            |                             |                       |
| REQ-EMG-005 |                             |                               | X                        |                              |                            |                             |                       |
| REQ-EMG-006 |                             |                               | X                        |                              |                            | X                           |                       |
| REQ-EMG-007 |                             |                               |                          |                              |                            |                             | X                     |
| REQ-EMG-008 |                             |                               |                          |                              |                            |                             | X                     |
| REQ-EMG-009 |                             |                               | X                        |                              |                            |                             |                       |
| REQ-EMG-010 |                             |                               |                          | X                            |                            |                             |                       |
| REQ-EMG-011 |                             |                               |                          |                              |                            |                             |                       |
| REQ-EMG-012 | X                           | X                             |                          | X                            |                            |                             |                       |
| REQ-EMG-013 |                             |                               | X                        |                              |                            |                             |                       |
| REQ-EMG-014 |                             | X                             | X                        |                              |                            |                             |                       |
| REQ-EMG-015 |                             |                               | X                        |                              | X                          |                             | X                     |
| REQ-EMG-016 |                             |                               |                          |                              |                            |                             |                       |
| REQ-EMG-017 |                             |                               |                          |                              |                            |                             |                       |
| REQ-EMG-018 |                             |                               |                          |                              |                            |                             |                       |
| REQ-EMG-019 | X                           |                               |                          |                              |                            |                             |                       |
| REQ-EMG-020 | X                           |                               |                          |                              |                            |                             |                       |
| REQ-EMG-021 | X                           |                               |                          |                              |                            |                             |                       |
| REQ-EMG-022 | X                           |                               |                          |                              |                            |                             |                       |
| REQ-EMG-023 | X                           |                               |                          |                              |                            |                             |                       |
| REQ-EMG-024 |                             | X                             |                          |                              |                            |                             |                       |
| REQ-EMG-025 | X                           | X                             |                          |                              |                            |                             |                       |
| REQ-EMG-026 | X                           | X                             |                          |                              |                            |                             |                       |
| REQ-EMG-027 |                             |                               |                          |                              |                            |                             |                       |

| Requirement | FU-ALLOCATION-TRACE | FU-ASCIIDOC-GENERATOR | FU-CLI-ORCHESTRATION | FU-CODEMAP-INFERENCE | FU-GEMARA-EXPORTER | FU-LOBSTER-EXPORTER | FU-MCP-SERVER |
|-------------|---------------------|-----------------------|----------------------|----------------------|--------------------|---------------------|---------------|
| REQ-EMG-028 | X                   |                       |                      | X                    |                    |                     |               |
| REQ-EMG-029 | X                   |                       |                      |                      |                    |                     |               |
| REQ-EMG-030 | X                   |                       |                      | X                    |                    |                     |               |

Functional Unit Columns 8-14

| Requirement | FU-MODEL-LOADER | FU-OSCAL-EXPORTER | FU-STRUCTURIZR-EXPORTER | FU-SYSTEM-COMPOSITION | FU-THREAT-EXPORTER | FU-TRLC-EXPORTER | FU-VALIDATION-ENGINE |
|-------------|-----------------|-------------------|-------------------------|-----------------------|--------------------|------------------|----------------------|
| REQ-EMG-001 | X               | X                 | X                       |                       | X                  |                  | X                    |
| REQ-EMG-003 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-004 |                 |                   |                         |                       | X                  |                  |                      |
| REQ-EMG-005 |                 |                   | X                       |                       |                    |                  |                      |
| REQ-EMG-006 |                 |                   |                         |                       |                    | X                |                      |
| REQ-EMG-007 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-008 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-009 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-010 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-011 |                 |                   | X                       |                       | X                  |                  | X                    |
| REQ-EMG-012 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-013 |                 | X                 |                         |                       |                    |                  |                      |
| REQ-EMG-014 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-015 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-016 | X               |                   |                         | X                     |                    |                  |                      |
| REQ-EMG-017 |                 |                   |                         | X                     |                    |                  |                      |
| REQ-EMG-018 |                 |                   |                         | X                     |                    |                  | X                    |
| REQ-EMG-019 |                 |                   |                         | X                     |                    |                  |                      |

| Requirement | FU-MODEL-LOADER | FU-OSCAL-EXPORTER | FU-STRUCTURIZR-EXPORTER | FU-SYSTEM-COMPOSITION | FU-THREAT-EXPORTER | FU-TRLC-EXPORTER | FU-VALIDATION-ENGINE |
|-------------|-----------------|-------------------|-------------------------|-----------------------|--------------------|------------------|----------------------|
| REQ-EMG-020 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-021 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-022 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-023 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-024 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-025 |                 |                   |                         |                       |                    |                  |                      |
| REQ-EMG-026 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-027 | X               |                   |                         | X                     |                    |                  |                      |
| REQ-EMG-028 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-029 |                 |                   |                         |                       |                    |                  | X                    |
| REQ-EMG-030 |                 |                   |                         |                       |                    |                  |                      |

Functional Unit Columns 15-15

| Requirement | FU-VIEW-PROJECTION |
|-------------|--------------------|
| REQ-EMG-001 |                    |
| REQ-EMG-003 | X                  |
| REQ-EMG-004 |                    |
| REQ-EMG-005 |                    |
| REQ-EMG-006 |                    |
| REQ-EMG-007 |                    |
| REQ-EMG-008 |                    |
| REQ-EMG-009 |                    |
| REQ-EMG-010 |                    |
| REQ-EMG-011 |                    |
| REQ-EMG-012 |                    |

| Requirement | FU-VIEW-PROJECTION |
|-------------|--------------------|
| REQ-EMG-013 |                    |
| REQ-EMG-014 |                    |
| REQ-EMG-015 |                    |
| REQ-EMG-016 |                    |
| REQ-EMG-017 |                    |
| REQ-EMG-018 |                    |
| REQ-EMG-019 |                    |
| REQ-EMG-020 |                    |
| REQ-EMG-021 |                    |
| REQ-EMG-022 |                    |
| REQ-EMG-023 |                    |
| REQ-EMG-024 | X                  |
| REQ-EMG-025 |                    |
| REQ-EMG-026 |                    |
| REQ-EMG-027 |                    |
| REQ-EMG-028 |                    |
| REQ-EMG-029 |                    |
| REQ-EMG-030 |                    |

### 10.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

| Check ID                                           | Requirement | Result  | Evidence / Notes                                               |
|----------------------------------------------------|-------------|---------|----------------------------------------------------------------|
| VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2 | REQ-COF-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8   | REQ-COF-002 | not-run | none; No parsed test-result artifact for this requirement yet. |



| Check ID                                           | Requirement | Result  | Evidence / Notes                                               |
|----------------------------------------------------|-------------|---------|----------------------------------------------------------------|
| VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC                | REQ-COF-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB              | REQ-COF-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB              | REQ-COF-004 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC                | REQ-COF-005 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8   | REQ-COF-007 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7           | REQ-EMG-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-E2E_TEST-BE56F8                       | REQ-EMG-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26            | REQ-EMG-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-LOAD_TEST-A58EB3                      | REQ-EMG-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-VALIDATE_TEST-498C59                  | REQ-EMG-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84    | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2    | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_TEST-F60765                  | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2            | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |

| Check ID                                       | Requirement | Result  | Evidence / Notes                                               |
|------------------------------------------------|-------------|---------|----------------------------------------------------------------|
| VER-INF-TEST-PROJECT_TEST-2ECF72               | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-RENDER_TEST-EECC23                | REQ-EMG-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D   | REQ-EMG-004 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374    | REQ-EMG-005 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E        | REQ-EMG-006 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-TRLC_EXPORT_TEST-DD1345           | REQ-EMG-006 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-MAIN_TEST-6B61AE                  | REQ-EMG-007 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-SERVER_TEST-389957                | REQ-EMG-007 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-MAIN_TEST-6B61AE                  | REQ-EMG-008 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-SERVER_TEST-389957                | REQ-EMG-008 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26        | REQ-EMG-009 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-VALIDATE_TEST-498C59              | REQ-EMG-009 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-INFERRED_CODE_TEST-3CA82E         | REQ-EMG-010 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF       | REQ-EMG-010 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6 | REQ-EMG-010 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-SCAN_TEST-B04F81                  | REQ-EMG-010 | not-run | none; No parsed test-result artifact for this requirement yet. |

| Check ID                                     | Requirement | Result  | Evidence / Notes                                               |
|----------------------------------------------|-------------|---------|----------------------------------------------------------------|
| VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D | REQ-EMG-011 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-VALIDATE_TEST-498C59            | REQ-EMG-011 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9         | REQ-EMG-013 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-OSCAL_SSP_TEST-4B08EF           | REQ-EMG-013 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-ASCIIDOC_TEST-F60765            | REQ-EMG-014 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-GEMARA_EXPORT_TEST-45C982       | REQ-EMG-015 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA        | REQ-EMG-015 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-016 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-017 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-020 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-022 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-023 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-025 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-COMPOSITION_TEST-8FE8F1         | REQ-EMG-027 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C        | REQ-EMG-028 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C        | REQ-EMG-029 | not-run | none; No parsed test-result artifact for this requirement yet. |

| Check ID                                                 | Requirement | Result  | Evidence / Notes                                               |
|----------------------------------------------------------|-------------|---------|----------------------------------------------------------------|
| VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C                    | REQ-EMG-030 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367              | REQ-PAY-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163               | REQ-PAY-002 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671       | REQ-PAY-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD         | REQ-PAY-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163               | REQ-PAY-004 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4 | REQ-PAY-005 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671       | REQ-PAY-006 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD         | REQ-PAY-008 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6                 | REQ-PRR-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6                 | REQ-PRR-002 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6                 | REQ-PRR-003 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003     | REQ-PRR-004 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6                 | REQ-PRR-005 | not-run | none; No parsed test-result artifact for this requirement yet. |

| Check ID                                             | Requirement    | Result  | Evidence / Notes                                               |
|------------------------------------------------------|----------------|---------|----------------------------------------------------------------|
| VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F   | REQ-PRR-005    | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003 | REQ-PRR-006    | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4      | REQ-PRR-007    | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE         | REQ-PRR-008    | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-SCAN_TEST-B04F81                        | REQ-SAMPLE-001 | not-run | none; No parsed test-result artifact for this requirement yet. |
| VER-INF-TEST-SCAN_TEST-B04F81                        | REQ-SAMPLE-002 | not-run | none; No parsed test-result artifact for this requirement yet. |

# 11. Traceability Appendix

## 11.1. Requirement Details

### 11.1.1. REQ-EMG-001

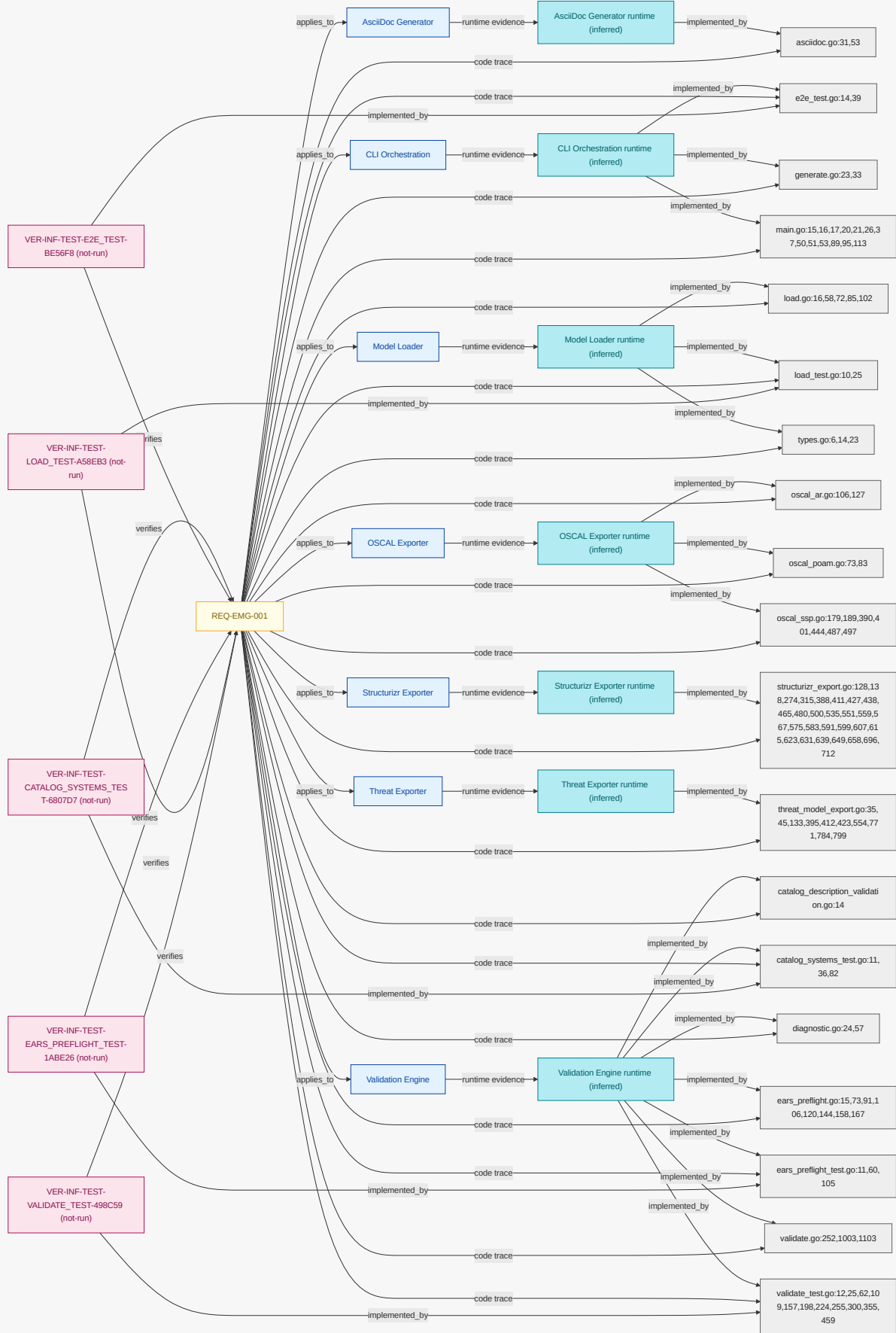
While local development mode is active , when model updated event is received , the engineering model go shall validate architecture model before generation run is requested .

#### NOTE

Validation must gate all generation flows.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 11.1.2. REQ-EMG-003

While model is valid , when generation run is requested , the engineering model go shall produce AsciiDoc architecture publication output.

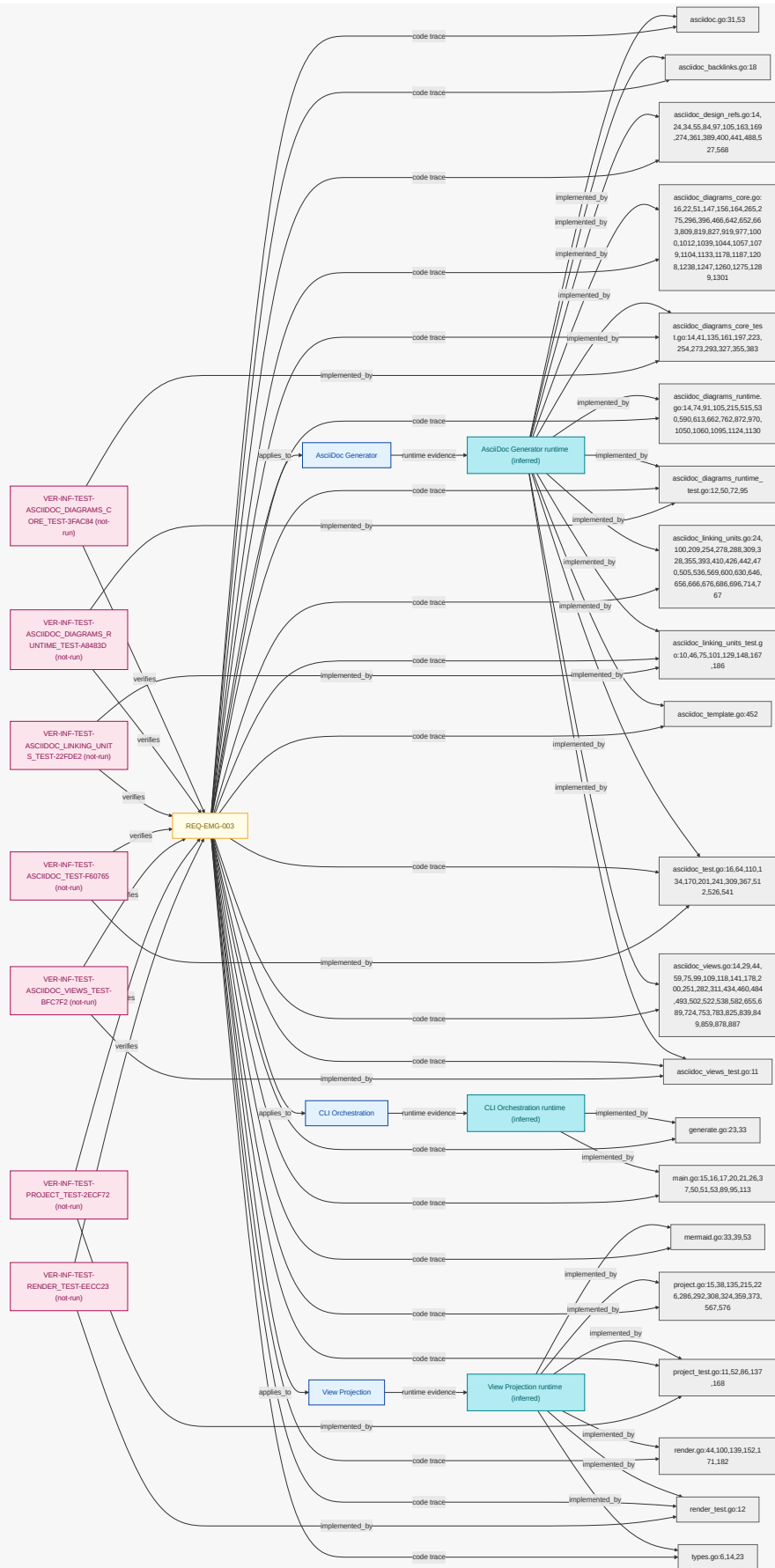
NOTE

Human-readable architecture output remains required.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.





### 11.1.3. REQ-EMG-004

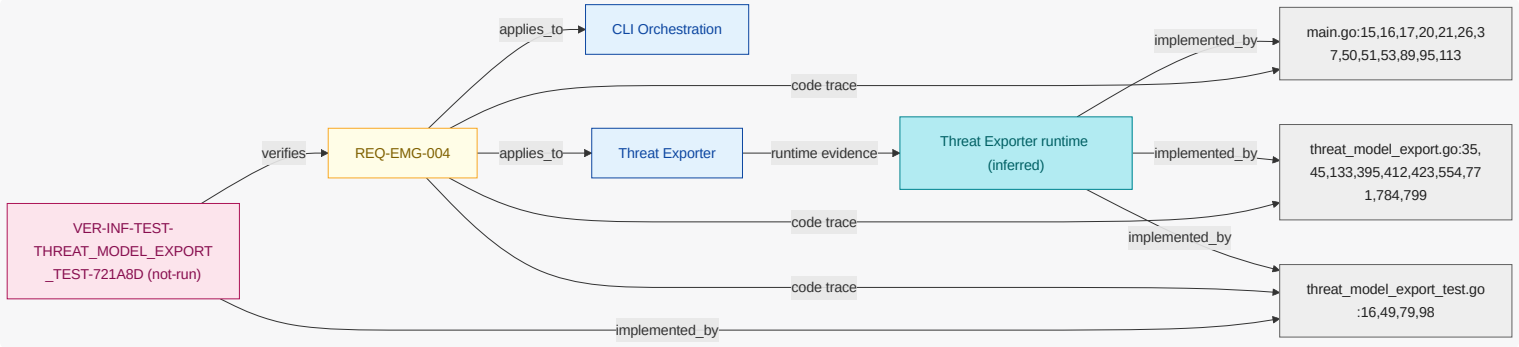
Where threat export is enabled , when generation run is requested , the engineering model go shall export threat dragon json artifact plus open threat model artifact.

NOTE

Threat model interchange output contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.4. REQ-EMG-005

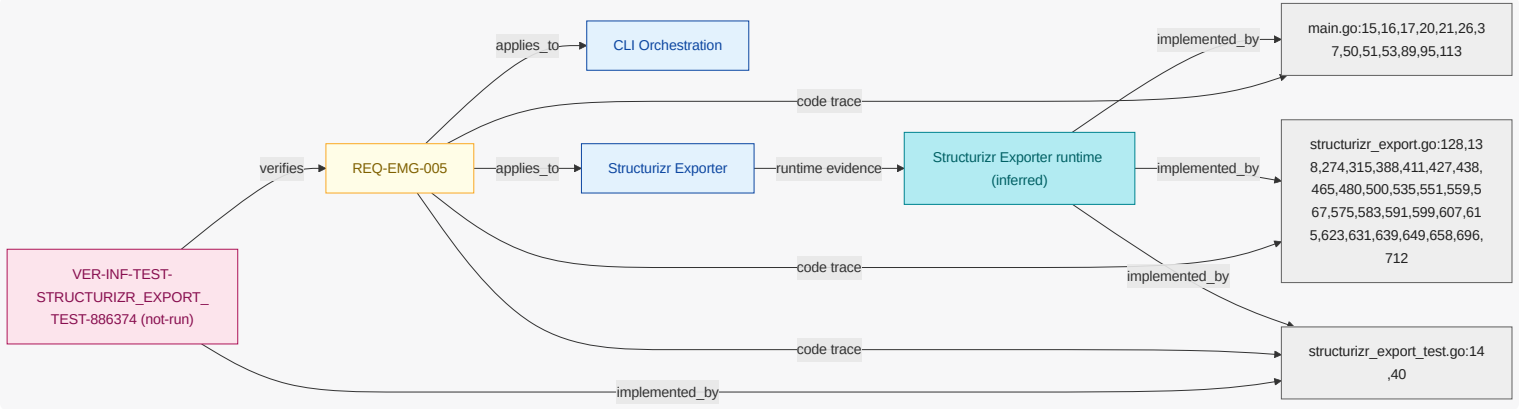
Where structurizr export is enabled , when generation run is requested , the engineering model go shall export structurizr dsl artifact .

NOTE

Structurizr DSL is exported for C4 visualization; the exported DSL is validated best-effort by the validate-all gauntlet when a Structurizr validator is available.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.5. REQ-EMG-006

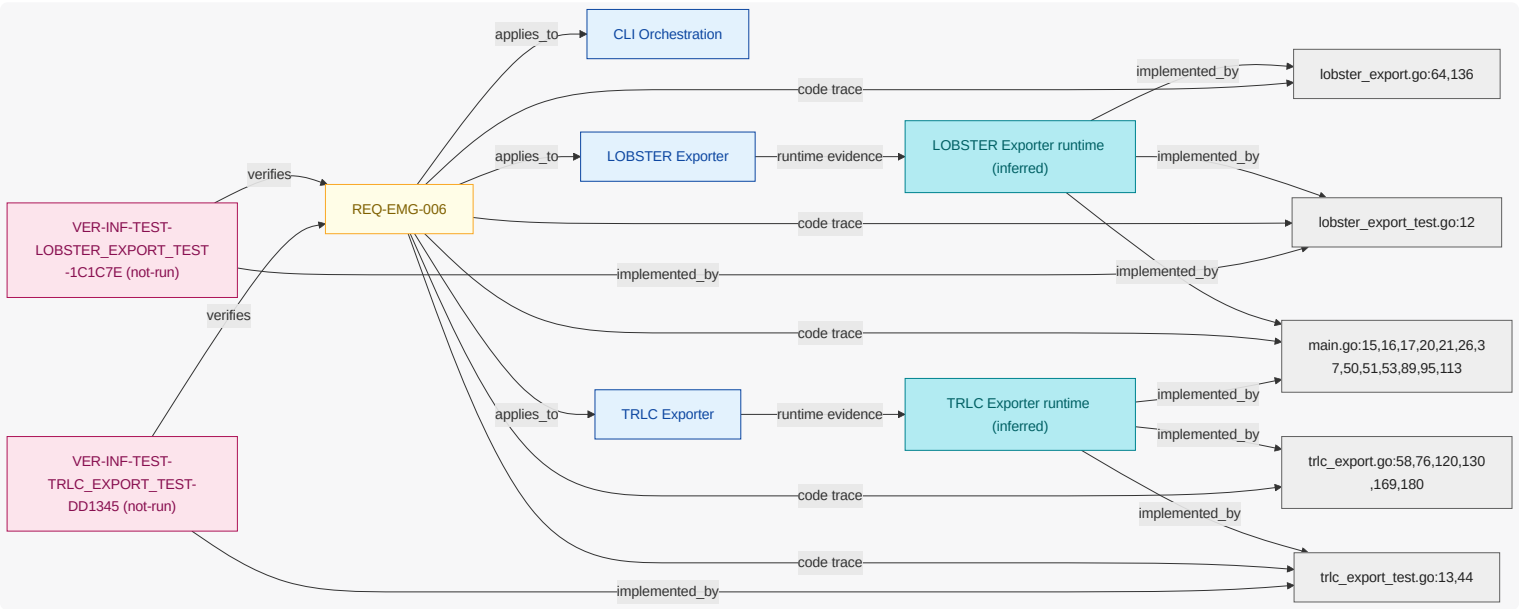
Where traceability export is enabled , when generation run is requested , the engineering model go shall produce trlc package and lobster trace report artifacts.

NOTE

Traceability outputs required for assurance workflows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



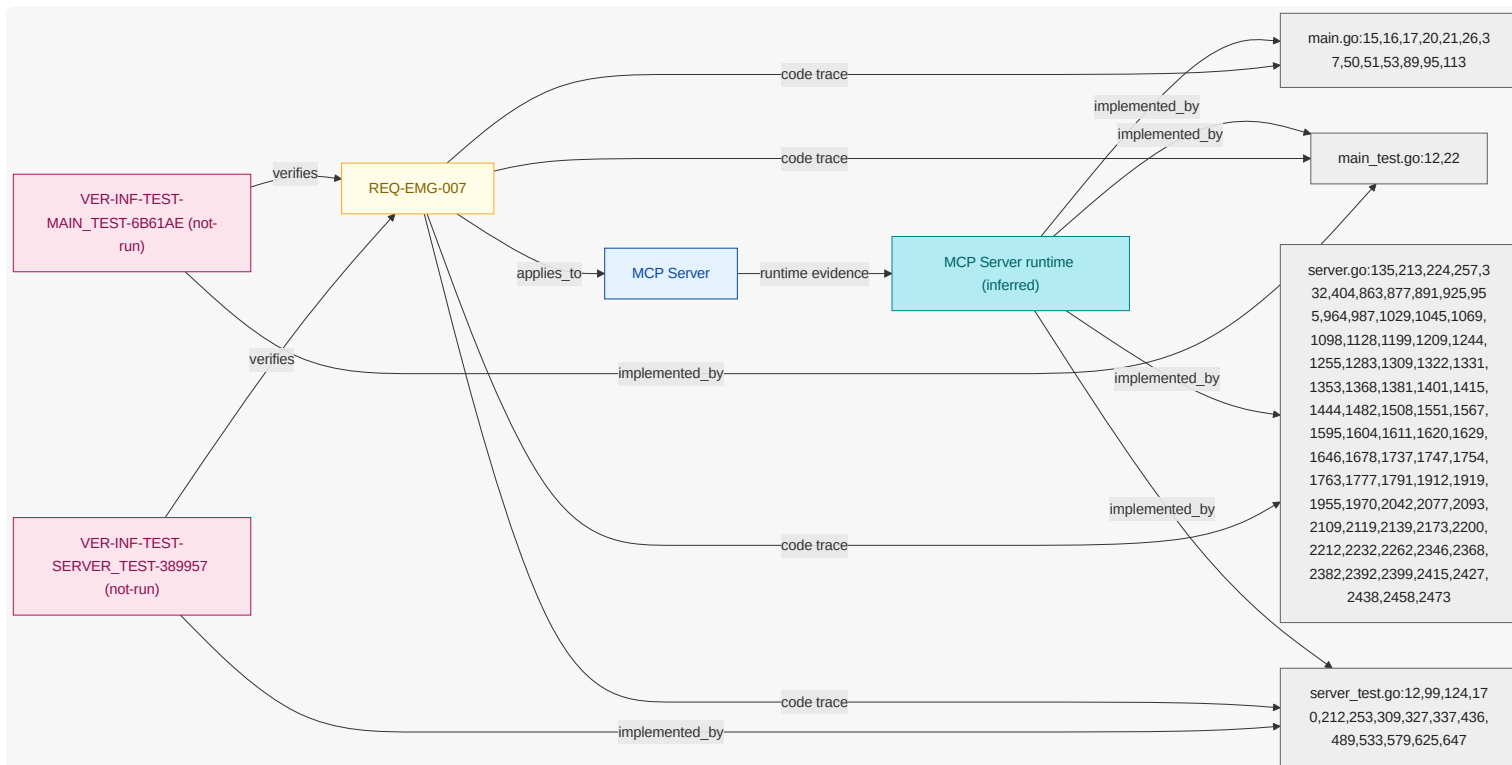
11.1.6. REQ-EMG-007

While mcp server is enabled , when mcp tool call is received and catalog references resolve successfully , the engineering model go shall return mcp tool result with schema version and structured error fields.

**NOTE** MCP response contract stability requirement .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 11.1.7. REQ-EMG-008

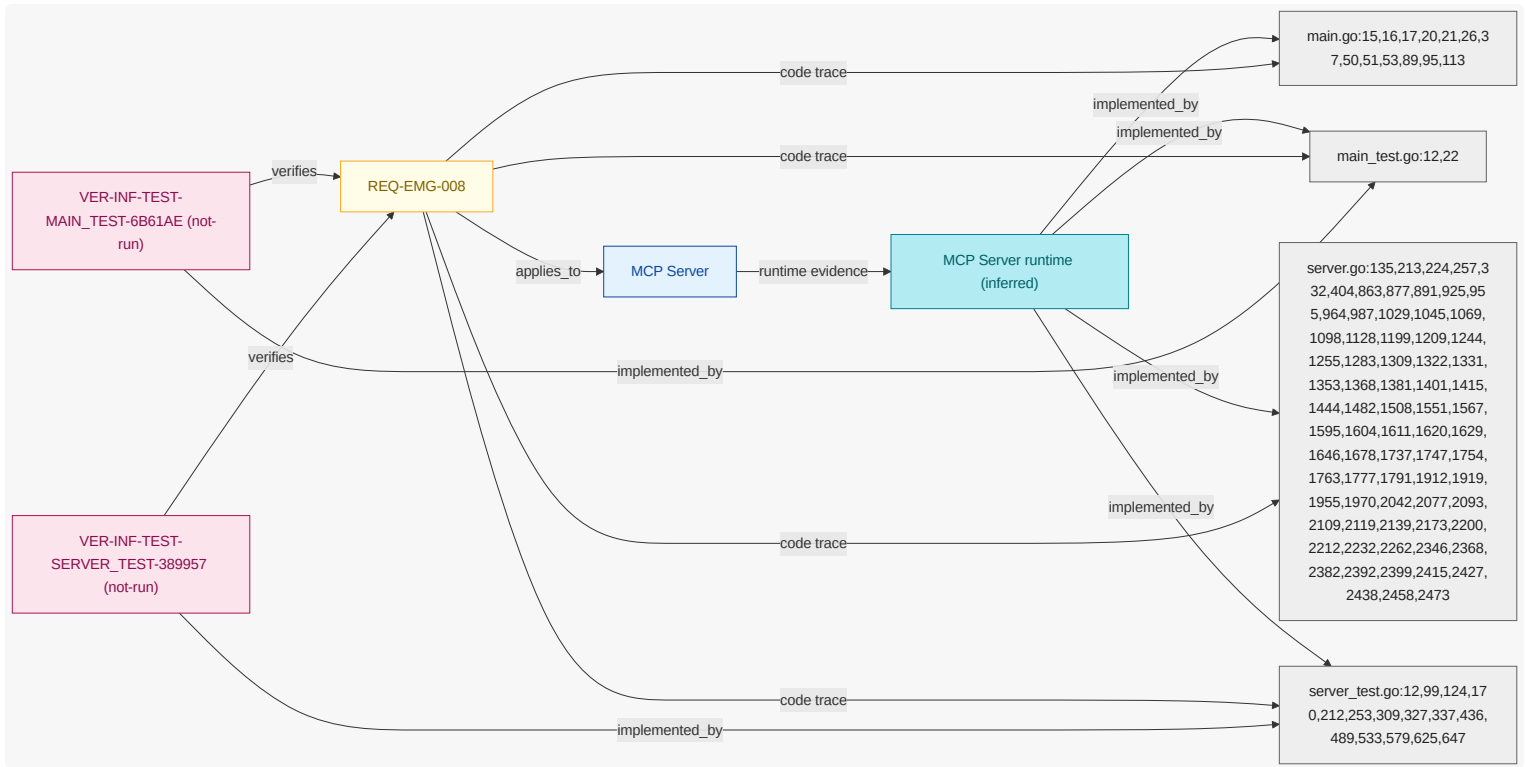
When `mcp tool call` is received, the `engineering model go` shall reject path arguments outside `repository workspace boundary`.

#### NOTE

Filesystem `trust boundary` guardrail.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



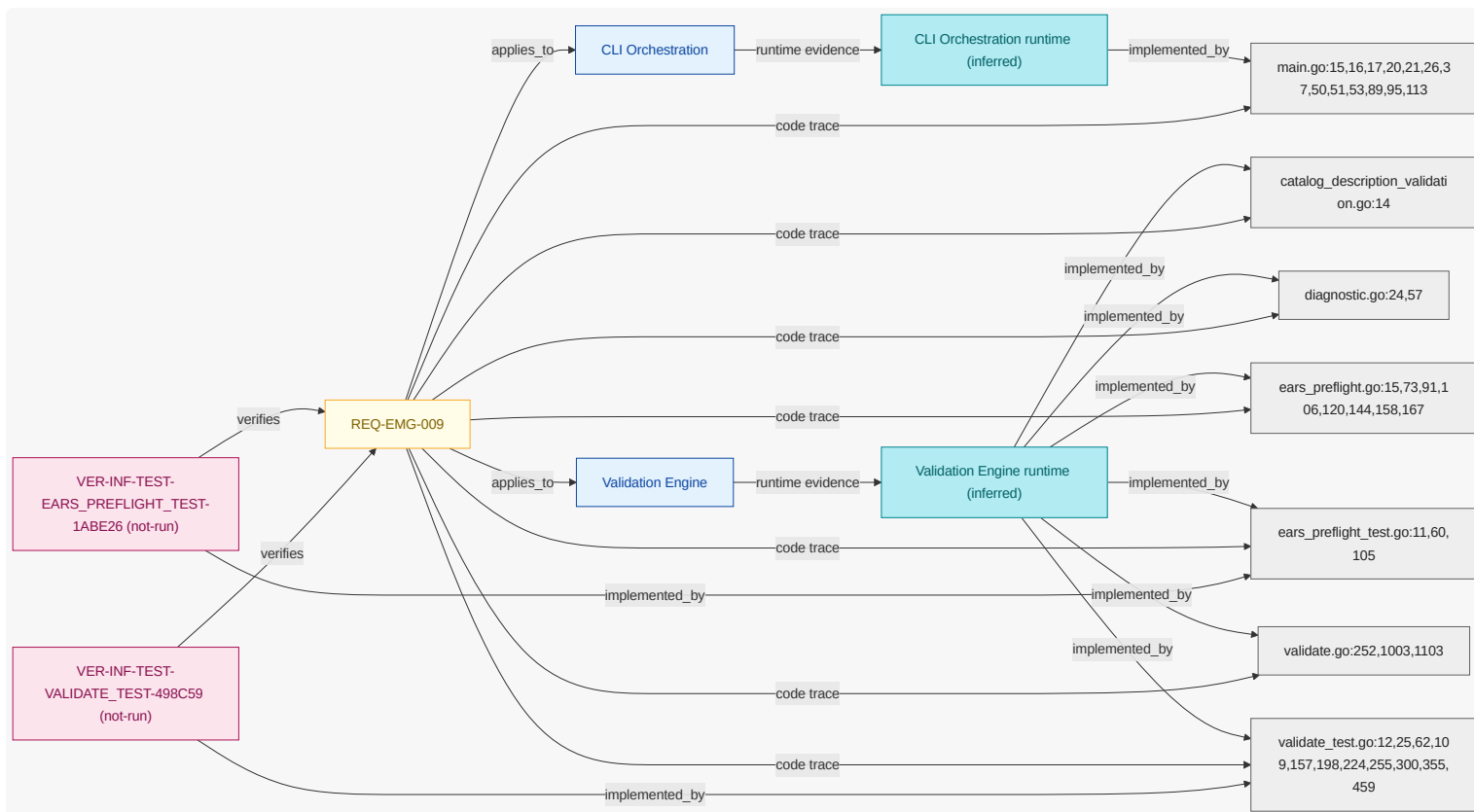
### 11.1.8. REQ-EMG-009

While `ci validation mode` is active, when `generation run` is requested, the `engineering model go` shall execute `validation plus tests` before `artifact generated event` is raised.

**NOTE** | CI gate behavior for deterministic quality.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 11.1.9. REQ-EMG-010

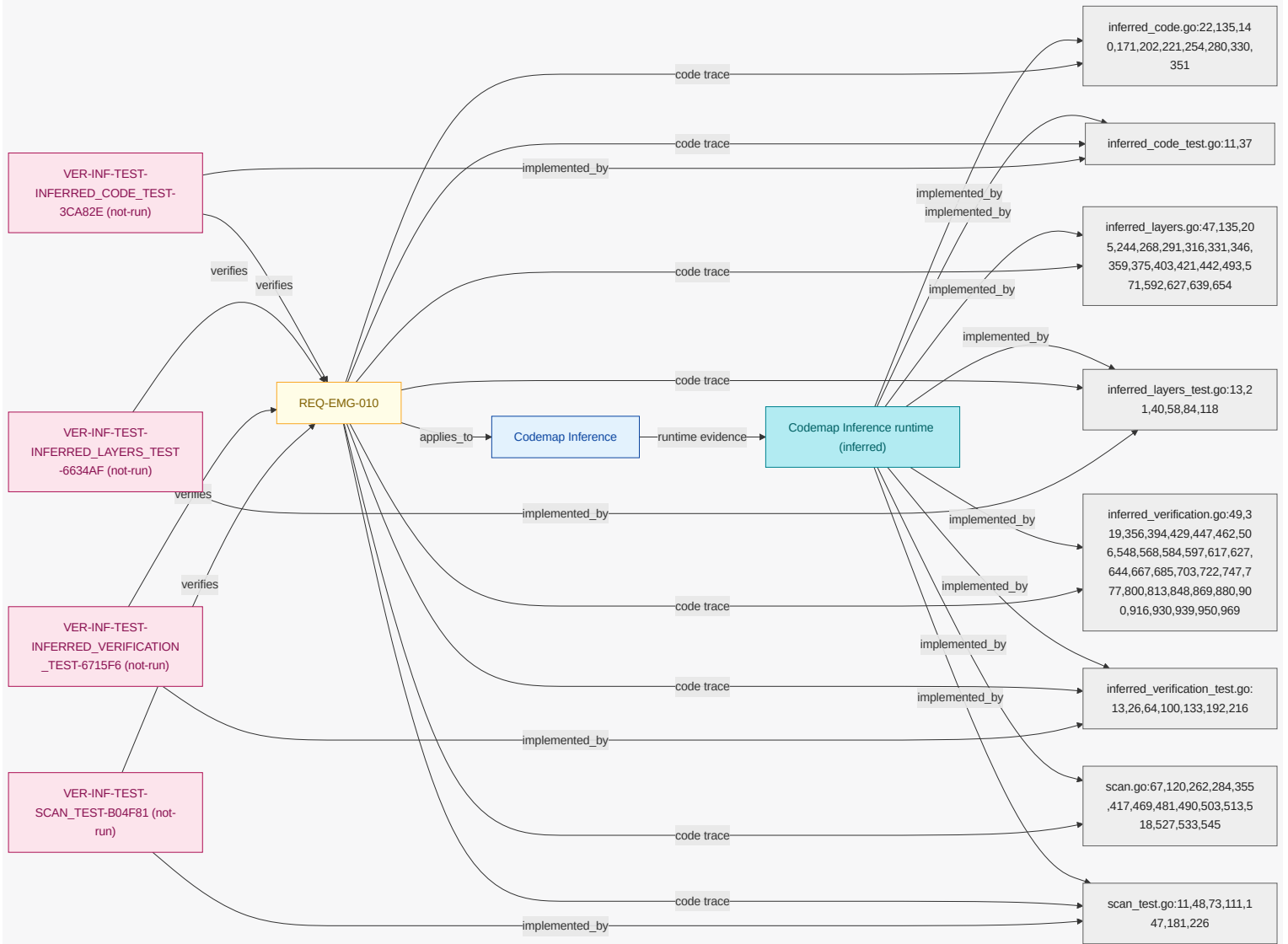
Where requirement trace is complete , when model updated event is received , the engineering model go shall infer traceability coverage from trace markers .

#### NOTE

Links runtime, code, and test markers to modeled requirement coverage. Code and metadata scanning excludes dependency caches (node\_modules, vendor) and dot-directories (.git, .engmod, tool caches) so coverage reflects only committed first-party source and stays reproducible.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.

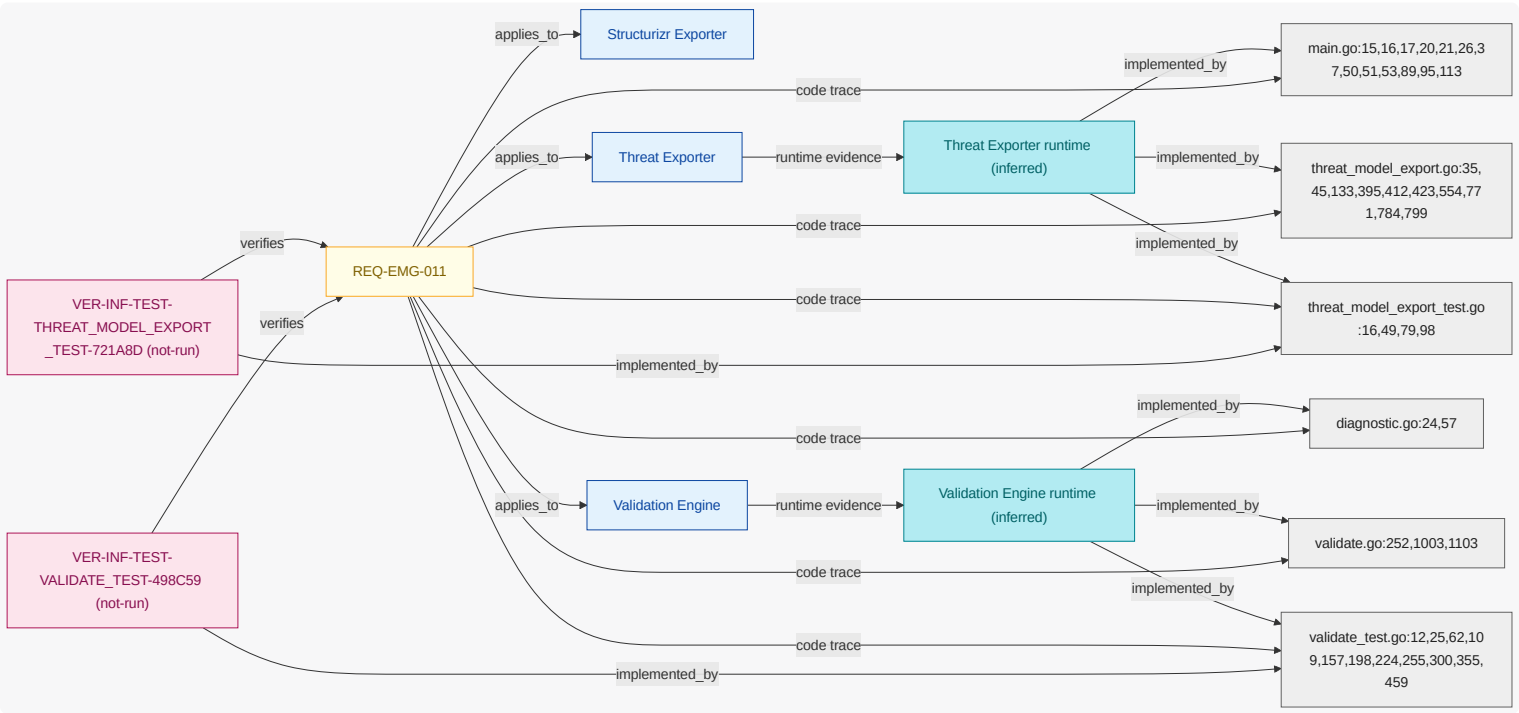


Where external schema dependencies are available, when generation run is requested, the engineering model go shall fail validation if schema checks do not match expected constraints.

**NOTE** Protect export integrity when schema trust fails.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.11. REQ-EMG-012

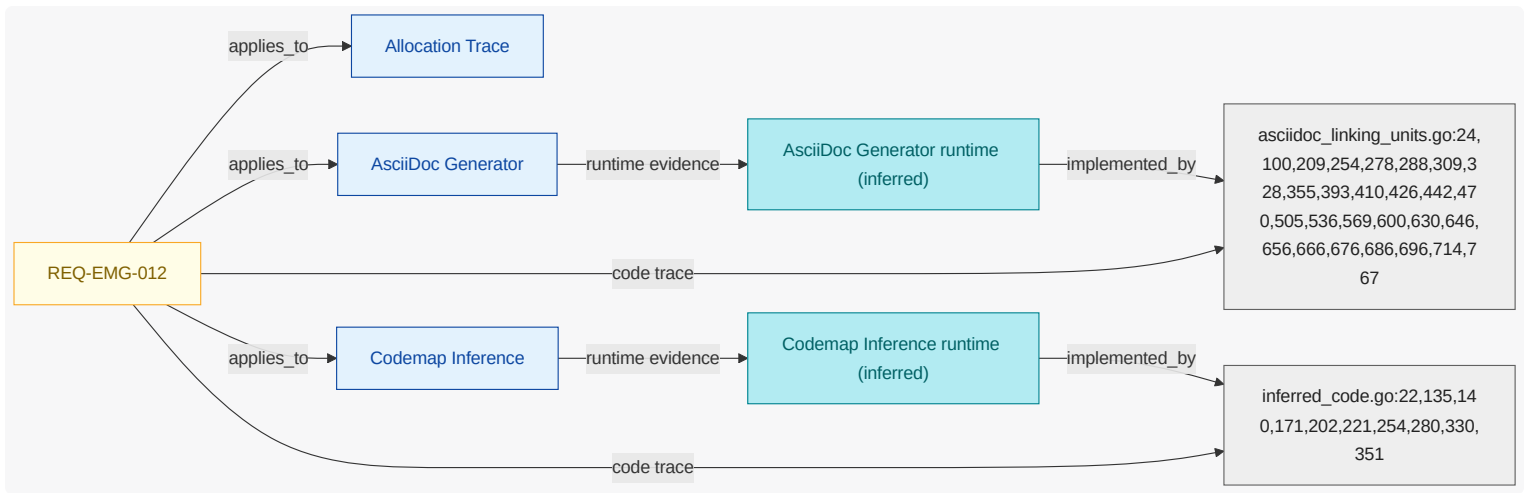
When generation run is requested, the engineering model go shall generate artifacts deterministically so that regeneration is reproducible.

**NOTE** Generation uses stable ordering (sorted diagnostics, link targets, scan results) and excludes environment-specific inputs, so regenerated documents are bit-for-bit reproducible and the CI freshness gate can detect drift.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.





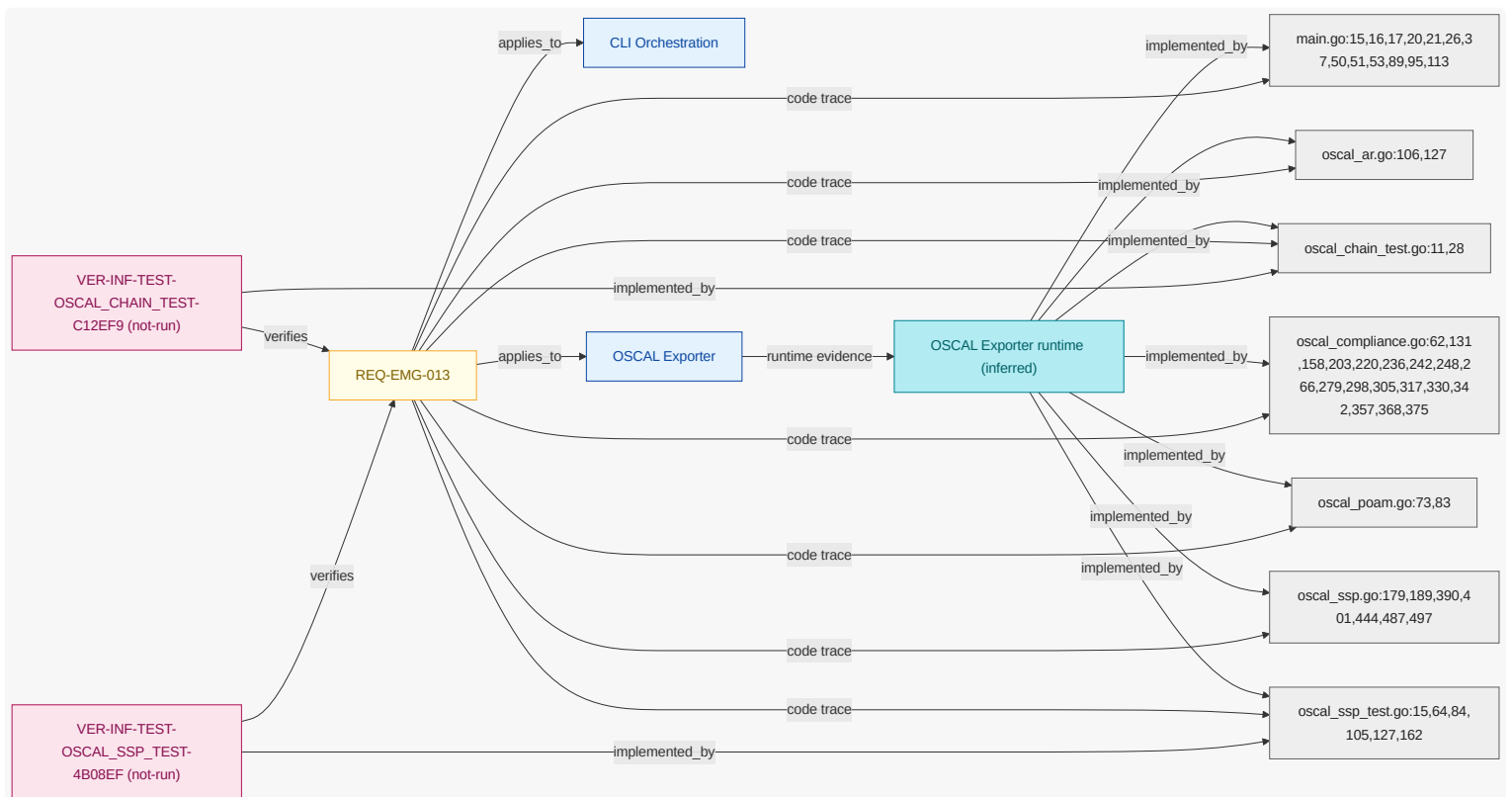
### 11.1.12. REQ-EMG-013

Where traceability export is enabled, when generation run is requested, the engineering model go shall export OSCAL SSP plus assessment results plus POA&M artifacts.

**NOTE** OSCAL export set for compliance workflows.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 11.1.13. REQ-EMG-014

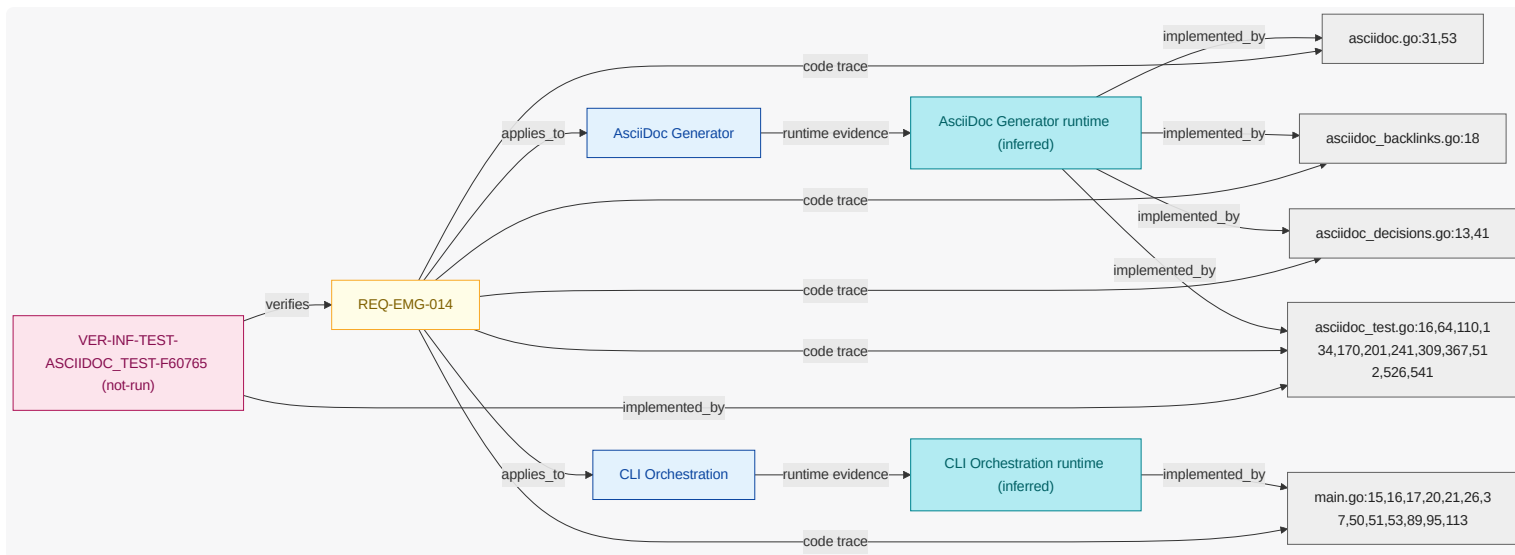
Where architecture decision records are authored, when generation run is requested, the engineering model go shall publish architecture decision records document linked from AsciiDoc architecture publication output.

## NOTE

ADRs are first-class `model` content but are published separately from the main architecture narrative.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 11.1.14. REQ-EMG-015

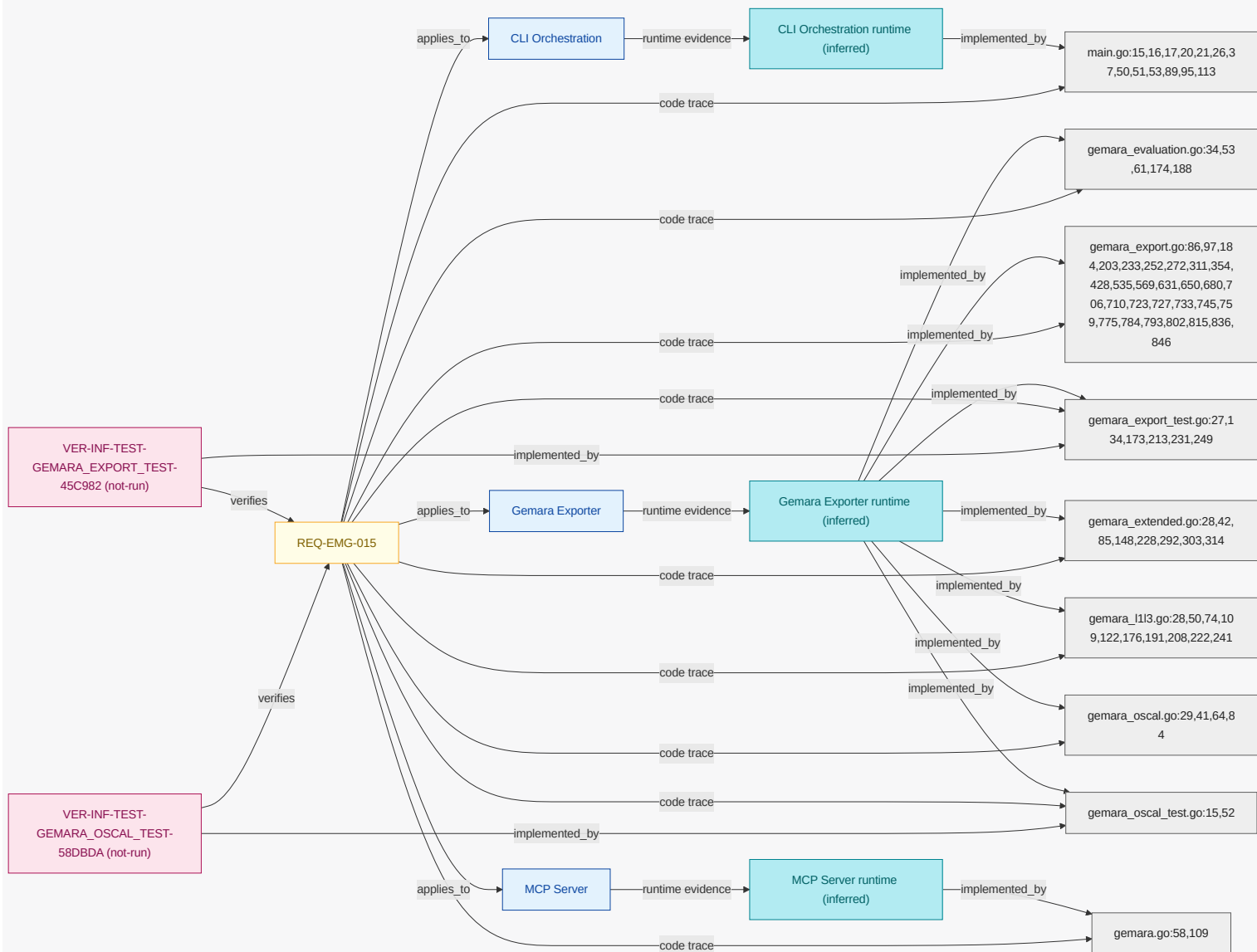
Where `gemara export` is enabled, when `generation run` is requested, the `engineering model go` shall render OpenSSF Gemara `model` artifacts.

## NOTE

Gemara is a first-class GRC rendering of the `model`, validated against the published Gemara schemas via the `go-gemara` SDK.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



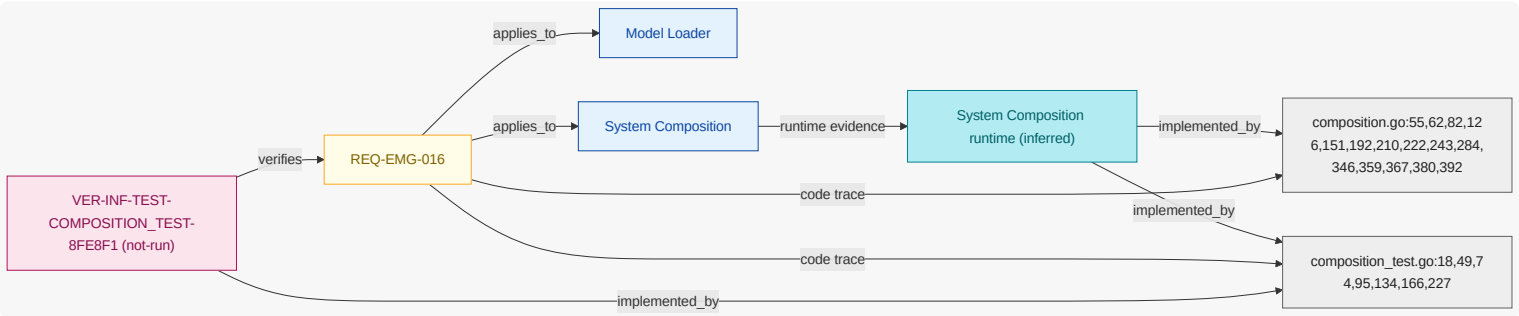
Where system composition is enabled , when generation run is requested , the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

NOTE

Recursive system-of-systems composition ; local subdirectory references, with external git repositories cloned into the .engmod cache ( REQ-EMG-027 , ADR-EMG-008, ADR-EMG-010).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.16. REQ-EMG-017

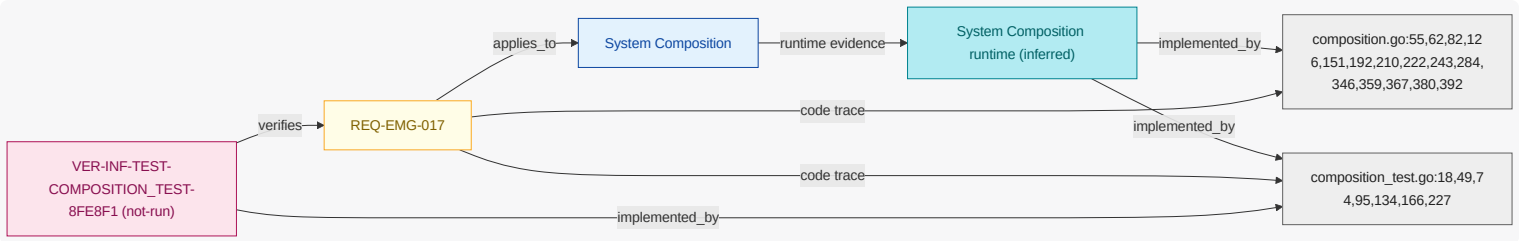
Where system composition is enabled , when generation run is requested , the engineering model go shall reject any subsystem reference that resolves outside the repository workspace boundary .

NOTE

Downward-only references remain within the workspace; mirrors the MCP path-boundary guardrail.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.17. REQ-EMG-018

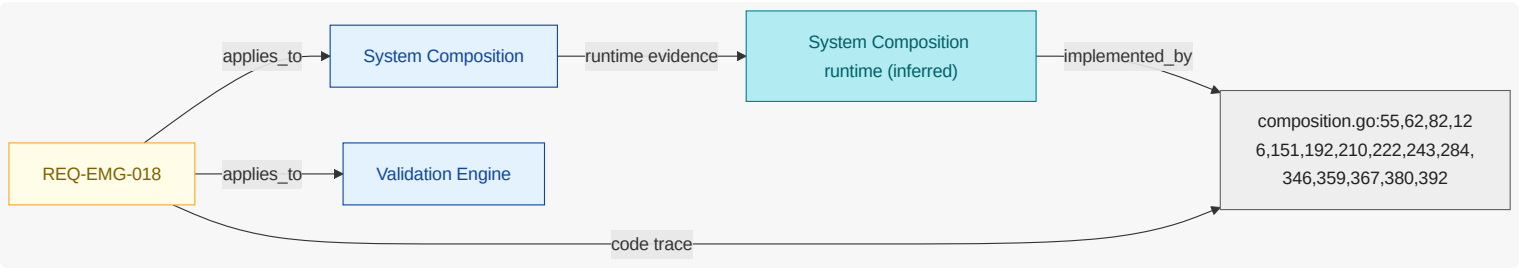
Where system composition is enabled , when generation run is requested , the engineering model go shall reject a subsystem reference graph that contains a cycle.

NOTE

Subsystem references must form a directed acyclic graph (ADR-EMG-009).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



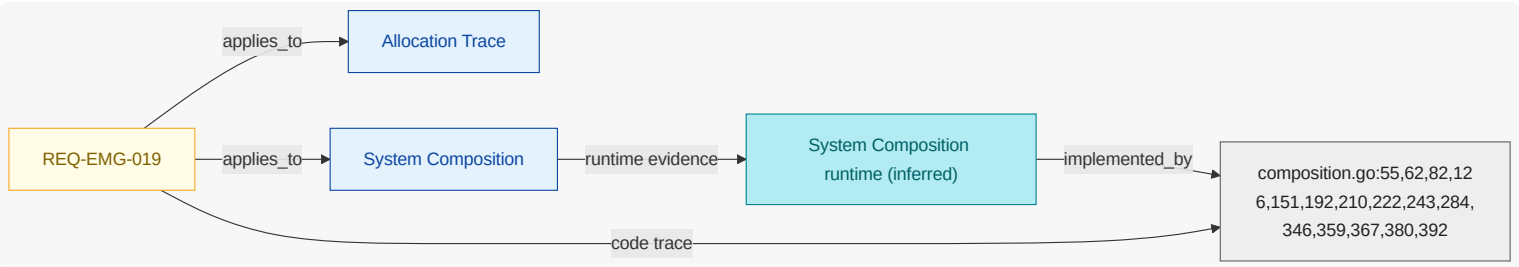
11.1.18. REQ-EMG-019

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve cross-system references using subsystem-qualified identifiers.

**NOTE** Identifiers are scoped per system; cross-system references are qualified by subsystem to avoid collisions.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



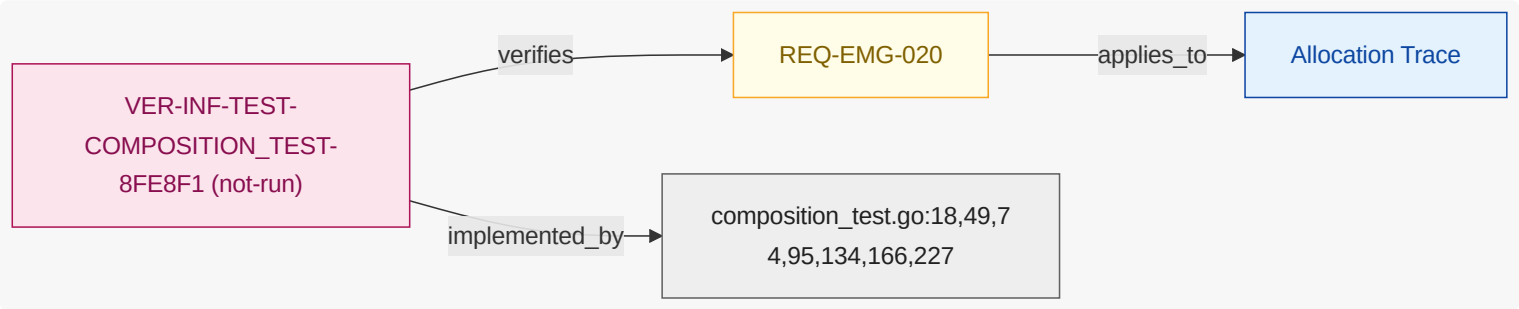
11.1.19. REQ-EMG-020

Where system composition is enabled, when generation run is requested, the engineering model go shall materialize parent-to-subsystem allocation trace ability in the composed output without modifying any subsystem model.

**NOTE** A subsystem is parent-agnostic; parent allocation is materialized only in composed artifacts, never written back to the child (ADR-EMG-009, ADR-EMG-011).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.20. REQ-EMG-021

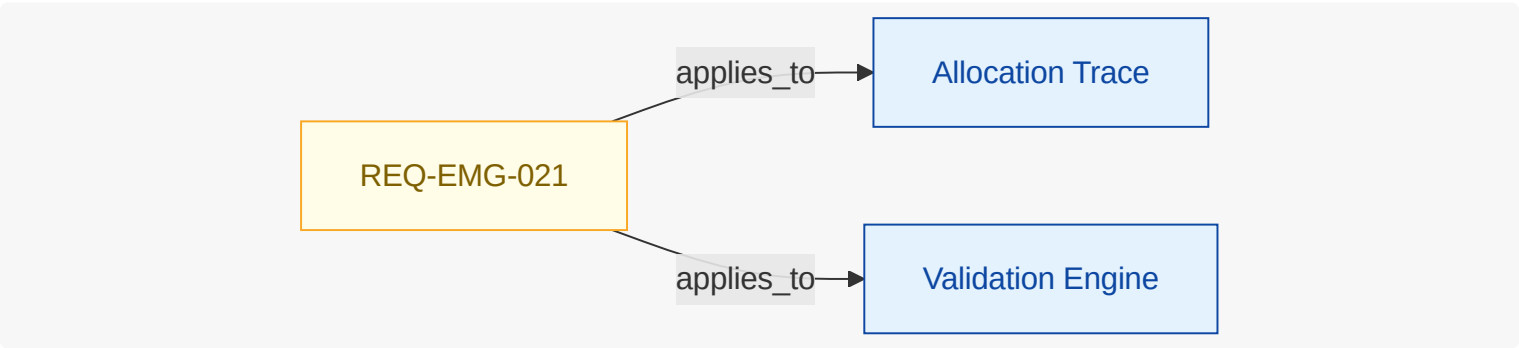
Where system composition is enabled , when generation run is requested , the engineering model go shall report each subsystem required interface that the parent system does not satisfy.

NOTE

Subsystems publish a provides-and-requires contract; the parent must satisfy each required interface .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.21. REQ-EMG-022

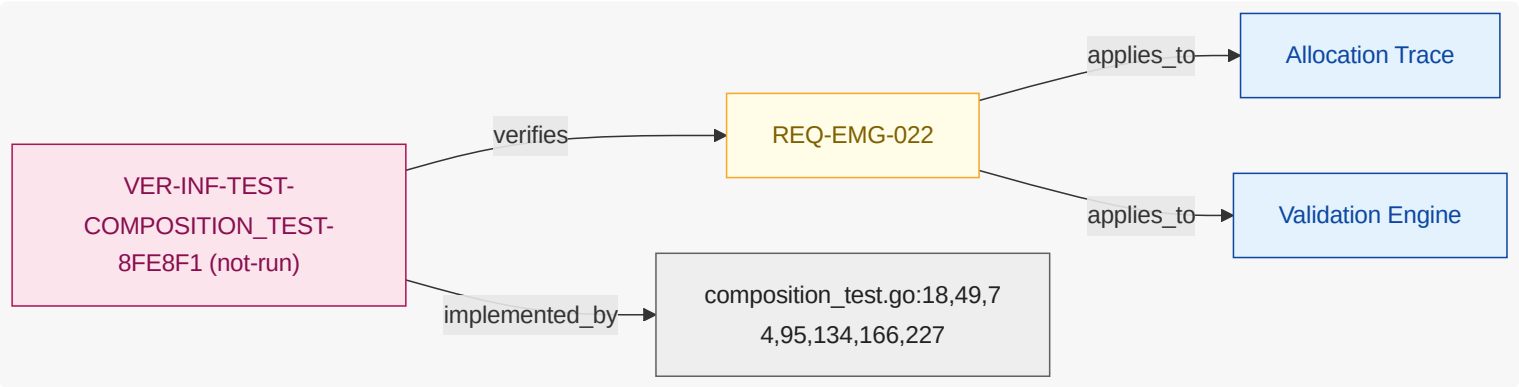
Where system composition is enabled , when generation run is requested , the engineering model go shall reject an allocation that targets a subsystem identifier not published in the subsystem contract.

NOTE

Parents may bind only to a subsystem’s public contract surface, not its internal identifiers.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.22. REQ-EMG-023

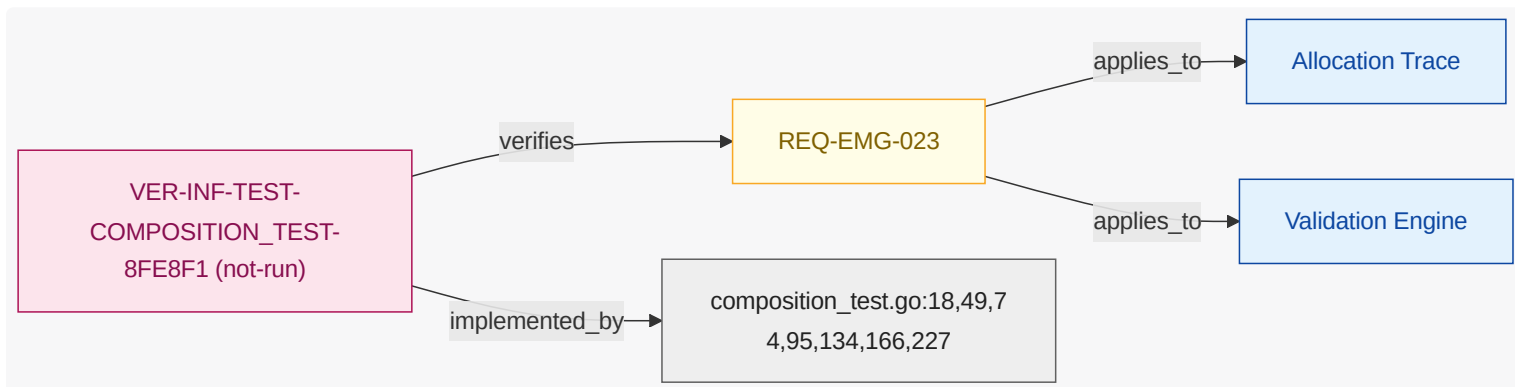
When generation run is requested , the engineering model go shall flag any requirement that links to another requirement in the same document.

## NOTE

Requirements carry no tiers. A requirement is either delegated to a subsystem through a downward composition allocation or implemented here, and the delegation link alone tells the two apart. Linking requirements to one another inside a document recreates tier coupling and is confusing across system boundaries, so it is flagged.

## Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



## 11.1.23. REQ-EMG-024

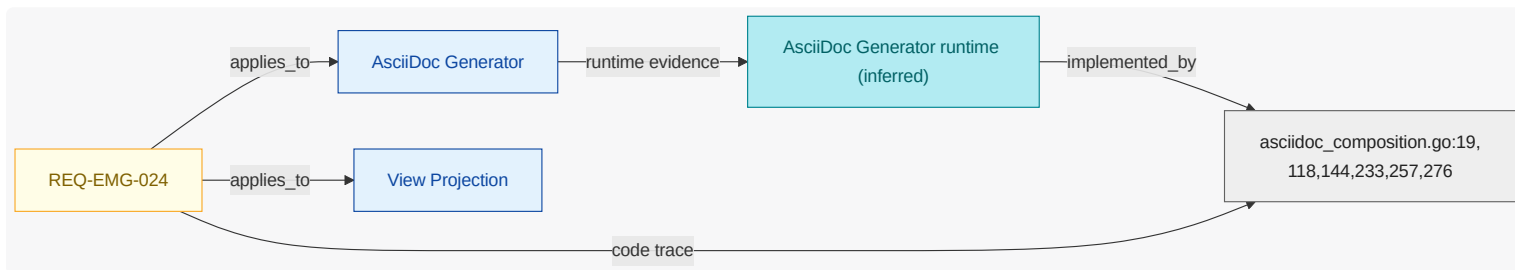
Where hardware interfaces are authored, when generation run is requested, the engineering model go shall render the hardware to software interface allocation view.

## NOTE

Hardware items and hardware interfaces capture the HW/SW interface (ICD/IRS) for the interface-control view.

## Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



## 11.1.24. REQ-EMG-025

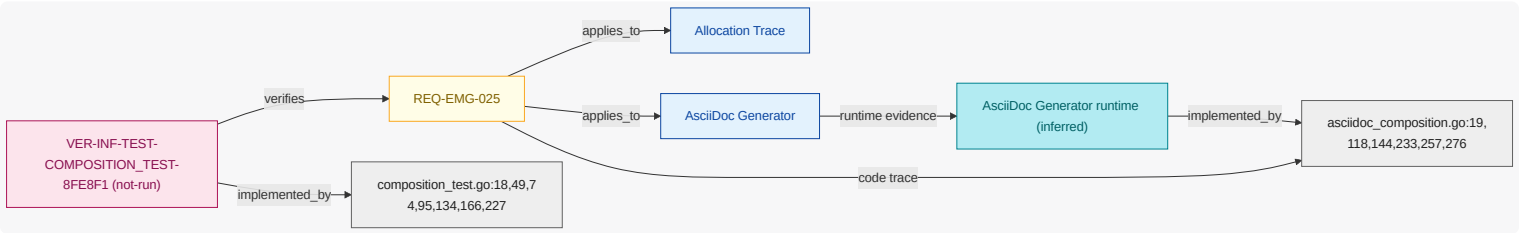
Where system composition is enabled, when generation run is requested, the engineering model go shall produce a system requirement allocation matrix aggregated across the composed subsystems.

## NOTE

Aggregate roll-up generation mode for the integrating system.

## Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.25. REQ-EMG-026

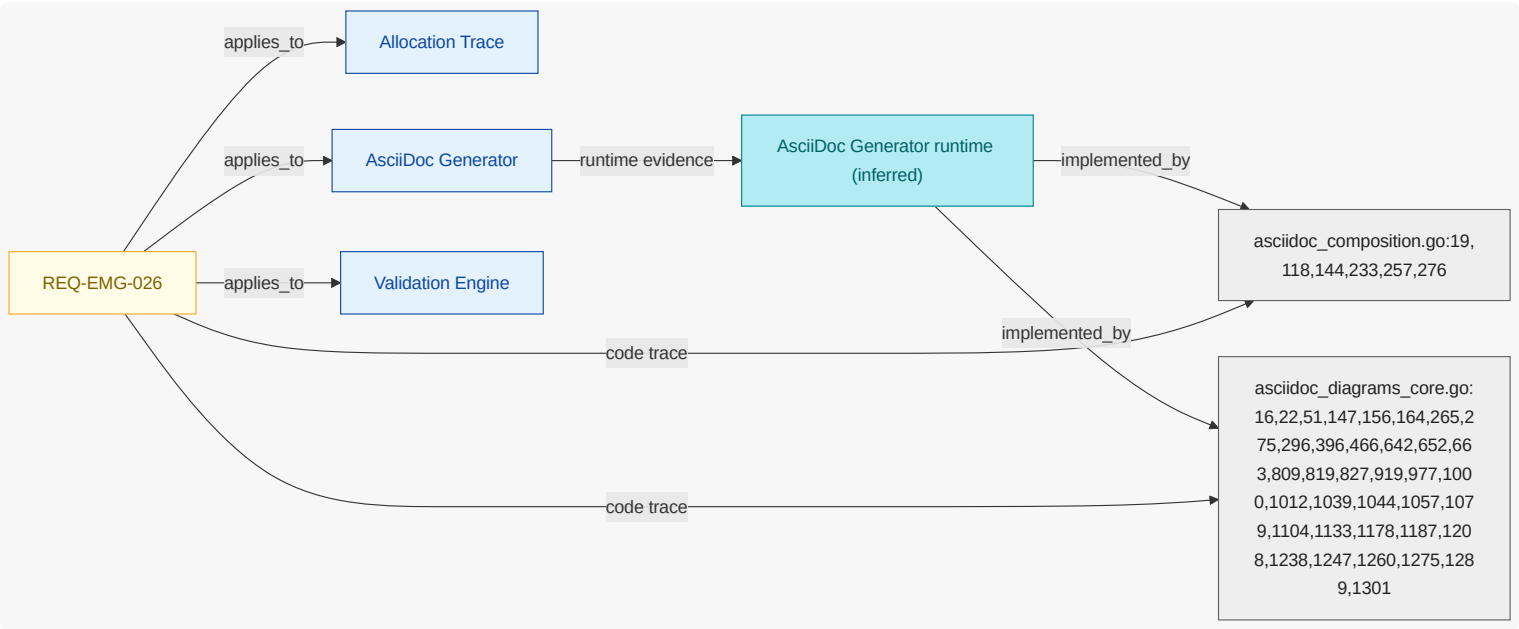
Where system composition is enabled, when generation run is requested, the engineering model go shall render in the requirement view the delegation link from each delegated requirement to the specific subsystem requirement that satisfies it.

NOTE

A provided contract entry names (ref) the subsystem requirement that realizes it, so a parent delegation traces to exactly the requirement that satisfies it. An allocated capability with no such ref is flagged as untraceable, and a ref to a requirement the subsystem does not define is rejected.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.26. REQ-EMG-027

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve an external git subsystem reference by cloning the repository into the workspace .engmod cache.

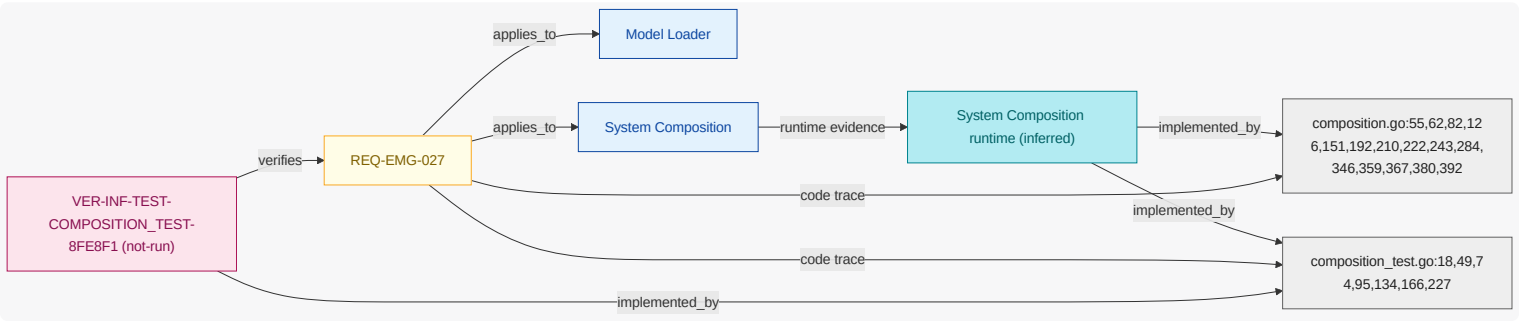
NOTE

A subsystem may reference an external repository with git (and an optional rev and path). The model is cloned once into .engmod/subsystems/<id> — a gitignored, temporary cache reused while it points at the same repository — and resolved like a local subsystem, giving a full cross-repository composed view. References stay downward-only.



Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.27. REQ-EMG-028

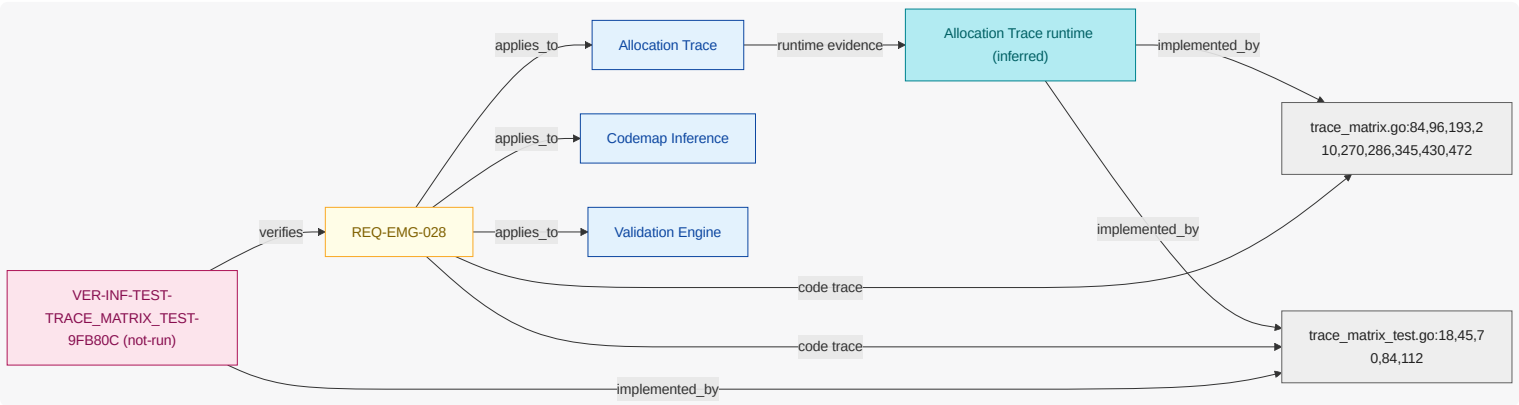
When generation run is requested , the engineering model go shall reject a code trace link that resolves to no requirement or model element.

NOTE

Code markers (TRLC-LINKS to a requirement , ENGMODEL-LINKS to a model element) are existence-checked against the model , not only shape-checked, so a link to a deleted or mistyped id is a dangling link and fails the build. A trace link is only checked for the model that owns its code (see REQ-EMG-030 scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.28. REQ-EMG-029

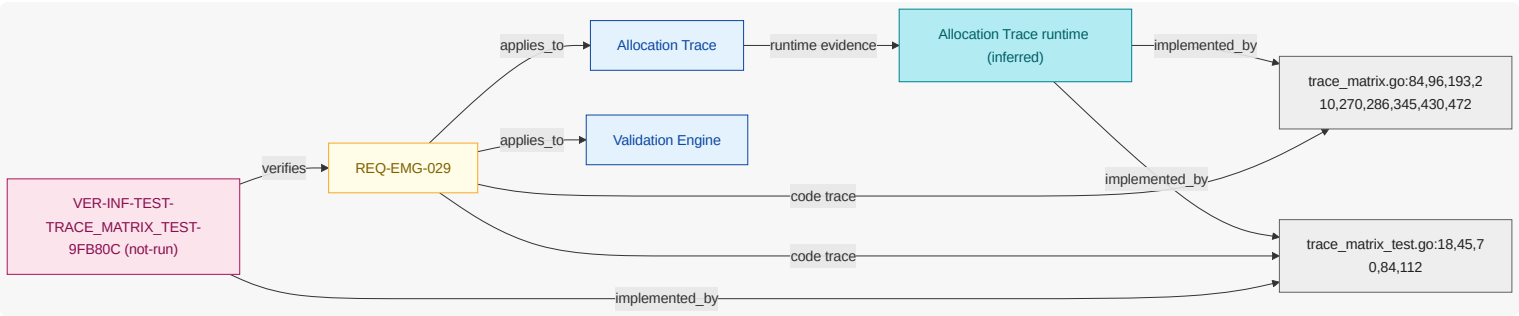
When generation run is requested , the engineering model go shall flag each requirement that is traced by neither code nor verification nor a downward delegation.

NOTE

An orphan requirement has no implementing code symbol, no verification, and is not delegated to a subsystem; it is a coverage gap surfaced as a warning so it is visible without breaking the build.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.29. REQ-EMG-030

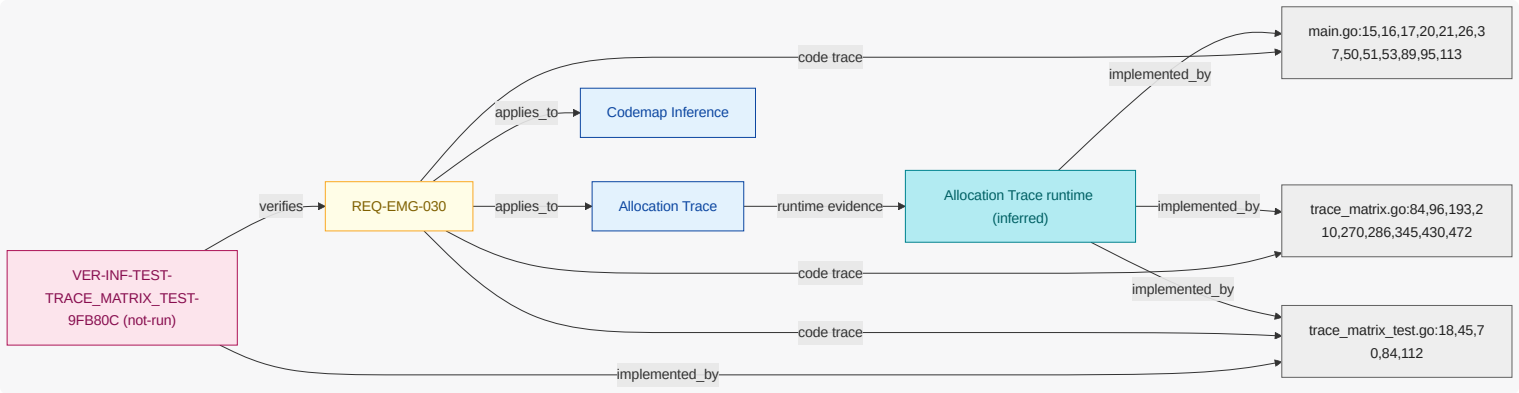
When generation run is requested , the engineering model go shall export a machine-readable traceability matrix that links each requirement to its implementing code and verification and delegated subsystem requirement .

NOTE

One consolidated matrix (JSON) replaces having to read it out of prose and diagrams. Code is attributed to the model whose architecture.yml is its nearest enclosing root, so a parent model never validates or counts a child model 's trace links (monorepo scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

11.2.1. Authorized Checkout Flow

| Field                 | Value                                                                        |
|-----------------------|------------------------------------------------------------------------------|
| ID                    | VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367                                  |
| Kind                  | e2e                                                                          |
| Status                | not-run                                                                      |
| Verifies Requirements | REQ-PAY-001                                                                  |
| Test Code Elements    | examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml |

| Field                     | Value                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| Derived Functional Owners | none                                                                                                    |
| Result Summary            | not-run:1                                                                                               |
| Evidence                  | examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml                            |
| Notes                     | executes <code>authorized checkout flow</code> from session start through payment authorization success |

### 11.2.2. Ota Rollout Flow

| Field                     | Value                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------|
| ID                        | VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC                                                          |
| Kind                      | e2e                                                                                          |
| Status                    | not-run                                                                                      |
| Verifies Requirements     | REQ-COF-003, REQ-COF-005                                                                     |
| Test Code Elements        | examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml                       |
| Derived Functional Owners | none                                                                                         |
| Result Summary            | not-run:2                                                                                    |
| Evidence                  | examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml                       |
| Notes                     | runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation |

### 11.2.3. Pr Opened Review Flow

| Field                     | Value                                                                             |
|---------------------------|-----------------------------------------------------------------------------------|
| ID                        | VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6                                          |
| Kind                      | e2e                                                                               |
| Status                    | not-run                                                                           |
| Verifies Requirements     | REQ-PRR-001, REQ-PRR-002, REQ-PRR-003, REQ-PRR-005                                |
| Test Code Elements        | examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml |
| Derived Functional Owners | none                                                                              |
| Result Summary            | not-run:4                                                                         |

| Field    | Value                                                                                 |
|----------|---------------------------------------------------------------------------------------|
| Evidence | examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml     |
| Notes    | exercises end-to-end pull request opened flow from webhook ingest to published review |

#### 11.2.4. Audit Record Persistence Test

| Field                     | Value                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------|
| ID                        | VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4                                |
| Kind                      | integration                                                                             |
| Status                    | not-run                                                                                 |
| Verifies Requirements     | REQ-PAY-005                                                                             |
| Test Code Elements        | audit_record_persistence_test.ts:5                                                      |
| Derived Functional Owners | none                                                                                    |
| Result Summary            | not-run:1                                                                               |
| Evidence                  | examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts |
| Notes                     | checks audit record persistence includes payment id and computed risk score             |

#### 11.2.5. Audit Redaction Test

| Field                     | Value                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------|
| ID                        | VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4                                        |
| Kind                      | integration                                                                            |
| Status                    | not-run                                                                                |
| Verifies Requirements     | REQ-PRR-007                                                                            |
| Test Code Elements        | audit_redaction_test.ts:3                                                              |
| Derived Functional Owners | none                                                                                   |
| Result Summary            | not-run:1                                                                              |
| Evidence                  | examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts |
| Notes                     | checks audit evidence redaction for secrets and sensitive payload fields               |

#### 11.2.6. Fraud Gate Test

| Field                     | Value                                                                               |
|---------------------------|-------------------------------------------------------------------------------------|
| ID                        | VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163                                          |
| Kind                      | integration                                                                         |
| Status                    | not-run                                                                             |
| Verifies Requirements     | REQ-PAY-002, REQ-PAY-004                                                            |
| Test Code Elements        | fraud_gate_test.rs:6                                                                |
| Derived Functional Owners | none                                                                                |
| Result Summary            | not-run:2                                                                           |
| Evidence                  | examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs           |
| Notes                     | validates fraud gate escalation and block decisions before authorization completion |

#### 11.2.7. Logging Pipeline Test

| Field                     | Value                                                                              |
|---------------------------|------------------------------------------------------------------------------------|
| ID                        | VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8                                   |
| Kind                      | integration                                                                        |
| Status                    | not-run                                                                            |
| Verifies Requirements     | REQ-COF-002, REQ-COF-007                                                           |
| Test Code Elements        | logging_pipeline_test.ts:4                                                         |
| Derived Functional Owners | none                                                                               |
| Result Summary            | not-run:2                                                                          |
| Evidence                  | examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts  |
| Notes                     | validates telemetry logging pipeline persistence and operational reporting signals |

#### 11.2.8. Policy Only Fallback Test

| Field  | Value                                                |
|--------|------------------------------------------------------|
| ID     | VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003 |
| Kind   | integration                                          |
| Status | not-run                                              |

| Field                     | Value                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------|
| Verifies Requirements     | REQ-PRR-004, REQ-PRR-006                                                                    |
| Test Code Elements        | policy_only_fallback_test.rs:5                                                              |
| Derived Functional Owners | none                                                                                        |
| Result Summary            | not-run:2                                                                                   |
| Evidence                  | examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs |
| Notes                     | verifies policy-only fallback when Bedrock inference is unavailable                         |

#### 11.2.9. AsciiDoc Diagrams Core Test

| Field                     | Value                                                                        |
|---------------------------|------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84                              |
| Kind                      | test                                                                         |
| Status                    | not-run                                                                      |
| Verifies Requirements     | REQ-EMG-003                                                                  |
| Test Code Elements        | asciidoc_diagrams_core_test.go:14,41,135,161,197,223,254,273,293,327,355,383 |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION            |
| Result Summary            | not-run:1                                                                    |
| Evidence                  | asciidoc_diagrams_core_test.go                                               |
| Notes                     | Inferred from test source artifact.                                          |

#### 11.2.10. AsciiDoc Diagrams Runtime Test

| Field                 | Value                                              |
|-----------------------|----------------------------------------------------|
| ID                    | VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D |
| Kind                  | test                                               |
| Status                | not-run                                            |
| Verifies Requirements | REQ-EMG-003                                        |
| Test Code Elements    | asciidoc_diagrams_runtime_test.go:12,50,72,95      |

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:1                                                         |
| Evidence                  | asciidoc_diagrams_runtime_test.go                                 |
| Notes                     | Inferred from test source artifact.                               |

#### 11.2.11. AsciiDoc Linking Units Test

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| ID                        | VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2                   |
| Kind                      | test                                                              |
| Status                    | not-run                                                           |
| Verifies Requirements     | REQ-EMG-003                                                       |
| Test Code Elements        | asciidoc_linking_units_test.go:10,46,75,101,129,148,167,186       |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:1                                                         |
| Evidence                  | asciidoc_linking_units_test.go                                    |
| Notes                     | Inferred from test source artifact.                               |

#### 11.2.12. AsciiDoc Test

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| ID                        | VER-INF-TEST-ASCIIDOC_TEST-F60765                                 |
| Kind                      | test                                                              |
| Status                    | not-run                                                           |
| Verifies Requirements     | REQ-EMG-003 , REQ-EMG-014                                         |
| Test Code Elements        | asciidoc_test.go:16,64,110,134,170,201,241,309,367,512,526,541    |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:2                                                         |

| Field    | Value                               |
|----------|-------------------------------------|
| Evidence | asciidoc_test.go                    |
| Notes    | Inferred from test source artifact. |

#### 11.2.13. AsciiDoc Views Test

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| ID                        | VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2                           |
| Kind                      | test                                                              |
| Status                    | not-run                                                           |
| Verifies Requirements     | REQ-EMG-003                                                       |
| Test Code Elements        | asciidoc_views_test.go:11                                         |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:1                                                         |
| Evidence                  | asciidoc_views_test.go                                            |
| Notes                     | Inferred from test source artifact.                               |

#### 11.2.14. Catalog Systems Test

| Field                     | Value                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7                                                                                                                 |
| Kind                      | test                                                                                                                                                     |
| Status                    | not-run                                                                                                                                                  |
| Verifies Requirements     | REQ-EMG-001                                                                                                                                              |
| Test Code Elements        | catalog_systems_test.go:11,36,82                                                                                                                         |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE |
| Result Summary            | not-run:1                                                                                                                                                |
| Evidence                  | catalog_systems_test.go                                                                                                                                  |
| Notes                     | Inferred from test source artifact.                                                                                                                      |

#### 11.2.15. Checkout Decline Response Test



| Field                     | Value                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671                                    |
| Kind                      | test                                                                                  |
| Status                    | not-run                                                                               |
| Verifies Requirements     | REQ-PAY-003, REQ-PAY-006                                                              |
| Test Code Elements        | checkout_decline_response_test.go:7                                                   |
| Derived Functional Owners | none                                                                                  |
| Result Summary            | not-run:2                                                                             |
| Evidence                  | examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go |
| Notes                     | validates checkout decline response contract includes reason and retry guidance       |

#### 11.2.16. Composition Test

| Field                     | Value                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-COMPOSITION_TEST-8FE8F1                                                                         |
| Kind                      | test                                                                                                         |
| Status                    | not-run                                                                                                      |
| Verifies Requirements     | REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-020 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-025 , REQ-EMG-027              |
| Test Code Elements        | composition_test.go:18,49,74,95,134,166,227                                                                  |
| Derived Functional Owners | FU-ALLOCATION-TRACE , FU-ASCIIDOC-GENERATOR , FU-MODEL-LOADER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE |
| Result Summary            | not-run:7                                                                                                    |
| Evidence                  | composition_test.go                                                                                          |
| Notes                     | Inferred from test source artifact.                                                                          |

#### 11.2.17. E2e Test

| Field | Value                        |
|-------|------------------------------|
| ID    | VER-INF-TEST-E2E_TEST-BE56F8 |
| Kind  | test                         |

| Field                     | Value                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Status                    | not-run                                                                                                                                            |
| Verifies Requirements     | REQ-EMG-001                                                                                                                                        |
| Test Code Elements        | e2e_test.go:14,39                                                                                                                                  |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE |
| Result Summary            | not-run:1                                                                                                                                          |
| Evidence                  | e2e_test.go                                                                                                                                        |
| Notes                     | Inferred from test source artifact.                                                                                                                |

#### 11.2.18. Ears Preflight Test

| Field                     | Value                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26                                                                                                            |
| Kind                      | test                                                                                                                                               |
| Status                    | not-run                                                                                                                                            |
| Verifies Requirements     | REQ-EMG-001, REQ-EMG-009                                                                                                                           |
| Test Code Elements        | ears_preflight_test.go:11,60,105                                                                                                                   |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE |
| Result Summary            | not-run:2                                                                                                                                          |
| Evidence                  | ears_preflight_test.go                                                                                                                             |
| Notes                     | Inferred from test source artifact.                                                                                                                |

#### 11.2.19. Gemara Export Test

| Field                 | Value                                        |
|-----------------------|----------------------------------------------|
| ID                    | VER-INF-TEST-GEMARA_EXPORT_TEST-45C982       |
| Kind                  | test                                         |
| Status                | not-run                                      |
| Verifies Requirements | REQ-EMG-015                                  |
| Test Code Elements    | gemara_export_test.go:27,134,173,213,231,249 |

| Field                     | Value                                                     |
|---------------------------|-----------------------------------------------------------|
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MCP-SERVER |
| Result Summary            | not-run:1                                                 |
| Evidence                  | gemara_export_test.go                                     |
| Notes                     | Inferred from test source artifact.                       |

#### 11.2.20. Gemara Oscal Test

| Field                     | Value                                                     |
|---------------------------|-----------------------------------------------------------|
| ID                        | VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA                     |
| Kind                      | test                                                      |
| Status                    | not-run                                                   |
| Verifies Requirements     | REQ-EMG-015                                               |
| Test Code Elements        | gemara_oscal_test.go:15,52                                |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MCP-SERVER |
| Result Summary            | not-run:1                                                 |
| Evidence                  | gemara_oscal_test.go                                      |
| Notes                     | Inferred from test source artifact.                       |

#### 11.2.21. Github Check Run Contract Test

| Field                     | Value                                              |
|---------------------------|----------------------------------------------------|
| ID                        | VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F |
| Kind                      | test                                               |
| Status                    | not-run                                            |
| Verifies Requirements     | REQ-PRR-005                                        |
| Test Code Elements        | github_check_run_contract_test.go:8                |
| Derived Functional Owners | none                                               |
| Result Summary            | not-run:1                                          |

| Field    | Value                                                                                         |
|----------|-----------------------------------------------------------------------------------------------|
| Evidence | examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go |
| Notes    | validates GitHub check-run contract shape and required review result fields                   |

#### 11.2.22. Inferred Code Test

| Field                     | Value                                  |
|---------------------------|----------------------------------------|
| ID                        | VER-INF-TEST-INFERRED_CODE_TEST-3CA82E |
| Kind                      | test                                   |
| Status                    | not-run                                |
| Verifies Requirements     | REQ-EMG-010                            |
| Test Code Elements        | inferred_code_test.go:11,37            |
| Derived Functional Owners | FU-CODEMAP-INFERENCE                   |
| Result Summary            | not-run:1                              |
| Evidence                  | inferred_code_test.go                  |
| Notes                     | Inferred from test source artifact.    |

#### 11.2.23. Inferred Layers Test

| Field                     | Value                                      |
|---------------------------|--------------------------------------------|
| ID                        | VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF   |
| Kind                      | test                                       |
| Status                    | not-run                                    |
| Verifies Requirements     | REQ-EMG-010                                |
| Test Code Elements        | inferred_layers_test.go:13,21,40,58,84,118 |
| Derived Functional Owners | FU-CODEMAP-INFERENCE                       |
| Result Summary            | not-run:1                                  |
| Evidence                  | inferred_layers_test.go                    |
| Notes                     | Inferred from test source artifact.        |

#### 11.2.24. Inferred Verification Test

| Field                     | Value                                                  |
|---------------------------|--------------------------------------------------------|
| ID                        | VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6         |
| Kind                      | test                                                   |
| Status                    | not-run                                                |
| Verifies Requirements     | REQ-EMG-010                                            |
| Test Code Elements        | inferred_verification_test.go:13,26,64,100,133,192,216 |
| Derived Functional Owners | FU-CODEMAP-INFERENCE                                   |
| Result Summary            | not-run:1                                              |
| Evidence                  | inferred_verification_test.go                          |
| Notes                     | Inferred from test source artifact.                    |

#### 11.2.25. Load Test

| Field                     | Value                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-LOAD_TEST-A58EB3                                                                                                                      |
| Kind                      | test                                                                                                                                               |
| Status                    | not-run                                                                                                                                            |
| Verifies Requirements     | REQ-EMG-001                                                                                                                                        |
| Test Code Elements        | load_test.go:10,25                                                                                                                                 |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE |
| Result Summary            | not-run:1                                                                                                                                          |
| Evidence                  | model/load_test.go                                                                                                                                 |
| Notes                     | Inferred from test source artifact.                                                                                                                |

#### 11.2.26. Lobster Export Test

| Field  | Value                                   |
|--------|-----------------------------------------|
| ID     | VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E |
| Kind   | test                                    |
| Status | not-run                                 |

| Field                     | Value                                                         |
|---------------------------|---------------------------------------------------------------|
| Verifies Requirements     | REQ-EMG-006                                                   |
| Test Code Elements        | lobster_export_test.go:12                                     |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-LOBSTER-EXPORTER , FU-TRLC-EXPORTER |
| Result Summary            | not-run:1                                                     |
| Evidence                  | lobster_export_test.go                                        |
| Notes                     | Inferred from test source artifact.                           |

#### 11.2.27. Main Test

| Field                     | Value                               |
|---------------------------|-------------------------------------|
| ID                        | VER-INF-TEST-MAIN_TEST-6B61AE       |
| Kind                      | test                                |
| Status                    | not-run                             |
| Verifies Requirements     | REQ-EMG-007 , REQ-EMG-008           |
| Test Code Elements        | main_test.go:12,22                  |
| Derived Functional Owners | FU-MCP-SERVER                       |
| Result Summary            | not-run:2                           |
| Evidence                  | cmd/engmcp/main_test.go             |
| Notes                     | Inferred from test source artifact. |

#### 11.2.28. Oscal Chain Test

| Field                 | Value                                |
|-----------------------|--------------------------------------|
| ID                    | VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9 |
| Kind                  | test                                 |
| Status                | not-run                              |
| Verifies Requirements | REQ-EMG-013                          |
| Test Code Elements    | oscal_chain_test.go:11,28            |

| Field                     | Value                                    |
|---------------------------|------------------------------------------|
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-OSCAL-EXPORTER |
| Result Summary            | not-run:1                                |
| Evidence                  | oscal_chain_test.go                      |
| Notes                     | Inferred from test source artifact.      |

#### 11.2.29. Oscal Ssp Test

| Field                     | Value                                    |
|---------------------------|------------------------------------------|
| ID                        | VER-INF-TEST-OSCAL_SSP_TEST-4B08EF       |
| Kind                      | test                                     |
| Status                    | not-run                                  |
| Verifies Requirements     | REQ-EMG-013                              |
| Test Code Elements        | oscal_ssp_test.go:15,64,84,105,127,162   |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-OSCAL-EXPORTER |
| Result Summary            | not-run:1                                |
| Evidence                  | oscal_ssp_test.go                        |
| Notes                     | Inferred from test source artifact.      |

#### 11.2.30. Project Test

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| ID                        | VER-INF-TEST-PROJECT_TEST-2ECF72                                  |
| Kind                      | test                                                              |
| Status                    | not-run                                                           |
| Verifies Requirements     | REQ-EMG-003                                                       |
| Test Code Elements        | project_test.go:11,52,86,137,168                                  |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:1                                                         |

| Field    | Value                               |
|----------|-------------------------------------|
| Evidence | view /project_test.go               |
| Notes    | Inferred from test source artifact. |

### 11.2.31. Render Test

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| ID                        | VER-INF-TEST-RENDER_TEST-EECC23                                   |
| Kind                      | test                                                              |
| Status                    | not-run                                                           |
| Verifies Requirements     | REQ-EMG-003                                                       |
| Test Code Elements        | render_test.go:12                                                 |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION |
| Result Summary            | not-run:1                                                         |
| Evidence                  | render/mermaid/render_test.go                                     |
| Notes                     | Inferred from test source artifact.                               |

### 11.2.32. Scan Test

| Field                     | Value                                        |
|---------------------------|----------------------------------------------|
| ID                        | VER-INF-TEST-SCAN_TEST-B04F81                |
| Kind                      | test                                         |
| Status                    | not-run                                      |
| Verifies Requirements     | REQ-EMG-010 , REQ-SAMPLE-001, REQ-SAMPLE-002 |
| Test Code Elements        | scan_test.go:11,48,73,111,147,181,226        |
| Derived Functional Owners | FU-CODEMAP-INFERENCE                         |
| Result Summary            | not-run:3                                    |
| Evidence                  | codemap/scan_test.go                         |
| Notes                     | Inferred from test source artifact.          |

### 11.2.33. Server Test



| Field                     | Value                                                                    |
|---------------------------|--------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-SERVER_TEST-389957                                          |
| Kind                      | test                                                                     |
| Status                    | not-run                                                                  |
| Verifies Requirements     | REQ-EMG-007 , REQ-EMG-008                                                |
| Test Code Elements        | server_test.go:12,99,124,170,212,253,309,327,337,436,489,533,579,625,647 |
| Derived Functional Owners | FU-MCP-SERVER                                                            |
| Result Summary            | not-run:2                                                                |
| Evidence                  | mcp/server_test.go                                                       |
| Notes                     | Inferred from test source artifact.                                      |

#### 11.2.34. Structurizr Export Test

| Field                     | Value                                          |
|---------------------------|------------------------------------------------|
| ID                        | VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374    |
| Kind                      | test                                           |
| Status                    | not-run                                        |
| Verifies Requirements     | REQ-EMG-005                                    |
| Test Code Elements        | structurizr_export_test.go:14,40               |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-STRUCTURIZR-EXPORTER |
| Result Summary            | not-run:1                                      |
| Evidence                  | structurizr_export_test.go                     |
| Notes                     | Inferred from test source artifact.            |

#### 11.2.35. Telemetry Ingest Contract Test

| Field  | Value                                              |
|--------|----------------------------------------------------|
| ID     | VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2 |
| Kind   | test                                               |
| Status | not-run                                            |

| Field                     | Value                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------|
| Verifies Requirements     | REQ-COF-001                                                                             |
| Test Code Elements        | telemetry_ingest_contract_test.go:7                                                     |
| Derived Functional Owners | none                                                                                    |
| Result Summary            | not-run:1                                                                               |
| Evidence                  | examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go |
| Notes                     | validates telemetry ingest contract fields and accepted request semantics               |

### 11.2.36. Threat Model Export Test

| Field                     | Value                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D                                               |
| Kind                      | test                                                                                       |
| Status                    | not-run                                                                                    |
| Verifies Requirements     | REQ-EMG-004 , REQ-EMG-011                                                                  |
| Test Code Elements        | threat_model_export_test.go:16,49,79,98                                                    |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-STRUCTURIZR-EXPORTER , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE |
| Result Summary            | not-run:2                                                                                  |
| Evidence                  | threat_model_export_test.go                                                                |
| Notes                     | Inferred from test source artifact.                                                        |

### 11.2.37. Trace Matrix Test

| Field                 | Value                                   |
|-----------------------|-----------------------------------------|
| ID                    | VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C   |
| Kind                  | test                                    |
| Status                | not-run                                 |
| Verifies Requirements | REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 |
| Test Code Elements    | trace_matrix_test.go:18,45,70,84,112    |

| Field                     | Value                                                             |
|---------------------------|-------------------------------------------------------------------|
| Derived Functional Owners | FU-ALLOCATION-TRACE , FU-CODEMAP-INFERENCE , FU-VALIDATION-ENGINE |
| Result Summary            | not-run:3                                                         |
| Evidence                  | trace_matrix_test.go                                              |
| Notes                     | Inferred from test source artifact.                               |

#### 11.2.38. Trlc Export Test

| Field                     | Value                                                         |
|---------------------------|---------------------------------------------------------------|
| ID                        | VER-INF-TEST-TRLC_EXPORT_TEST-DD1345                          |
| Kind                      | test                                                          |
| Status                    | not-run                                                       |
| Verifies Requirements     | REQ-EMG-006                                                   |
| Test Code Elements        | trlc_export_test.go:13,44                                     |
| Derived Functional Owners | FU-CLI-ORCHESTRATION , FU-LOBSTER-EXPORTER , FU-TRLC-EXPORTER |
| Result Summary            | not-run:1                                                     |
| Evidence                  | trlc_export_test.go                                           |
| Notes                     | Inferred from test source artifact.                           |

#### 11.2.39. Validate Test

| Field                     | Value                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                        | VER-INF-TEST-VALIDATE_TEST-498C59                                                                                                                        |
| Kind                      | test                                                                                                                                                     |
| Status                    | not-run                                                                                                                                                  |
| Verifies Requirements     | REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-011                                                                                                                  |
| Test Code Elements        | validate_test.go:12,25,62,109,157,198,224,255,300,355,459                                                                                                |
| Derived Functional Owners | FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE |
| Result Summary            | not-run:3                                                                                                                                                |

| Field    | Value                               |
|----------|-------------------------------------|
| Evidence | validate/validate_test.go           |
| Notes    | Inferred from test source artifact. |

#### 11.2.40. Checkout Decline Reason Test

| Field                     | Value                                                                           |
|---------------------------|---------------------------------------------------------------------------------|
| ID                        | VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD                                |
| Kind                      | unit                                                                            |
| Status                    | not-run                                                                         |
| Verifies Requirements     | REQ-PAY-003, REQ-PAY-008                                                        |
| Test Code Elements        | checkout_decline_reason_test.go:7,15,20                                         |
| Derived Functional Owners | none                                                                            |
| Result Summary            | not-run:2                                                                       |
| Evidence                  | examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go |
| Notes                     | validates checkout decline reason normalization and retry policy messaging      |

#### 11.2.41. Ota Campaign Test

| Field                     | Value                                                                   |
|---------------------------|-------------------------------------------------------------------------|
| ID                        | VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB                                   |
| Kind                      | unit                                                                    |
| Status                    | not-run                                                                 |
| Verifies Requirements     | REQ-COF-003, REQ-COF-004                                                |
| Test Code Elements        | ota_campaign_test.go:8                                                  |
| Derived Functional Owners | none                                                                    |
| Result Summary            | not-run:2                                                               |
| Evidence                  | examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go  |
| Notes                     | verifies OTA campaign signature validation and rollout trigger behavior |

#### 11.2.42. Rate Limit Deferral Test

| Field                     | Value                                                                               |
|---------------------------|-------------------------------------------------------------------------------------|
| ID                        | VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE                                        |
| Kind                      | unit                                                                                |
| Status                    | not-run                                                                             |
| Verifies Requirements     | REQ-PRR-008                                                                         |
| Test Code Elements        | rate_limit_deferral_test.go:8                                                       |
| Derived Functional Owners | none                                                                                |
| Result Summary            | not-run:1                                                                           |
| Evidence                  | examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go |
| Notes                     | verifies rate-limit deferral behavior for pull request review processing            |

## 12. Gemara GRC Model

This chapter renders the model as OpenSSF Gemara (<https://gemara.openssf.org>) documents. Each artifact is built with the official `go-gemara` SDK types and validated against the published Gemara CUE schemas. Generate the full YAML set with `enggemara` ; query it via the `gemara.*` MCP tools.

| Gemara Layer | Artifact                 | Entries |
|--------------|--------------------------|---------|
| L1           | Vector Catalog           | 4       |
| L1           | Principle Catalog        | 5       |
| L1           | Guidance Catalog         | 2       |
| L2           | Capability Catalog       | 16      |
| L2           | Control Catalog          | 5       |
| L2           | Threat Catalog           | 2       |
| L3           | Risk Catalog             | 3       |
| L3           | Policy                   | 2       |
| -            | Control → Threat Mapping | 3       |
| -            | Lexicon                  | 59      |
| L7           | Audit Log                | 5       |
| L6           | Enforcement Log          | 2       |

### 12.1. Controls (L2)

| Control                      | Objective                                                                                                                                    | Group            | Assessment Requirements |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------|
| CTRL-MCP-PATH-BOUNDARY       | Enforce repo-root path constraints and reject traversal paths in MCP tools.                                                                  | input-validation | 1                       |
| CTRL-STRICT-MCP-INPUT-SCHEMA | Enforce per-tool input schema and reject unknown arguments.                                                                                  | input-validation | 1                       |
| CTRL-TRACEABILITY-COVERAGE   | Require requirement-linked verification evidence for modeled behavior.                                                                       | assurance        | 1                       |
| CTRL-TRACE-LINK-INTEGRITY    | Reject code trace links (TRLINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability... | assurance        | 1                       |

| Control                      | Objective                                                                                                                            | Group     | Assessment Requirements |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------|
| CTRL-ARTIFACT-FRESHNESS-GATE | Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts. | assurance | 1                       |

12.2. Threats (L2)

| Threat                | Title                                               | Group              |
|-----------------------|-----------------------------------------------------|--------------------|
| TS-MCP-PATH-TRAVERSAL | MCP path traversal accesses non-repo files          | stride-tampering   |
| TS-TRACEABILITY-DRIFT | Requirement traceability drifts from implementation | stride-repudiation |

12.3. Risks (L3)

| Risk                    | Title                                             | Severity |
|-------------------------|---------------------------------------------------|----------|
| RISK-MCP-PATH-ESCAPE    | MCP path traversal exposes non-repository files   | High     |
| RISK-TRACEABILITY-DRIFT | Traceability drift weakens assurance evidence     | Medium   |
| RISK-STALE-ARTIFACTS    | Stale generated artifacts misrepresent the system | Low      |

# 13. Reference Index

## 13.1. Authored References

### CTRL-ARTIFACT-FRESHNESS-GATE

category=assurance; description=Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Control                           |
| Name         | Generated Artifact Freshness Gate |
| Mentioned In | n/a                               |

### CTRL-MCP-PATH-BOUNDARY

category=input-validation; description=Enforce repo-root path constraints and reject traversal paths in MCP tools.

| Key          | Value                         |
|--------------|-------------------------------|
| Kind         | Control                       |
| Name         | MCP Path Boundary Enforcement |
| Mentioned In | Security View                 |

### CTRL-STRICT-MCP-INPUT-SCHEMA

category=input-validation; description=Enforce per-tool input schema and reject unknown arguments.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Control                  |
| Name         | Strict MCP Input Schemas |
| Mentioned In | Security View            |

### CTRL-TRACE-LINK-INTEGRITY

category=assurance; description=Reject code trace links (TRLC-LINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability matrix with an implemented, verified, delegated, and orphan rollup.

| Key  | Value                |
|------|----------------------|
| Kind | Control              |
| Name | Trace Link Integrity |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

CTRL-TRACEABILITY-COVERAGE

category=assurance; description=Require requirement-linked verification evidence for modeled behavior.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Control                           |
| Name         | Requirement Traceability Coverage |
| Mentioned In | Security View                     |

DO-ARCHITECTURE-MODEL

termRef=DATA-ARCHITECTURE-MODEL; schemaRef=architecture.yml; sensitivity=internal

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Data Object                 |
| Name         | Architecture Model          |
| Mentioned In | Security View , REQ-EMG-001 |

DO-MCP-TOOL-RESULT

termRef=DATA-MCP-TOOL-RESULT; schemaRef=mcp.tool-response.v1; sensitivity=internal

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Data Object                 |
| Name         | MCP Tool Result             |
| Mentioned In | Security View , REQ-EMG-007 |

DO-STRUCTURIZR-DSL

termRef=DATA-STRUCTURIZR-DSL; schemaRef=generated/STRUCTURIZR.dsl; sensitivity=internal

| Key          | Value                                                  |
|--------------|--------------------------------------------------------|
| Kind         | Data Object                                            |
| Name         | Structurizr DSL                                        |
| Mentioned In | Architecture Intent View , Security View , REQ-EMG-005 |

DO-THREAT-DRAGON-JSON

termRef=DATA-THREAT-DRAGON-JSON; schemaRef=generated/threat-dragon-v2.json; sensitivity=internal

| Key          | Value              |
|--------------|--------------------|
| Kind         | Data Object        |
| Name         | Threat Dragon JSON |
| Mentioned In | Security View      |

DEP-CI-PIPELINE

environment=ci; region=hosted; account=github-actions; cluster=shared-runner; namespace=engineering-model-go; trustZone=automation

| Key          | Value              |
|--------------|--------------------|
| Kind         | Deployment Target  |
| Name         | CI Pipeline Runner |
| Mentioned In | n/a                |

DEP-LOCAL-WORKSPACE

environment=dev; region=local; account=workstation; cluster=local-shell; namespace=engineering-model-go; trustZone=developer

| Key          | Value             |
|--------------|-------------------|
| Kind         | Deployment Target |
| Name         | Local Workspace   |
| Mentioned In | n/a               |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS

entry=edit-model; exits=publish-artifacts; steps=5

| Key          | Value                                   |
|--------------|-----------------------------------------|
| Kind         | Flow                                    |
| Name         | Model Change to Verified Artifacts Flow |
| Mentioned In | Security View                           |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::edit-model

kind=user\_action; ref=ACT-IMPLEMENTATION-ENGINEER; action=Update model and requirements artifacts; dataIn=; dataOut=architecture\_update; async=false

| Key          | Value                                   |
|--------------|-----------------------------------------|
| Kind         | Flow Step                               |
| Name         | Update model and requirements artifacts |
| Mentioned In | n/a                                     |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::fail-validation

kind=system\_action; ref=FU-VALIDATION-ENGINE; action=Emit validation diagnostics and block publication; dataIn=; dataOut=; async=false

| Key          | Value                                             |
|--------------|---------------------------------------------------|
| Kind         | Flow Step                                         |
| Name         | Emit validation diagnostics and block publication |
| Mentioned In | n/a                                               |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::generate-artifacts

kind=system\_action; ref=FU-CLI-ORCHESTRATION; action=Run generation and export commands; dataIn=; dataOut=generated\_artifacts; async=false

| Key          | Value                              |
|--------------|------------------------------------|
| Kind         | Flow Step                          |
| Name         | Run generation and export commands |
| Mentioned In | n/a                                |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::publish-artifacts

kind=system\_action; ref=FU-ASCIIDOC-GENERATOR; action=Persist generated outputs for downstream use; dataIn=; dataOut=; async=false

| Key          | Value                                        |
|--------------|----------------------------------------------|
| Kind         | Flow Step                                    |
| Name         | Persist generated outputs for downstream use |
| Mentioned In | n/a                                          |

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::validate-model

kind=system\_action; ref=FU-VALIDATION-ENGINE; action=Validate authored model references and relationships;  
dataIn=architecture\_update; dataOut=; async=false

| Key          | Value                                                |
|--------------|------------------------------------------------------|
| Kind         | Flow Step                                            |
| Name         | Validate authored model references and relationships |
| Mentioned In | n/a                                                  |

IF-CLI-ENGDOC

protocol=cli; endpoint=cmd/engdoc; schemaRef=docs/asciidoc-architecture-generator-design.md; owner=FU-ASCIIDOC-GENERATOR

| Key          | Value      |
|--------------|------------|
| Kind         | Interface  |
| Name         | engdoc CLI |
| Mentioned In | n/a        |

IF-CLI-ENGDRAGON

protocol=cli; endpoint=cmd/engdragon; schemaRef=threat\_model\_export.go; owner=FU-THREAT-EXPORTER

| Key          | Value         |
|--------------|---------------|
| Kind         | Interface     |
| Name         | engdragon CLI |
| Mentioned In | n/a           |

IF-CLI-ENGGEMARA

protocol=cli; endpoint=cmd/enggemara; schemaRef=gemara\_export.go; owner=FU-GEMARA-EXPORTER

| Key          | Value         |
|--------------|---------------|
| Kind         | Interface     |
| Name         | enggemara CLI |
| Mentioned In | n/a           |

IF-CLI-ENGLOBSTER

protocol=cli; endpoint=cmd/englobster; schemaRef=lobster\_export.go; owner=FU-LOBSTER-EXPORTER

| Key          | Value          |
|--------------|----------------|
| Kind         | Interface      |
| Name         | englobster CLI |
| Mentioned In | n/a            |

IF-CLI-ENGMCP

protocol=stdio-jsonrpc; endpoint=cmd/engmcp; schemaRef=mcp/server.go; owner=FU-MCP-SERVER

| Key          | Value      |
|--------------|------------|
| Kind         | Interface  |
| Name         | engmcp CLI |
| Mentioned In | n/a        |

IF-CLI-ENGOSCAL

protocol=cli; endpoint=cmd/engoscal; schemaRef=oscal\_ssp.go; owner=FU-OSCAL-EXPORTER

| Key          | Value        |
|--------------|--------------|
| Kind         | Interface    |
| Name         | engoscal CLI |
| Mentioned In | n/a          |

IF-CLI-ENGSTRUCT

protocol=cli; endpoint=cmd/engstruct; schemaRef=structurizr\_export.go; owner=FU-STRUCTURIZR-EXPORTER

| Key          | Value         |
|--------------|---------------|
| Kind         | Interface     |
| Name         | engstruct CLI |
| Mentioned In | n/a           |

IF-CLI-ENGTRACE

protocol=cli; endpoint=cmd/engtrace; schemaRef=trace\_matrix.go; owner=FU-ALLOCATION-TRACE

| Key  | Value     |
|------|-----------|
| Kind | Interface |

| Key          | Value        |
|--------------|--------------|
| Name         | engtrace CLI |
| Mentioned In | n/a          |

IF-CLI-ENGTRLC

protocol=cli; endpoint=cmd/engtrlc; schemaRef=trlc\_export.go; owner=FU-TRLC-EXPORTER

| Key          | Value       |
|--------------|-------------|
| Kind         | Interface   |
| Name         | engtrlc CLI |
| Mentioned In | n/a         |

IF-CLI-ENGVIEW

protocol=cli; endpoint=cmd/engview; schemaRef=view/project.go; owner=FU-VIEW-PROJECTION

| Key          | Value       |
|--------------|-------------|
| Kind         | Interface   |
| Name         | engview CLI |
| Mentioned In | n/a         |

REQ-EMG-001

While local development mode is active, when model updated event is received, the engineering model go shall validate architecture model before generation run is requested.

| Key          | Value                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                                                                          |
| Name         | REQ-EMG-001                                                                                                                          |
| Mentioned In | Architecture Intent View , Interaction Flow View , Catalog Systems Test , E2e Test , Ears Preflight Test , Load Test , Validate Test |

REQ-EMG-003

While model is valid, when generation run is requested, the engineering model go shall produce AsciiDoc architecture publication output.

| Key  | Value       |
|------|-------------|
| Kind | Requirement |

| Key          | Value                                                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name         | REQ-EMG-003                                                                                                                                                                                                      |
| Mentioned In | Architecture Intent View , Interaction Flow View , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Linking Units Test , AsciiDoc Test , AsciiDoc Views Test , Project Test , Render Test |

#### REQ-EMG-004

Where threat export is enabled, when generation run is requested, the engineering model go shall export threat dragon json artifact plus open threat model artifact.

| Key          | Value                                                                       |
|--------------|-----------------------------------------------------------------------------|
| Kind         | Requirement                                                                 |
| Name         | REQ-EMG-004                                                                 |
| Mentioned In | Architecture Intent View , Interaction Flow View , Threat Model Export Test |

#### REQ-EMG-005

Where structurizr export is enabled, when generation run is requested, the engineering model go shall export structurizr dsl artifact.

| Key          | Value                                                                      |
|--------------|----------------------------------------------------------------------------|
| Kind         | Requirement                                                                |
| Name         | REQ-EMG-005                                                                |
| Mentioned In | Architecture Intent View , Interaction Flow View , Structurizr Export Test |

#### REQ-EMG-006

Where traceability export is enabled, when generation run is requested, the engineering model go shall produce trlc package and lobster trace report artifacts.

| Key          | Value                                                                                     |
|--------------|-------------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                               |
| Name         | REQ-EMG-006                                                                               |
| Mentioned In | Architecture Intent View , Interaction Flow View , Lobster Export Test , Trlc Export Test |

#### REQ-EMG-007

While mcp server is enabled, when mcp tool call is received and catalog references resolve successfully, the engineering model go shall return mcp tool result with schema version and structured error fields.

| Key          | Value                                                                      |
|--------------|----------------------------------------------------------------------------|
| Kind         | Requirement                                                                |
| Name         | REQ-EMG-007                                                                |
| Mentioned In | Architecture Intent View , Interaction Flow View , Main Test , Server Test |

REQ-EMG-008

When mcp tool call is received, the engineering model go shall reject path arguments outside repository workspace boundary.

| Key          | Value                                                                      |
|--------------|----------------------------------------------------------------------------|
| Kind         | Requirement                                                                |
| Name         | REQ-EMG-008                                                                |
| Mentioned In | Architecture Intent View , Interaction Flow View , Main Test , Server Test |

REQ-EMG-009

While ci validation mode is active, when generation run is requested, the engineering model go shall execute validation plus tests before artifact generated event is raised.

| Key          | Value                                                                                  |
|--------------|----------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                            |
| Name         | REQ-EMG-009                                                                            |
| Mentioned In | Architecture Intent View , Interaction Flow View , Ears Preflight Test , Validate Test |

REQ-EMG-010

Where requirement trace is complete, when model updated event is received, the engineering model go shall infer traceability coverage from trace markers.

| Key          | Value                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                                                                           |
| Name         | REQ-EMG-010                                                                                                                           |
| Mentioned In | Architecture Intent View , Interaction Flow View , Inferred Code Test , Inferred Layers Test , Inferred Verification Test , Scan Test |

REQ-EMG-011

Where external schema dependencies are available, when generation run is requested, the engineering model go shall fail validation if schema checks do not match expected constraints.



| Key          | Value                                                                                       |
|--------------|---------------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                                 |
| Name         | REQ-EMG-011                                                                                 |
| Mentioned In | Architecture Intent View , Interaction Flow View , Threat Model Export Test , Validate Test |

#### REQ-EMG-012

When generation run is requested, the engineering model go shall generate artifacts deterministically so that regeneration is reproducible.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-012              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-013

Where traceability export is enabled, when generation run is requested, the engineering model go shall export OSCAL SSP plus assessment results plus POA&M artifacts.

| Key          | Value                                                                                |
|--------------|--------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                          |
| Name         | REQ-EMG-013                                                                          |
| Mentioned In | Architecture Intent View , Interaction Flow View , Oscal Chain Test , Oscal Ssp Test |

#### REQ-EMG-014

Where architecture decision records are authored, when generation run is requested, the engineering model go shall publish architecture decision records document linked from AsciiDoc architecture publication output.

| Key          | Value                                                            |
|--------------|------------------------------------------------------------------|
| Kind         | Requirement                                                      |
| Name         | REQ-EMG-014                                                      |
| Mentioned In | Architecture Intent View , Interaction Flow View , AsciiDoc Test |

#### REQ-EMG-015

Where gemara export is enabled, when generation run is requested, the engineering model go shall render OpenSSF Gemara model artifacts.

| Key          | Value                                                                                     |
|--------------|-------------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                               |
| Name         | REQ-EMG-015                                                                               |
| Mentioned In | Architecture Intent View , Interaction Flow View , Gemara Export Test , Gemara Oscal Test |

#### REQ-EMG-016

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-016                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-017

Where system composition is enabled, when generation run is requested, the engineering model go shall reject any subsystem reference that resolves outside the repository workspace boundary.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-017                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-018

Where system composition is enabled, when generation run is requested, the engineering model go shall reject a subsystem reference graph that contains a cycle.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-018              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-019

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve cross-system references using subsystem-qualified identifiers.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-019              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-020

Where system composition is enabled, when generation run is requested, the engineering model go shall materialize parent-to-subsystem allocation traceability in the composed output without modifying any subsystem model.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-020                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-021

Where system composition is enabled, when generation run is requested, the engineering model go shall report each subsystem required interface that the parent system does not satisfy.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-021              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-022

Where system composition is enabled, when generation run is requested, the engineering model go shall reject an allocation that targets a subsystem identifier not published in the subsystem contract.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-022                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-023

When generation run is requested, the engineering model go shall flag any requirement that links to another requirement in the same document.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-023                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-024

Where hardware interfaces are authored, when generation run is requested, the engineering model go shall render the hardware to software interface allocation view.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-024              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-025

Where system composition is enabled, when generation run is requested, the engineering model go shall produce a system requirement allocation matrix aggregated across the composed subsystems.

| Key          | Value                                                               |
|--------------|---------------------------------------------------------------------|
| Kind         | Requirement                                                         |
| Name         | REQ-EMG-025                                                         |
| Mentioned In | Architecture Intent View , Interaction Flow View , Composition Test |

#### REQ-EMG-026

Where system composition is enabled, when generation run is requested, the engineering model go shall render in the requirement view the delegation link from each delegated requirement to the specific subsystem requirement that satisfies it.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Requirement              |
| Name         | REQ-EMG-026              |
| Mentioned In | Architecture Intent View |

#### REQ-EMG-027

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve an external git subsystem reference by cloning the repository into the workspace .engmod cache.

| Key          | Value                                                                             |
|--------------|-----------------------------------------------------------------------------------|
| Kind         | Requirement                                                                       |
| Name         | REQ-EMG-027                                                                       |
| Mentioned In | Architecture Intent View , Interaction Flow View , REQ-EMG-016 , Composition Test |

REQ-EMG-028

When generation run is requested, the engineering model go shall reject a code trace link that resolves to no requirement or model element.

| Key          | Value                                                                |
|--------------|----------------------------------------------------------------------|
| Kind         | Requirement                                                          |
| Name         | REQ-EMG-028                                                          |
| Mentioned In | Architecture Intent View , Interaction Flow View , Trace Matrix Test |

REQ-EMG-029

When generation run is requested, the engineering model go shall flag each requirement that is traced by neither code nor verification nor a downward delegation.

| Key          | Value                                                                |
|--------------|----------------------------------------------------------------------|
| Kind         | Requirement                                                          |
| Name         | REQ-EMG-029                                                          |
| Mentioned In | Architecture Intent View , Interaction Flow View , Trace Matrix Test |

REQ-EMG-030

When generation run is requested, the engineering model go shall export a machine-readable traceability matrix that links each requirement to its implementing code and verification and delegated subsystem requirement.

| Key          | Value                                                                              |
|--------------|------------------------------------------------------------------------------------|
| Kind         | Requirement                                                                        |
| Name         | REQ-EMG-030                                                                        |
| Mentioned In | Architecture Intent View , Interaction Flow View , REQ-EMG-028 , Trace Matrix Test |

TB-EXTERNAL-VALIDATION-TOOLS

Boundary for external validators, schemas, and compliance toolchains.

| Key          | Value                              |
|--------------|------------------------------------|
| Kind         | Trust Boundary                     |
| Name         | External Validation Tools Boundary |
| Mentioned In | n/a                                |

TB-REPO-WORKSPACE

Boundary limiting file operations to repository-root owned paths.

| Key          | Value                         |
|--------------|-------------------------------|
| Kind         | Trust Boundary                |
| Name         | Repository Workspace Boundary |
| Mentioned In | REQ-EMG-008 , REQ-EMG-017     |

13.2. Catalog References

ACT-AI-AGENT

Consumes MCP outputs to plan and execute model-guided edits.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Actor |
| Name         | ai agent      |
| Mentioned In | n/a           |

ACT-ARCHITECTURE-AUTHOR

Maintains model structure, mappings, and architecture intent.

| Key          | Value               |
|--------------|---------------------|
| Kind         | Catalog Actor       |
| Name         | architecture author |
| Mentioned In | Security View       |

ACT-CI-PIPELINE

Runs automated validation and export checks for pull requests.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Actor |
| Name         | ci pipeline   |
| Mentioned In | n/a           |

ACT-COMPLIANCE-ENGINEER

Reviews controls, risks, and traceability outputs for compliance needs.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Catalog Actor               |
| Name         | compliance engineer         |
| Mentioned In | Security View , REQ-EMG-006 |

ACT-IMPLEMENTATION-ENGINEER

Implements code changes guided by model requirements and support paths.

| Key          | Value                   |
|--------------|-------------------------|
| Kind         | Catalog Actor           |
| Name         | implementation engineer |
| Mentioned In | Security View           |

adr document

Generated document containing authored architecture decision records.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | adr document  |
| Mentioned In | n/a           |

adrs authored

Feature indicating the architecture model includes first-class ADR entries.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |

| Key          | Value         |
|--------------|---------------|
| Name         | adrs authored |
| Mentioned In | n/a           |

allocation

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

| Key          | Value                                                                                          |
|--------------|------------------------------------------------------------------------------------------------|
| Kind         | Catalog Alias                                                                                  |
| Name         | allocation                                                                                     |
| Mentioned In | Architecture Intent View , REQ-EMG-020 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 |

analysis

Validation, inference, and evidence analysis over authored and inferred model content.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | analysis      |
| Mentioned In | Security View |

architecture.yml

Authored architecture source describing functional intent and mappings.

| Key          | Value            |
|--------------|------------------|
| Kind         | Catalog Alias    |
| Name         | architecture.yml |
| Mentioned In | n/a              |

artifact generated

Event indicating one or more generated outputs were produced successfully.

| Key  | Value              |
|------|--------------------|
| Kind | Catalog Alias      |
| Name | artifact generated |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

assurance

Reviews controls, risks, and traceability outputs for compliance needs.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | assurance     |
| Mentioned In | n/a           |

authoring

Authoring and loading of architecture, catalog, requirements, and design inputs.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | authoring     |
| Mentioned In | n/a           |

bundle loader

Loads and normalizes architecture, catalog, requirements, and design documents.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | bundle loader |
| Mentioned In | n/a           |

catalog resolved

Condition requiring all authored references to match known catalog entries.

| Key          | Value            |
|--------------|------------------|
| Kind         | Catalog Alias    |
| Name         | catalog resolved |
| Mentioned In | n/a              |

ci

Runs automated validation and export checks for pull requests.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | ci            |
| Mentioned In | n/a           |

ci mode

Automated pipeline mode focused on deterministic checks and artifacts.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | ci mode       |
| Mentioned In | n/a           |

cmd layer

Command entrypoints that orchestrate loading, validation, generation, and export flows.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | cmd layer     |
| Mentioned In | n/a           |

coding agent

Consumes MCP outputs to plan and execute model-guided edits.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | coding agent  |
| Mentioned In | n/a           |

compliance

Requirement trace exports and compliance-oriented output generation.

| Key          | Value                                  |
|--------------|----------------------------------------|
| Kind         | Catalog Alias                          |
| Name         | compliance                             |
| Mentioned In | Architecture Intent View , REQ-EMG-013 |

composition

Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Catalog Alias             |
| Name         | composition               |
| Mentioned In | REQ-EMG-016 , REQ-EMG-023 |

decisions document

Generated document containing authored architecture decision records.

| Key          | Value              |
|--------------|--------------------|
| Kind         | Catalog Alias      |
| Name         | decisions document |
| Mentioned In | n/a                |

dsl

Generated Structurizr workspace DSL document.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | dsl           |
| Mentioned In | n/a           |

dsl export

Feature enabling Structurizr DSL generation and validation.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |

| Key          | Value      |
|--------------|------------|
| Name         | dsl export |
| Mentioned In | n/a        |

engdoc

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engdoc        |
| Mentioned In | n/a           |

engdragon

Exports threat model views to Threat Dragon and Open OTM representations.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engdragon     |
| Mentioned In | n/a           |

enggemara

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | enggemara     |
| Mentioned In | n/a           |

engineer

Implements code changes guided by model requirements and support paths.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |
| Name | engineer      |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

engineering model toolchain

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Catalog Alias               |
| Name         | engineering model toolchain |
| Mentioned In | n/a                         |

englobster

Generates LOBSTER traceability inputs and reports from model and verification links.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | englobster    |
| Mentioned In | n/a           |

engmcp

Serves model-backed MCP tools with validated inputs and structured responses.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Catalog Alias             |
| Name         | engmcp                    |
| Mentioned In | Security View , Main Test |

engmodel

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engmodel      |
| Mentioned In | n/a           |

engoscal

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engoscal      |
| Mentioned In | n/a           |

engstruct

Exports Structurizr DSL and supports validation/static-site workflows.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engstruct     |
| Mentioned In | n/a           |

engtrlc

Exports requirements and model structures to TRLC artifacts.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | engtrlc       |
| Mentioned In | n/a           |

gemara generation

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

| Key          | Value             |
|--------------|-------------------|
| Kind         | Catalog Alias     |
| Name         | gemara generation |
| Mentioned In | n/a               |

generation

Generation of AsciiDoc and exchange artifacts from model inputs.

| Key          | Value                                                              |
|--------------|--------------------------------------------------------------------|
| Kind         | Catalog Alias                                                      |
| Name         | generation                                                         |
| Mentioned In | Architecture Intent View , REQ-EMG-001 , REQ-EMG-012 , REQ-EMG-025 |

generation requested

Event indicating documentation or export generation execution starts.

| Key          | Value                |
|--------------|----------------------|
| Kind         | Catalog Alias        |
| Name         | generation requested |
| Mentioned In | n/a                  |

go

Go compiler, test runner, and module tooling used by the repository.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | go            |
| Mentioned In | n/a           |

hw interfaces

Feature enabling hardware items and hardware/software interface modeling.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | hw interfaces |
| Mentioned In | n/a           |

inference

Infers runtime, code, and verification evidence from repository sources and test markers.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |

| Key          | Value                                                                |
|--------------|----------------------------------------------------------------------|
| Name         | inference                                                            |
| Mentioned In | Architecture Intent View , Security View , Policy Only Fallback Test |

invalid

State where validation diagnostics include one or more errors.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | invalid       |
| Mentioned In | Security View |

invalid model

Invalid or ambiguous model content that causes incorrect graph construction.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | invalid model |
| Mentioned In | n/a           |

lobster

LOBSTER core and TRLC integration tools used for traceability reporting.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Alias            |
| Name         | lobster                  |
| Mentioned In | Architecture Intent View |

lobster report

Traceability report output generated from TRLC and test links.

| Key  | Value          |
|------|----------------|
| Kind | Catalog Alias  |
| Name | lobster report |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

local mode

Local developer mode for iterative model and code updates.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | local mode    |
| Mentioned In | n/a           |

mcp

MCP server interfaces that expose model-aware tooling for AI agents.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | mcp           |
| Mentioned In | n/a           |

mcp enabled

Feature enabling MCP runtime APIs for model-aware tooling.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | mcp enabled   |
| Mentioned In | n/a           |

model author

Maintains model structure, mappings, and architecture intent.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | model author  |
| Mentioned In | n/a           |

model change

Event indicating architecture or requirement model source changed.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | model change  |
| Mentioned In | n/a           |

otm schema

Open Threat Model schema used to validate OTM output artifacts.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | otm schema    |
| Mentioned In | n/a           |

path traversal

Host file access attempts through untrusted path arguments in MCP tools.

| Key          | Value          |
|--------------|----------------|
| Kind         | Catalog Alias  |
| Name         | path traversal |
| Mentioned In | Security View  |

projection

Builds view-specific projections for architecture communication and traceability outputs.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Alias            |
| Name         | projection               |
| Mentioned In | Architecture Intent View |

schema available

Condition requiring external schema artifacts to be present for validation.

| Key          | Value            |
|--------------|------------------|
| Kind         | Catalog Alias    |
| Name         | schema available |
| Mentioned In | n/a              |

schema tamper

Tampering or drift in external schemas used for export validation.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | schema tamper |
| Mentioned In | n/a           |

ssp json

Generated OSCAL system security plan JSON output.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | ssp json      |
| Mentioned In | n/a           |

structurizr

Structurizr DSL validation runtime used to verify generated DSL.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | structurizr   |
| Mentioned In | n/a           |

system of systems

Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |

| Key          | Value             |
|--------------|-------------------|
| Name         | system of systems |
| Mentioned In | n/a               |

threat dragon

External JSON schemas used to validate Threat Dragon exports.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Alias            |
| Name         | threat dragon            |
| Mentioned In | Architecture Intent View |

threat dragon output

Generated threat model artifact for OWASP Threat Dragon.

| Key          | Value                |
|--------------|----------------------|
| Kind         | Catalog Alias        |
| Name         | threat dragon output |
| Mentioned In | n/a                  |

threat modeling export

Feature enabling Threat Dragon and Open OTM exports.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Catalog Alias          |
| Name         | threat modeling export |
| Mentioned In | n/a                    |

tool call

Event indicating the MCP server received a tool invocation request.

| Key  | Value         |
|------|---------------|
| Kind | Catalog Alias |
| Name | tool call     |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

tool result

Structured MCP tool response payload carrying model-aware query results.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | tool result   |
| Mentioned In | n/a           |

trace complete

Condition requiring requirements to map to implementation and verification evidence.

| Key          | Value          |
|--------------|----------------|
| Kind         | Catalog Alias  |
| Name         | trace complete |
| Mentioned In | n/a            |

trace drift

Requirement links or tests drifting from implementation over time.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | trace drift   |
| Mentioned In | n/a           |

trace export

Feature enabling TRLC and LOBSTER generation flows.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | trace export  |
| Mentioned In | n/a           |

trlc

TRLC parser and runtime used for requirements language validation.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Alias            |
| Name         | trlc                     |
| Mentioned In | Architecture Intent View |

trlc output

Generated TRLC requirements package and model metadata.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | trlc output   |
| Mentioned In | n/a           |

up to date

State where generated outputs reflect the latest authored model inputs.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | up to date    |
| Mentioned In | n/a           |

valid

State where authored model passes validation checks.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | valid         |
| Mentioned In | n/a           |

validation failure

Event indicating model or export validation failed.

| Key          | Value              |
|--------------|--------------------|
| Kind         | Catalog Alias      |
| Name         | validation failure |
| Mentioned In | n/a                |

validator

Validates IDs, references, and model consistency across authored entities and mappings.

| Key          | Value         |
|--------------|---------------|
| Kind         | Catalog Alias |
| Name         | validator     |
| Mentioned In | n/a           |

AV-MALFORMED-MODEL-INPUT

Invalid or ambiguous model content that causes incorrect graph construction.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Catalog Attack Vector |
| Name         | malformed model input |
| Mentioned In | n/a                   |

AV-PATH-TRAVERSAL-IN-MCP

Host file access attempts through untrusted path arguments in MCP tools.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Catalog Attack Vector       |
| Name         | path traversal in mcp calls |
| Mentioned In | <div>Security View</div>    |

AV-SCHEMA-SUPPLY-CHAIN-TAMPER

Tampering or drift in external schemas used for export validation.

| Key  | Value                 |
|------|-----------------------|
| Kind | Catalog Attack Vector |

| Key          | Value                      |
|--------------|----------------------------|
| Name         | schema supply chain tamper |
| Mentioned In | n/a                        |

AV-TRACEABILITY-GAP-DRIFT

Requirement links or tests drifting from implementation over time.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Catalog Attack Vector  |
| Name         | traceability gap drift |
| Mentioned In | Security View          |

COND-CATALOG-REFERENCES-RESOLVE

Condition requiring all authored references to match known catalog entries.

| Key          | Value                                   |
|--------------|-----------------------------------------|
| Kind         | Catalog Condition                       |
| Name         | catalog references resolve successfully |
| Mentioned In | REQ-EMG-007                             |

COND-EXTERNAL-SCHEMA-AVAILABLE

Condition requiring external schema artifacts to be present for validation.

| Key          | Value                                      |
|--------------|--------------------------------------------|
| Kind         | Catalog Condition                          |
| Name         | external schema dependencies are available |
| Mentioned In | REQ-EMG-011                                |

COND-REQUIREMENT-TRACE-COMPLETE

Condition requiring requirements to map to implementation and verification evidence.

| Key  | Value                         |
|------|-------------------------------|
| Kind | Catalog Condition             |
| Name | requirement trace is complete |



| Key          | Value       |
|--------------|-------------|
| Mentioned In | REQ-EMG-010 |

DATA-ARCHITECTURE-DECISION-RECORDS

Generated document containing authored architecture decision records.

| Key          | Value                                  |
|--------------|----------------------------------------|
| Kind         | Catalog Data Term                      |
| Name         | architecture decision records document |
| Mentioned In | REQ-EMG-014                            |

DATA-ARCHITECTURE-MODEL

Authored architecture source describing functional intent and mappings.

| Key          | Value              |
|--------------|--------------------|
| Kind         | Catalog Data Term  |
| Name         | architecture model |
| Mentioned In | n/a                |

DATA-LOBSTER-TRACE-REPORT

Traceability report output generated from TRLC and test links.

| Key          | Value                |
|--------------|----------------------|
| Kind         | Catalog Data Term    |
| Name         | lobster trace report |
| Mentioned In | REQ-EMG-006          |

DATA-MCP-TOOL-RESULT

Structured MCP tool response payload carrying model-aware query results.

| Key          | Value             |
|--------------|-------------------|
| Kind         | Catalog Data Term |
| Name         | mcp tool result   |
| Mentioned In | n/a               |

DATA-OSCAL-SSP

Generated OSCAL system security plan JSON output.

| Key          | Value              |
|--------------|--------------------|
| Kind         | Catalog Data Term  |
| Name         | oscal ssp artifact |
| Mentioned In | n/a                |

DATA-STRUCTURIZR-DSL

Generated Structurizr workspace DSL document.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Data Term        |
| Name         | structurizr dsl artifact |
| Mentioned In | REQ-EMG-005              |

DATA-THREAT-DRAGON-JSON

Generated threat model artifact for OWASP Threat Dragon.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Catalog Data Term           |
| Name         | threat dragon json artifact |
| Mentioned In | REQ-EMG-004                 |

DATA-TRLC-PACKAGE

Generated TRLC requirements package and model metadata.

| Key          | Value             |
|--------------|-------------------|
| Kind         | Catalog Data Term |
| Name         | trlc package      |
| Mentioned In | n/a               |

EVT-ARTIFACT-GENERATED

Event indicating one or more generated outputs were produced successfully.

| Key          | Value                              |
|--------------|------------------------------------|
| Kind         | Catalog Event                      |
| Name         | artifact generated event is raised |
| Mentioned In | REQ-EMG-009                        |

EVT-GENERATION-RUN-REQUESTED

Event indicating documentation or export generation execution starts.

| Key          | Value                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Event                                                                                                                                                                                                                                                                                                                                                             |
| Name         | generation run is requested                                                                                                                                                                                                                                                                                                                                               |
| Mentioned In | REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-009 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 |

EVT-MCP-TOOL-CALL-RECEIVED

Event indicating the MCP server received a tool invocation request.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Catalog Event             |
| Name         | mcp tool call is received |
| Mentioned In | REQ-EMG-007 , REQ-EMG-008 |

EVT-MODEL-UPDATED

Event indicating architecture or requirement model source changed.

| Key          | Value                           |
|--------------|---------------------------------|
| Kind         | Catalog Event                   |
| Name         | model updated event is received |
| Mentioned In | REQ-EMG-001 , REQ-EMG-010       |

EVT-VALIDATION-FAILED

Event indicating model or export validation failed.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Catalog Event                     |
| Name         | validation failed event is raised |
| Mentioned In | n/a                               |

FEAT-ARCHITECTURE-DECISION-RECORDS

Feature indicating the architecture model includes first-class ADR entries.

| Key          | Value                                      |
|--------------|--------------------------------------------|
| Kind         | Catalog Feature                            |
| Name         | architecture decision records are authored |
| Mentioned In | REQ-EMG-014                                |

FEAT-GEMARA-EXPORT

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Feature          |
| Name         | gemara export is enabled |
| Mentioned In | REQ-EMG-015              |

FEAT-HARDWARE-INTERFACES

Feature enabling hardware items and hardware/software interface modeling.

| Key          | Value                            |
|--------------|----------------------------------|
| Kind         | Catalog Feature                  |
| Name         | hardware interfaces are authored |
| Mentioned In | REQ-EMG-024                      |

FEAT-MCP-SERVER

Feature enabling MCP runtime APIs for model-aware tooling.

| Key  | Value           |
|------|-----------------|
| Kind | Catalog Feature |

| Key          | Value                 |
|--------------|-----------------------|
| Name         | mcp server is enabled |
| Mentioned In | REQ-EMG-007           |

FEAT-STRUCTURIZR-EXPORT

Feature enabling Structurizr DSL generation and validation.

| Key          | Value                         |
|--------------|-------------------------------|
| Kind         | Catalog Feature               |
| Name         | structurizr export is enabled |
| Mentioned In | REQ-EMG-005                   |

FEAT-SYSTEM-COMPOSITION

Feature enabling recursive composition of subsystem models (local or external git) into a system-of-systems.

| Key          | Value                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Feature                                                                                                                           |
| Name         | system composition is enabled                                                                                                             |
| Mentioned In | REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 |

FEAT-THREAT-EXPORT

Feature enabling Threat Dragon and Open OTM exports.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Feature          |
| Name         | threat export is enabled |
| Mentioned In | REQ-EMG-004              |

FEAT-TRACEABILITY-EXPORT

Feature enabling TRLC and LOBSTER generation flows.

| Key  | Value                          |
|------|--------------------------------|
| Kind | Catalog Feature                |
| Name | traceability export is enabled |

| Key          | Value                     |
|--------------|---------------------------|
| Mentioned In | REQ-EMG-006 , REQ-EMG-013 |

FG-ARTIFACT-GENERATION

Generation of AsciiDoc and exchange artifacts from model inputs.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Functional Group |
| Name         | artifact generation      |
| Mentioned In | Architecture Intent View |

FG-MCP-INTEGRATION

MCP server interfaces that expose model-aware tooling for AI agents.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Functional Group |
| Name         | mcp integration          |
| Mentioned In | n/a                      |

FG-MODEL-AUTHORING

Authoring and loading of architecture, catalog, requirements, and design inputs.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Functional Group |
| Name         | model authoring          |
| Mentioned In | n/a                      |

FG-TRACEABILITY-COMPLIANCE

Requirement trace exports and compliance-oriented output generation.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Catalog Functional Group    |
| Name         | traceability and compliance |
| Mentioned In | n/a                         |

FG-VALIDATION-ANALYSIS

Validation, inference, and evidence analysis over authored and inferred model content.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Functional Group |
| Name         | validation and analysis  |
| Mentioned In | n/a                      |

FU-ALLOCATION-TRACE

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

| Key          | Value                                                                         |
|--------------|-------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                       |
| Name         | allocation trace                                                              |
| Mentioned In | Architecture Intent View , REQ-EMG-020 , Composition Test , Trace Matrix Test |

FU-ASCIIDOC-GENERATOR

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

| Key          | Value                                                                                                                                                                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                                                                                                                                                                                             |
| Name         | asciidoc generator                                                                                                                                                                                                                                                                                                                  |
| Mentioned In | Architecture Intent View , Security View , Traceability View , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Linking Units Test , AsciiDoc Test , AsciiDoc Views Test , Catalog Systems Test , Composition Test , E2e Test , Ears Preflight Test , Load Test , Project Test , Render Test , Validate Test |

FU-CLI-ORCHESTRATION

Command entrypoints that orchestrate loading, validation, generation, and export flows.

| Key  | Value                   |
|------|-------------------------|
| Kind | Catalog Functional Unit |
| Name | cli orchestration       |

| Key          | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mentioned In | Architecture Intent View , Security View , Traceability View , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Linking Units Test , AsciiDoc Test , AsciiDoc Views Test , Catalog Systems Test , E2e Test , Ears Preflight Test , Gemara Export Test , Gemara Oscal Test , Load Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Project Test , Render Test , Structurizr Export Test , Threat Model Export Test , Trlc Export Test , Validate Test |

FU-CODEMAP-INFERENCE

Infers runtime, code, and verification evidence from repository sources and test markers.

| Key          | Value                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                               |
| Name         | codemap inference                                                                                                                                                     |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Inferred Code Test , Inferred Layers Test , Inferred Verification Test , Scan Test , Trace Matrix Test |

FU-GEMARA-EXPORTER

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

| Key          | Value                                                             |
|--------------|-------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                           |
| Name         | gemara exporter                                                   |
| Mentioned In | Architecture Intent View , Gemara Export Test , Gemara Oscal Test |

FU-LOBSTER-EXPORTER

Generates LOBSTER traceability inputs and reports from model and verification links.

| Key          | Value                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                               |
| Name         | lobster exporter                                                                                      |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Lobster Export Test , Trlc Export Test |

FU-MCP-SERVER

Serves model-backed MCP tools with validated inputs and structured responses.



| Key          | Value                                                            |
|--------------|------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                          |
| Name         | mcp server                                                       |
| Mentioned In | Gemara Export Test , Gemara Oscal Test , Main Test , Server Test |

FU-MODEL-LOADER

Loads and normalizes architecture, catalog, requirements, and design documents.

| Key          | Value                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                             |
| Name         | model loader                                                                                                                                                        |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Catalog Systems Test , Composition Test , E2e Test , Ears Preflight Test , Load Test , Validate Test |

FU-OSCAL-EXPORTER

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

| Key          | Value                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                          |
| Name         | oscal exporter                                                                                                                                   |
| Mentioned In | Architecture Intent View , Catalog Systems Test , E2e Test , Ears Preflight Test , Load Test , Oscal Chain Test , Oscal Ssp Test , Validate Test |

FU-STRUCTURIZR-EXPORTER

Exports Structurizr DSL and supports validation/static-site workflows.

| Key          | Value                                                                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                                           |
| Name         | structurizr exporter                                                                                                                                                              |
| Mentioned In | Architecture Intent View , Security View , Catalog Systems Test , E2e Test , Ears Preflight Test , Load Test , Structurizr Export Test , Threat Model Export Test , Validate Test |

FU-SYSTEM-COMPOSITION

Resolves downward subsystem references from local subdirectories or external git repositories and composes a federated system-of-systems model.

| Key          | Value                                       |
|--------------|---------------------------------------------|
| Kind         | Catalog Functional Unit                     |
| Name         | system composition                          |
| Mentioned In | Architecture Intent View , Composition Test |

FU-THREAT-EXPORTER

Exports threat model views to Threat Dragon and Open OTM representations.

| Key          | Value                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                 |
| Name         | threat exporter                                                                                                                                         |
| Mentioned In | Architecture Intent View , Security View , Catalog Systems Test , E2e Test , Ears Preflight Test , Load Test , Threat Model Export Test , Validate Test |

FU-TRLC-EXPORTER

Exports requirements and model structures to TRLC artifacts.

| Key          | Value                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                               |
| Name         | trlc exporter                                                                                         |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Lobster Export Test , Trlc Export Test |

FU-VALIDATION-ENGINE

Validates IDs, references, and model consistency across authored entities and mappings.

| Key          | Value                                                                                                                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                                                                            |
| Name         | validation engine                                                                                                                                                                                                  |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Catalog Systems Test , Composition Test , E2e Test , Ears Preflight Test , Load Test , Threat Model Export Test , Trace Matrix Test , Validate Test |

FU-VIEW-PROJECTION

Builds view-specific projections for architecture communication and traceability outputs.

| Key          | Value                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Catalog Functional Unit                                                                                                                                                                  |
| Name         | view projection                                                                                                                                                                          |
| Mentioned In | Architecture Intent View , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Linking Units Test , AsciiDoc Test , AsciiDoc Views Test , Project Test , Render Test |

MODE-CI-VALIDATION

Automated pipeline mode focused on deterministic checks and artifacts.

| Key          | Value                        |
|--------------|------------------------------|
| Kind         | Catalog Mode                 |
| Name         | ci validation mode is active |
| Mentioned In | REQ-EMG-009                  |

MODE-LOCAL-DEVELOPMENT

Local developer mode for iterative model and code updates.

| Key          | Value                            |
|--------------|----------------------------------|
| Kind         | Catalog Mode                     |
| Name         | local development mode is active |
| Mentioned In | REQ-EMG-001                      |

REF-GO-TOOLCHAIN

Go compiler, test runner, and module tooling used by the repository.

| Key          | Value                      |
|--------------|----------------------------|
| Kind         | Catalog Referenced Element |
| Name         | go toolchain               |
| Mentioned In | Architecture Intent View   |

REF-LOBSTER-TOOLCHAIN

LOBSTER core and TRLC integration tools used for traceability reporting.

| Key  | Value                      |
|------|----------------------------|
| Kind | Catalog Referenced Element |

| Key          | Value                    |
|--------------|--------------------------|
| Name         | lobster toolchain        |
| Mentioned In | Architecture Intent View |

REF-OPEN-OTM-SCHEMA

Open Threat Model schema used to validate OTM output artifacts.

| Key          | Value                      |
|--------------|----------------------------|
| Kind         | Catalog Referenced Element |
| Name         | open otm schema            |
| Mentioned In | Architecture Intent View   |

REF-STRUCTURIZR-VALIDATOR

Structurizr DSL validation runtime used to verify generated DSL.

| Key          | Value                                  |
|--------------|----------------------------------------|
| Kind         | Catalog Referenced Element             |
| Name         | structurizr validator                  |
| Mentioned In | Architecture Intent View , REQ-EMG-005 |

REF-THREAT-DRAGON-SCHEMAS

External JSON schemas used to validate Threat Dragon exports.

| Key          | Value                      |
|--------------|----------------------------|
| Kind         | Catalog Referenced Element |
| Name         | threat dragon schemas      |
| Mentioned In | Architecture Intent View   |

REF-TRLC-TOOLCHAIN

TRLC parser and runtime used for requirements language validation.

| Key  | Value                      |
|------|----------------------------|
| Kind | Catalog Referenced Element |
| Name | trlc toolchain             |

| Key          | Value                    |
|--------------|--------------------------|
| Mentioned In | Architecture Intent View |

STATE-ARTIFACTS-FRESH

State where generated outputs reflect the latest authored model inputs.

| Key          | Value                         |
|--------------|-------------------------------|
| Kind         | Catalog State                 |
| Name         | generated artifacts are fresh |
| Mentioned In | n/a                           |

STATE-MODEL-INVALID

State where validation diagnostics include one or more errors.

| Key          | Value            |
|--------------|------------------|
| Kind         | Catalog State    |
| Name         | model is invalid |
| Mentioned In | n/a              |

STATE-MODEL-VALID

State where authored model passes validation checks.

| Key          | Value          |
|--------------|----------------|
| Kind         | Catalog State  |
| Name         | model is valid |
| Mentioned In | REQ-EMG-003    |

SYS-ENGINEERING-MODEL-GO

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

| Key  | Value                |
|------|----------------------|
| Kind | Catalog System       |
| Name | engineering model go |

| Key          | Value                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mentioned In | REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-009 , REQ-EMG-010 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 |

TERM-ACTOR

A person or operational role that interacts with functional units.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | actor                  |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | AsciiDoc diagram       |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | AsciiDoc document      |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | AsciiDoc section       |
| Mentioned In | n/a                    |

TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | attack vector          |
| Mentioned In | n/a                    |

TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | authored mapping       |
| Mentioned In | Security View          |

TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | model bundle           |
| Mentioned In | n/a                    |

TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value                    |
|--------------|--------------------------|
| Name         | catalog                  |
| Mentioned In | Architecture Intent View |

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | catalog entry          |
| Mentioned In | n/a                    |

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | CLI command            |
| Mentioned In | n/a                    |

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | code element           |
| Mentioned In | n/a                    |

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | compliance mapping     |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

| Key          | Value                                    |
|--------------|------------------------------------------|
| Kind         | Engineering Model Term                   |
| Name         | control                                  |
| Mentioned In | Architecture Intent View , Security View |

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | control verification   |
| Mentioned In | n/a                    |

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | data object            |
| Mentioned In | n/a                    |

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | architecture decision  |
| Mentioned In | n/a                    |

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | deployment target      |
| Mentioned In | n/a                    |

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Engineering Model Term   |
| Name         | design document          |
| Mentioned In | Architecture Intent View |

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | design view            |
| Mentioned In | n/a                    |

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

| Key          | Value                                |
|--------------|--------------------------------------|
| Kind         | Engineering Model Term               |
| Name         | event                                |
| Mentioned In | Security View , State Lifecycle View |

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

| Key          | Value                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                                                                             |
| Name         | evidence                                                                                                                           |
| Mentioned In | Architecture Intent View , Security View , Traceability View , State Lifecycle View , Interaction Flow View , Audit Redaction Test |

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

| Key          | Value                                                                    |
|--------------|--------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                   |
| Name         | flow                                                                     |
| Mentioned In | Architecture Intent View , Interaction Flow View , Pr Opened Review Flow |

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | flow step              |
| Mentioned In | n/a                    |

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | functional group       |
| Mentioned In | n/a                    |

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value                |
|--------------|----------------------|
| Name         | functional unit      |
| Mentioned In | State Lifecycle View |

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | generation workflow    |
| Mentioned In | n/a                    |

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | inference hint         |
| Mentioned In | n/a                    |

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | inferred mapping       |
| Mentioned In | n/a                    |

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | interface              |

| Key          | Value                     |
|--------------|---------------------------|
| Mentioned In | REQ-EMG-021 , REQ-EMG-024 |

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | lint run               |
| Mentioned In | n/a                    |

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | LOBSTER activity trace |
| Mentioned In | n/a                    |

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

| Key          | Value                                                        |
|--------------|--------------------------------------------------------------|
| Kind         | Engineering Model Term                                       |
| Name         | MCP server                                                   |
| Mentioned In | Architecture Intent View , Security View , Traceability View |

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP stdio transport    |
| Mentioned In | n/a                    |

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP tool               |
| Mentioned In | n/a                    |

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP tool response      |
| Mentioned In | n/a                    |

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

| Key          | Value                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                                                                                                 |
| Name         | model                                                                                                                                                  |
| Mentioned In | Architecture Intent View , Security View , REQ-EMG-004 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-020 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-030 , Load Test |

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Open OTM document      |
| Mentioned In | n/a                    |

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Engineering Model Term   |
| Name         | OSCAL assessment results |
| Mentioned In | n/a                      |

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | OSCAL POA&M            |
| Mentioned In | n/a                    |

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | OSCAL SSP              |
| Mentioned In | REQ-EMG-013            |

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | POA&M item             |
| Mentioned In | n/a                    |

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value           |
|--------------|-----------------|
| Name         | reference index |
| Mentioned In | n/a             |

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | referenced element     |
| Mentioned In | n/a                    |

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | repository index       |
| Mentioned In | n/a                    |

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

| Key          | Value                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                                                                                                                                               |
| Name         | requirement                                                                                                                                                                                          |
| Mentioned In | Architecture Intent View , Security View , Traceability View , Interaction Flow View , REQ-EMG-007 , REQ-EMG-010 , REQ-EMG-023 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 |

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |



| Key          | Value                                                                    |
|--------------|--------------------------------------------------------------------------|
| Name         | risk                                                                     |
| Mentioned In | Architecture Intent View , Security View , Audit Record Persistence Test |

#### TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | runtime element        |
| Mentioned In | n/a                    |

#### TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | security context       |
| Mentioned In | n/a                    |

#### TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | source block           |
| Mentioned In | n/a                    |

#### TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | state                  |

| Key          | Value                                |
|--------------|--------------------------------------|
| Mentioned In | Security View , State Lifecycle View |

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Structurizr DSL        |
| Mentioned In | n/a                    |

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat assumption      |
| Mentioned In | n/a                    |

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Threat Dragon document |
| Mentioned In | n/a                    |

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat mitigation      |
| Mentioned In | n/a                    |

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat model export    |
| Mentioned In | n/a                    |

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

| Key          | Value                        |
|--------------|------------------------------|
| Kind         | Engineering Model Term       |
| Name         | threat out-of-scope decision |
| Mentioned In | n/a                          |

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat scenario        |
| Mentioned In | n/a                    |

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | trace marker           |
| Mentioned In | REQ-EMG-010            |

TERM-TRLIC-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | TRLC package           |
| Mentioned In | REQ-EMG-006            |

## TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Engineering Model Term      |
| Name         | trust boundary              |
| Mentioned In | Security View , REQ-EMG-008 |

## TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | upward linking         |
| Mentioned In | n/a                    |

## TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

| Key          | Value                                                                                  |
|--------------|----------------------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                                 |
| Name         | validation                                                                             |
| Mentioned In | Architecture Intent View , REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-011 , Ota Campaign Test |

## TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value                 |
|--------------|-----------------------|
| Name         | validation diagnostic |
| Mentioned In | n/a                   |

### TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | verification check     |
| Mentioned In | n/a                    |

### TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

| Key          | Value                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind         | Engineering Model Term                                                                                                                                               |
| Name         | view                                                                                                                                                                 |
| Mentioned In | Architecture Intent View , Security View , Traceability View , State Lifecycle View , Interaction Flow View , REQ-EMG-024 , REQ-EMG-026 , REQ-EMG-027 , Project Test |

## 13.3. Inferred Runtime References

## 13.4. Inferred Code References

[aws-sdk-go-v2/secretsmanager \(secrets\\_configuration.ts\)](#)

Inferred library\_external dependency owned by FU-SECRETS-CONFIGURATION from src/secrets\_configuration.ts.

| Key          | Value                        |
|--------------|------------------------------|
| Kind         | Code library_external        |
| Owner        | FU-SECRETS-CONFIGURATION     |
| Source       | src/secrets_configuration.ts |
| Mentioned In | n/a                          |

[github-rest-sdk \(pr\\_context\\_assembly.ts\)](#)

Inferred library\_external dependency owned by FU-PR-CONTEXT-ASSEMBLY from src/pr\_context\_assembly.ts.

| Key          | Value                      |
|--------------|----------------------------|
| Kind         | Code library_external      |
| Owner        | FU-PR-CONTEXT-ASSEMBLY     |
| Source       | src/pr_context_assembly.ts |
| Mentioned In | n/a                        |

[github-rest-sdk/publication-client \(review\\_publication.ts\)](#)

Inferred library\_external dependency owned by FU-REVIEW-PUBLICATION from src/review\_publication.ts.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Code library_external     |
| Owner        | FU-REVIEW-PUBLICATION     |
| Source       | src/review_publication.ts |
| Mentioned In | n/a                       |

[github.com/defenseunicorns/go-oscal/src/types/oscal-1-1-3 \(gemara\\_oscal.go\)](#)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_oscal.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_oscal.go       |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara \(gemara.go\)](#)

Inferred library\_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-MCP-SERVER         |
| Source       | mcp/gemara.go         |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara](https://github.com/gemaraproj/go-gemara) (gemara\_evaluation.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_evaluation.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_evaluation.go  |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara](https://github.com/gemaraproj/go-gemara) (gemara\_export.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_export.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_export.go      |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara](https://github.com/gemaraproj/go-gemara) (gemara\_export\_test.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_export\_test.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_export_test.go |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara](https://github.com/gemaraproj/go-gemara) (gemara\_extended.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_extended.go.

| Key    | Value                 |
|--------|-----------------------|
| Kind   | Code library_external |
| Owner  | FU-GEMARA-EXPORTER    |
| Source | gemara_extended.go    |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/gemaraproj/go-gemara](https://github.com/gemaraproj/go-gemara) (gemara\_l1l3.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_l1l3.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_l1l3.go        |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara/fetcher](https://github.com/gemaraproj/go-gemara/fetcher) (gemara\_export\_test.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_export\_test.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_export_test.go |
| Mentioned In | n/a                   |

[github.com/gemaraproj/go-gemara/gemaraconv](https://github.com/gemaraproj/go-gemara/gemaraconv) (gemara\_oscal.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_oscal.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_oscal.go       |
| Mentioned In | n/a                   |

[github.com/goccy/go-yaml](https://github.com/goccy/go-yaml) (gemara\_export.go)

Inferred library\_external dependency owned by FU-GEMARA-EXPORTER from gemara\_export.go.



| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-GEMARA-EXPORTER    |
| Source       | gemara_export.go      |
| Mentioned In | n/a                   |

[github.com/hashicorp/hcl/v2](https://github.com/hashicorp/hcl/v2) (inferred\_layers.go)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | inferred_layers.go    |
| Mentioned In | n/a                   |

[github.com/hashicorp/hcl/v2/hclparse](https://github.com/hashicorp/hcl/v2/hclparse) (inferred\_layers.go)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | inferred_layers.go    |
| Mentioned In | n/a                   |

[github.com/labeth/ears-lint-go](https://github.com/labeth/ears-lint-go) (ears\_preflight.go)

Inferred library\_external dependency owned by FU-VALIDATION-ENGINE from ears\_preflight.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-VALIDATION-ENGINE  |
| Source       | ears_preflight.go     |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go \(gemara.go\)](#)

Inferred library\_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-MCP-SERVER         |
| Source       | mcp/gemara.go         |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engdoc/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CLI-ORCHESTRATION  |
| Source       | cmd/engdoc/main.go    |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-THREAT-EXPORTER from cmd/engdragon/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-THREAT-EXPORTER    |
| Source       | cmd/engdragon/main.go |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-CLI-ORCHESTRATION from cmd/enggemara/main.go.

| Key    | Value                 |
|--------|-----------------------|
| Kind   | Code library_external |
| Owner  | FU-CLI-ORCHESTRATION  |
| Source | cmd/enggemara/main.go |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-LOBSTER-EXPORTER from cmd/englobster/main.go.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Code library_external  |
| Owner        | FU-LOBSTER-EXPORTER    |
| Source       | cmd/englobster/main.go |
| Mentioned In | n/a                    |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-OSCAL-EXPORTER from cmd/engoscal/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-OSCAL-EXPORTER     |
| Source       | cmd/engoscal/main.go  |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-STRUCTURIZR-EXPORTER from cmd/engstruct/main.go.

| Key          | Value                   |
|--------------|-------------------------|
| Kind         | Code library_external   |
| Owner        | FU-STRUCTURIZR-EXPORTER |
| Source       | cmd/engstruct/main.go   |
| Mentioned In | n/a                     |

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library\_external dependency owned by FU-ALLOCATION-TRACE from cmd/engtrace/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-ALLOCATION-TRACE   |
| Source       | cmd/engtrace/main.go  |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go](https://github.com/labeth/engineering-model-go) (main.go)

Inferred library\_external dependency owned by FU-TRLC-EXPORTER from cmd/engtrlc/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-TRLC-EXPORTER      |
| Source       | cmd/engtrlc/main.go   |
| Mentioned In | n/a                   |

[github.com/labeth/engineering-model-go](https://github.com/labeth/engineering-model-go) (main.go)

Inferred library\_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engview/main.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CLI-ORCHESTRATION  |
| Source       | cmd/engview/main.go   |
| Mentioned In | n/a                   |

[github.com/santhosh-tekuri/jsonschema/v5](https://github.com/santhosh-tekuri/jsonschema/v5) (threat\_model\_export\_test.go)

Inferred library\_external dependency owned by FU-THREAT-EXPORTER from threat\_model\_export\_test.go.

| Key          | Value                       |
|--------------|-----------------------------|
| Kind         | Code library_external       |
| Owner        | FU-THREAT-EXPORTER          |
| Source       | threat_model_export_test.go |
| Mentioned In | n/a                         |

[github.com/tree-sitter/go-tree-sitter \(scan.go\)](#)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | codemap/scan.go       |
| Mentioned In | n/a                   |

[github.com/tree-sitter/tree-sitter-go/bindings/go \(scan.go\)](#)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | codemap/scan.go       |
| Mentioned In | n/a                   |

[github.com/tree-sitter/tree-sitter-rust/bindings/go \(scan.go\)](#)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | codemap/scan.go       |
| Mentioned In | n/a                   |

[github.com/tree-sitter/tree-sitter-typescript/bindings/go \(scan.go\)](#)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

| Key    | Value                 |
|--------|-----------------------|
| Kind   | Code library_external |
| Owner  | FU-CODEMAP-INFERENCE  |
| Source | codemap/scan.go       |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[gopkg.in/yaml.v3 \(checkout\\_api.go\)](#)

Inferred library\_external dependency owned by FU-CHECKOUT from src/checkout\_api.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CHECKOUT           |
| Source       | src/checkout_api.go   |
| Mentioned In | n/a                   |

[gopkg.in/yaml.v3 \(inferred\\_layers.go\)](#)

Inferred library\_external dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-CODEMAP-INFERENCE  |
| Source       | inferred_layers.go    |
| Mentioned In | n/a                   |

[gopkg.in/yaml.v3 \(load.go\)](#)

Inferred library\_external dependency owned by FU-MODEL-LOADER from model/load.go.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-MODEL-LOADER       |
| Source       | model/load.go         |
| Mentioned In | n/a                   |

[serde\\_json::json \(payment\\_engine.rs\)](#)

Inferred library\_external dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment\_engine.rs.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_external    |
| Owner        | FU-PAYMENT-AUTHORIZATION |
| Source       | src/payment_engine.rs    |
| Mentioned In | n/a                      |

serde\_json::json (review\_orchestrator.rs)

Inferred library\_external dependency owned by FU-REVIEW-ORCHESTRATION from src/review\_orchestrator.rs.

| Key          | Value                      |
|--------------|----------------------------|
| Kind         | Code library_external      |
| Owner        | FU-REVIEW-ORCHESTRATION    |
| Source       | src/review_orchestrator.rs |
| Mentioned In | n/a                        |

zod (risk\_scorer.ts)

Inferred library\_external dependency owned by FU-RISK-SCORING from src/risk\_scorer.ts.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Code library_external |
| Owner        | FU-RISK-SCORING       |
| Source       | src/risk_scorer.ts    |
| Mentioned In | n/a                   |

./support/audit\_envelope (risk\_scorer.ts)

Inferred library\_first\_party dependency owned by FU-RISK-SCORING from src/risk\_scorer.ts.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-RISK-SCORING          |
| Source       | src/risk_scorer.ts       |
| Mentioned In | n/a                      |

crate::domain::events::PaymentEvent (payment\_engine.rs)

Inferred library\_first\_party dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment\_engine.rs.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-PAYMENT-AUTHORIZATION |
| Source       | src/payment_engine.rs    |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/codemap (inferred\_code.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_code.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_code.go         |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/codemap (inferred\_verification.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_verification.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_verification.go |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/codemap (server.go)

Inferred library\_first\_party dependency owned by FU-MCP-SERVER from mcp/server.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-MCP-SERVER            |
| Source | mcp/server.go            |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/mcp](https://github.com/labeth/engineering-model-go/mcp) (main.go)

Inferred library\_first\_party dependency owned by FU-MCP-SERVER from cmd/engmcp/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-MCP-SERVER            |
| Source       | cmd/engmcp/main.go       |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc.go              |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_composition.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_composition.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_composition.go  |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_decisions.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_decisions.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_decisions.go    |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(asciidoc\\_design\\_refs.go\)](#)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_design\_refs.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_design_refs.go  |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(asciidoc\\_diagrams\\_core.go\)](#)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_diagrams\_core.go.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Code library_first_party  |
| Owner        | FU-ASCIIDOC-GENERATOR     |
| Source       | asciidoc_diagrams_core.go |
| Mentioned In | n/a                       |

[github.com/labeth/engineering-model-go/model \(asciidoc\\_diagrams\\_core\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_diagrams\_core\_test.go.

| Key          | Value                          |
|--------------|--------------------------------|
| Kind         | Code library_first_party       |
| Owner        | FU-ASCIIDOC-GENERATOR          |
| Source       | asciidoc_diagrams_core_test.go |
| Mentioned In | n/a                            |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_diagrams\_runtime.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_diagrams\_runtime.go.

| Key          | Value                        |
|--------------|------------------------------|
| Kind         | Code library_first_party     |
| Owner        | FU-ASCIIDOC-GENERATOR        |
| Source       | asciidoc_diagrams_runtime.go |
| Mentioned In | n/a                          |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_diagrams\_runtime\_test.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_diagrams\_runtime\_test.go.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Code library_first_party          |
| Owner        | FU-ASCIIDOC-GENERATOR             |
| Source       | asciidoc_diagrams_runtime_test.go |
| Mentioned In | n/a                               |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_gemara.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_gemara.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_gemara.go       |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (asciidoc\_linking\_units.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_linking\_units.go.

| Key    | Value                     |
|--------|---------------------------|
| Kind   | Code library_first_party  |
| Owner  | FU-ASCIIDOC-GENERATOR     |
| Source | asciidoc_linking_units.go |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/model\(asciidoc\\_test.go\)](https://github.com/labeth/engineering-model-go/model(asciidoc_test.go))

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_test.go         |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model\(asciidoc\\_views.go\)](https://github.com/labeth/engineering-model-go/model(asciidoc_views.go))

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_views.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_views.go        |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model\(asciidoc\\_views\\_test.go\)](https://github.com/labeth/engineering-model-go/model(asciidoc_views_test.go))

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_views\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_views_test.go   |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model\(catalog\\_description\\_validation.go\)](https://github.com/labeth/engineering-model-go/model(catalog_description_validation.go))

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from catalog\_description\_validation.go.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Code library_first_party          |
| Owner        | FU-VALIDATION-ENGINE              |
| Source       | catalog_description_validation.go |
| Mentioned In | n/a                               |

[github.com/labeth/engineering-model-go/model \(catalog\\_systems\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from catalog\_systems\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VALIDATION-ENGINE     |
| Source       | catalog_systems_test.go  |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(composition.go\)](#)

Inferred library\_first\_party dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-SYSTEM-COMPOSITION    |
| Source       | composition.go           |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(composition\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-SYSTEM-COMPOSITION from composition\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-SYSTEM-COMPOSITION    |
| Source       | composition_test.go      |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/model (ears\_preflight.go)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from ears\_preflight.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VALIDATION-ENGINE     |
| Source       | ears_preflight.go        |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/model (ears\_preflight\_test.go)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from ears\_preflight\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VALIDATION-ENGINE     |
| Source       | ears_preflight_test.go   |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/model (gemara.go)

Inferred library\_first\_party dependency owned by FU-MCP-SERVER from mcp/gemara.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-MCP-SERVER            |
| Source       | mcp/gemara.go            |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/model (gemara\_evaluation.go)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_evaluation.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-GEMARA-EXPORTER       |
| Source | gemara_evaluation.go     |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (gemara\_export.go)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_export.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_export.go         |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (gemara\_export\_test.go)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_export\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_export_test.go    |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (gemara\_extended.go)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_extended.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_extended.go       |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (gemara\_l1l3.go)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_l1l3.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_l1l3.go           |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(gemara\\_oscal.go\)](#)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_oscal.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_oscal.go          |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(gemara\\_oscal\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-GEMARA-EXPORTER from gemara\_oscal\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-GEMARA-EXPORTER       |
| Source       | gemara_oscal_test.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(generate.go\)](#)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | generate.go              |
| Mentioned In | n/a                      |



[github.com/labeth/engineering-model-go/model \(inferred\\_code.go\)](#)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_code.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_code.go         |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(inferred\\_layers.go\)](#)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_layers.go       |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(inferred\\_layers\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_layers_test.go  |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(inferred\\_verification.go\)](#)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_verification.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-CODEMAP-INFERENCE     |
| Source | inferred_verification.go |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (inferred\_verification\_test.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_verification\_test.go.

| Key          | Value                         |
|--------------|-------------------------------|
| Kind         | Code library_first_party      |
| Owner        | FU-CODEMAP-INFERENCE          |
| Source       | inferred_verification_test.go |
| Mentioned In | n/a                           |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (oscal\_ar.go)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_ar.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_ar.go              |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (oscal\_compliance.go)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_compliance.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_compliance.go      |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (oscal\_poam.go)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_poam.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_poam.go            |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(oscal\\_ssp.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_ssp.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_ssp.go             |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(oscal\\_ssp\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_ssp\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_ssp_test.go        |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(project.go\)](#)

Inferred library\_first\_party dependency owned by FU-VIEW-PROJECTION from view/project.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VIEW-PROJECTION       |
| Source       | view/project.go          |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(project\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-VIEW-PROJECTION from view/project\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VIEW-PROJECTION       |
| Source       | view/project_test.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(server.go\)](#)

Inferred library\_first\_party dependency owned by FU-MCP-SERVER from mcp/server.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-MCP-SERVER            |
| Source       | mcp/server.go            |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(structurizr\\_export.go\)](#)

Inferred library\_first\_party dependency owned by FU-STRUCTURIZR-EXPORTER from structurizr\_export.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-STRUCTURIZR-EXPORTER  |
| Source       | structurizr_export.go    |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(threat\\_model\\_export.go\)](#)

Inferred library\_first\_party dependency owned by FU-THREAT-EXPORTER from threat\_model\_export.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-THREAT-EXPORTER       |
| Source | threat_model_export.go   |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (trace\_matrix.go)

Inferred library\_first\_party dependency owned by FU-ALLOCATION-TRACE from trace\_matrix.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ALLOCATION-TRACE      |
| Source       | trace_matrix.go          |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (trace\_matrix\_test.go)

Inferred library\_first\_party dependency owned by FU-ALLOCATION-TRACE from trace\_matrix\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ALLOCATION-TRACE      |
| Source       | trace_matrix_test.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (trlc\_export.go)

Inferred library\_first\_party dependency owned by FU-TRLC-EXPORTER from trlc\_export.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-TRLC-EXPORTER         |
| Source       | trlc_export.go           |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model](https://github.com/labeth/engineering-model-go/model) (validate.go)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from validate/validate.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VALIDATION-ENGINE     |
| Source       | validate/validate.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/model \(validate\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from validate/validate\_test.go.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Code library_first_party  |
| Owner        | FU-VALIDATION-ENGINE      |
| Source       | validate/validate_test.go |
| Mentioned In | n/a                       |

[github.com/labeth/engineering-model-go/render/diagramstyle \(asciidoc.go\)](#)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc.go              |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/render/diagramstyle \(asciidoc\\_diagrams\\_core.go\)](#)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_diagrams\_core.go.

| Key          | Value                     |
|--------------|---------------------------|
| Kind         | Code library_first_party  |
| Owner        | FU-ASCIIDOC-GENERATOR     |
| Source       | asciidoc_diagrams_core.go |
| Mentioned In | n/a                       |

github.com/labeth/engineering-model-go/render/diagramstyle (render.go)

Inferred library\_first\_party dependency owned by FU-VIEW-PROJECTION from render/mermaid/render.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VIEW-PROJECTION       |
| Source       | render/mermaid/render.go |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/render/mermaid (generate.go)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | generate.go              |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (asciidoc.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc.go              |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (asciidoc\_composition.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_composition.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-ASCIIDOC-GENERATOR    |
| Source | asciidoc_composition.go  |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/validate](#) (asciidoc\_test.go)

Inferred library\_first\_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ASCIIDOC-GENERATOR    |
| Source       | asciidoc_test.go         |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate](#) (catalog\_description\_validation.go)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from catalog\_description\_validation.go.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | Code library_first_party          |
| Owner        | FU-VALIDATION-ENGINE              |
| Source       | catalog_description_validation.go |
| Mentioned In | n/a                               |

[github.com/labeth/engineering-model-go/validate](#) (checkout\_api.go)

Inferred library\_first\_party dependency owned by FU-CHECKOUT from src/checkout\_api.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CHECKOUT              |
| Source       | src/checkout_api.go      |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate](#) (composition.go)

Inferred library\_first\_party dependency owned by FU-SYSTEM-COMPOSITION from composition.go.



| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-SYSTEM-COMPOSITION    |
| Source       | composition.go           |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(composition\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-SYSTEM-COMPOSITION from composition\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-SYSTEM-COMPOSITION    |
| Source       | composition_test.go      |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(e2e\\_test.go\)](#)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from e2e\_test.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | e2e_test.go              |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(ears\\_preflight.go\)](#)

Inferred library\_first\_party dependency owned by FU-VALIDATION-ENGINE from ears\_preflight.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-VALIDATION-ENGINE     |
| Source       | ears_preflight.go        |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (generate.go)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | generate.go              |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (inferred\_code.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_code.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_code.go         |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (inferred\_layers.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_layers.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | inferred_layers.go       |
| Mentioned In | n/a                      |

github.com/labeth/engineering-model-go/validate (inferred\_verification.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from inferred\_verification.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-CODEMAP-INFERENCE     |
| Source | inferred_verification.go |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from cmd/engdoc/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | cmd/engdoc/main.go       |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-THREAT-EXPORTER from cmd/engdragon/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-THREAT-EXPORTER       |
| Source       | cmd/engdragon/main.go    |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from cmd/engoscal/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | cmd/engoscal/main.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-STRUCTURIZR-EXPORTER from cmd/engstruct/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-STRUCTURIZR-EXPORTER  |
| Source       | cmd/engstruct/main.go    |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-ALLOCATION-TRACE from cmd/engtrace/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-ALLOCATION-TRACE      |
| Source       | cmd/engtrace/main.go     |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(main.go\)](#)

Inferred library\_first\_party dependency owned by FU-CLI-ORCHESTRATION from cmd/engview/main.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CLI-ORCHESTRATION     |
| Source       | cmd/engview/main.go      |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(oscal\\_ar.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_ar.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_ar.go              |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(oscal\\_compliance.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_compliance.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_compliance.go      |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(oscal\\_poam.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_poam.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_poam.go            |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(oscal\\_ssp.go\)](#)

Inferred library\_first\_party dependency owned by FU-OSCAL-EXPORTER from oscal\_ssp.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-OSCAL-EXPORTER        |
| Source       | oscal_ssp.go             |
| Mentioned In | n/a                      |

[github.com/labeth/engineering-model-go/validate \(project.go\)](#)

Inferred library\_first\_party dependency owned by FU-VIEW-PROJECTION from view/project.go.

| Key    | Value                    |
|--------|--------------------------|
| Kind   | Code library_first_party |
| Owner  | FU-VIEW-PROJECTION       |
| Source | view/project.go          |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

[github.com/labeth/engineering-model-go/validate](https://github.com/labeth/engineering-model-go/validate) (scan.go)

Inferred library\_first\_party dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Code library_first_party |
| Owner        | FU-CODEMAP-INFERENCE     |
| Source       | codemap/scan.go          |
| Mentioned In | n/a                      |

### 13.5. Verification References

[VER-INF-E2E-AUTHORIZED\\_CHECKOUT\\_FLOW-1B5367](#)

executes authorized checkout flow from session start through payment authorization success

| Key          | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Kind         | e2e                                                                          |
| Status       | not-run                                                                      |
| Source       | examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml |
| Name         | Authorized Checkout Flow                                                     |
| Mentioned In | Interaction Flow View , Authorized Checkout Flow                             |

[VER-INF-E2E-OTA\\_ROLLOUT\\_FLOW-45BCCC](#)

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

| Key          | Value                                                                  |
|--------------|------------------------------------------------------------------------|
| Kind         | e2e                                                                    |
| Status       | not-run                                                                |
| Source       | examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml |
| Name         | Ota Rollout Flow                                                       |
| Mentioned In | Interaction Flow View , Ota Rollout Flow                               |

VER-INF-E2E-PR\_OPENED\_REVIEW\_FLOW-7E76F6

exercises end-to-end pull request opened flow from webhook ingest to published review

| Key          | Value                                                                             |
|--------------|-----------------------------------------------------------------------------------|
| Kind         | e2e                                                                               |
| Status       | not-run                                                                           |
| Source       | examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml |
| Name         | Pr Opened Review Flow                                                             |
| Mentioned In | Interaction Flow View , Pr Opened Review Flow                                     |

VER-INF-INTEGRATION-AUDIT\_RECORD\_PERSISTENCE\_TEST-7F02A4

checks audit record persistence includes payment id and computed risk score

| Key          | Value                                                                                   |
|--------------|-----------------------------------------------------------------------------------------|
| Kind         | integration                                                                             |
| Status       | not-run                                                                                 |
| Source       | examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts |
| Name         | Audit Record Persistence Test                                                           |
| Mentioned In | Interaction Flow View , Audit Record Persistence Test                                   |

VER-INF-INTEGRATION-AUDIT\_REDACTION\_TEST-75E1E4

checks audit evidence redaction for secrets and sensitive payload fields

| Key          | Value                                                                                  |
|--------------|----------------------------------------------------------------------------------------|
| Kind         | integration                                                                            |
| Status       | not-run                                                                                |
| Source       | examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts |
| Name         | Audit Redaction Test                                                                   |
| Mentioned In | Interaction Flow View , Audit Redaction Test                                           |

VER-INF-INTEGRATION-FRAUD\_GATE\_TEST-C0B163

validates fraud gate escalation and block decisions before authorization completion

| Key          | Value                                                                     |
|--------------|---------------------------------------------------------------------------|
| Kind         | integration                                                               |
| Status       | not-run                                                                   |
| Source       | examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs |
| Name         | Fraud Gate Test                                                           |
| Mentioned In | Interaction Flow View , Fraud Gate Test                                   |

VER-INF-INTEGRATION-LOGGING\_PIPELINE\_TEST-6E4BE8

validates telemetry logging pipeline persistence and operational reporting signals

| Key          | Value                                                                             |
|--------------|-----------------------------------------------------------------------------------|
| Kind         | integration                                                                       |
| Status       | not-run                                                                           |
| Source       | examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts |
| Name         | Logging Pipeline Test                                                             |
| Mentioned In | Interaction Flow View , Logging Pipeline Test                                     |

VER-INF-INTEGRATION-POLICY\_ONLY\_FALLBACK\_TEST-99C003

verifies policy-only fallback when Bedrock inference is unavailable

| Key          | Value                                                                                       |
|--------------|---------------------------------------------------------------------------------------------|
| Kind         | integration                                                                                 |
| Status       | not-run                                                                                     |
| Source       | examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs |
| Name         | Policy Only Fallback Test                                                                   |
| Mentioned In | Interaction Flow View , Policy Only Fallback Test                                           |

VER-INF-TEST-ASCIIDOC\_DIAGRAMS\_CORE\_TEST-3FAC84

Inferred from test source artifact.

| Key  | Value |
|------|-------|
| Kind | test  |



| Key          | Value                                               |
|--------------|-----------------------------------------------------|
| Status       | not-run                                             |
| Source       | asciidoc_diagrams_core_test.go                      |
| Name         | Asciidoc Diagrams Core Test                         |
| Mentioned In | Interaction Flow View , Asciidoc Diagrams Core Test |

VER-INF-TEST-ASCIIDOC\_DIAGRAMS\_RUNTIME\_TEST-A8483D

Inferred from test source artifact.

| Key          | Value                                                  |
|--------------|--------------------------------------------------------|
| Kind         | test                                                   |
| Status       | not-run                                                |
| Source       | asciidoc_diagrams_runtime_test.go                      |
| Name         | Asciidoc Diagrams Runtime Test                         |
| Mentioned In | Interaction Flow View , Asciidoc Diagrams Runtime Test |

VER-INF-TEST-ASCIIDOC\_LINKING\_UNITS\_TEST-22FDE2

Inferred from test source artifact.

| Key          | Value                                               |
|--------------|-----------------------------------------------------|
| Kind         | test                                                |
| Status       | not-run                                             |
| Source       | asciidoc_linking_units_test.go                      |
| Name         | Asciidoc Linking Units Test                         |
| Mentioned In | Interaction Flow View , Asciidoc Linking Units Test |

VER-INF-TEST-ASCIIDOC\_TEST-F60765

Inferred from test source artifact.

| Key    | Value   |
|--------|---------|
| Kind   | test    |
| Status | not-run |

| Key          | Value                                 |
|--------------|---------------------------------------|
| Source       | asciidoc_test.go                      |
| Name         | Asciidoc Test                         |
| Mentioned In | Interaction Flow View , Asciidoc Test |

VER-INF-TEST-ASCIIDOC\_VIEWS\_TEST-BFC7F2

Inferred from test source artifact.

| Key          | Value                                       |
|--------------|---------------------------------------------|
| Kind         | test                                        |
| Status       | not-run                                     |
| Source       | asciidoc_views_test.go                      |
| Name         | Asciidoc Views Test                         |
| Mentioned In | Interaction Flow View , Asciidoc Views Test |

VER-INF-TEST-CATALOG\_SYSTEMS\_TEST-6807D7

Inferred from test source artifact.

| Key          | Value                                        |
|--------------|----------------------------------------------|
| Kind         | test                                         |
| Status       | not-run                                      |
| Source       | catalog_systems_test.go                      |
| Name         | Catalog Systems Test                         |
| Mentioned In | Interaction Flow View , Catalog Systems Test |

VER-INF-TEST-CHECKOUT\_DECLINE\_RESPONSE\_TEST-A14671

validates checkout decline response contract includes reason and retry guidance

| Key    | Value                                                                                 |
|--------|---------------------------------------------------------------------------------------|
| Kind   | test                                                                                  |
| Status | not-run                                                                               |
| Source | examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go |

| Key          | Value                                                  |
|--------------|--------------------------------------------------------|
| Name         | Checkout Decline Response Test                         |
| Mentioned In | Interaction Flow View , Checkout Decline Response Test |

VER-INF-TEST-COMPOSITION\_TEST-8FE8F1

Inferred from test source artifact.

| Key          | Value                                    |
|--------------|------------------------------------------|
| Kind         | test                                     |
| Status       | not-run                                  |
| Source       | composition_test.go                      |
| Name         | Composition Test                         |
| Mentioned In | Interaction Flow View , Composition Test |

VER-INF-TEST-E2E\_TEST-BE56F8

Inferred from test source artifact.

| Key          | Value                            |
|--------------|----------------------------------|
| Kind         | test                             |
| Status       | not-run                          |
| Source       | e2e_test.go                      |
| Name         | E2e Test                         |
| Mentioned In | Interaction Flow View , E2e Test |

VER-INF-TEST-EARS\_PREFLIGHT\_TEST-1ABE26

Inferred from test source artifact.

| Key    | Value                  |
|--------|------------------------|
| Kind   | test                   |
| Status | not-run                |
| Source | ears_preflight_test.go |
| Name   | Ears Preflight Test    |

| Key          | Value                                       |
|--------------|---------------------------------------------|
| Mentioned In | Interaction Flow View , Ears Preflight Test |

VER-INF-TEST-GEMARA\_EXPORT\_TEST-45C982

Inferred from test source artifact.

| Key          | Value                                      |
|--------------|--------------------------------------------|
| Kind         | test                                       |
| Status       | not-run                                    |
| Source       | gemara_export_test.go                      |
| Name         | Gemara Export Test                         |
| Mentioned In | Interaction Flow View , Gemara Export Test |

VER-INF-TEST-GEMARA\_OSCAL\_TEST-58DBDA

Inferred from test source artifact.

| Key          | Value                                     |
|--------------|-------------------------------------------|
| Kind         | test                                      |
| Status       | not-run                                   |
| Source       | gemara_oscal_test.go                      |
| Name         | Gemara Oscal Test                         |
| Mentioned In | Interaction Flow View , Gemara Oscal Test |

VER-INF-TEST-GITHUB\_CHECK\_RUN\_CONTRACT\_TEST-3F4D3F

validates GitHub check-run contract shape and required review result fields

| Key          | Value                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------|
| Kind         | test                                                                                          |
| Status       | not-run                                                                                       |
| Source       | examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go |
| Name         | Github Check Run Contract Test                                                                |
| Mentioned In | Interaction Flow View , Github Check Run Contract Test                                        |

VER-INF-TEST-INFERRED\_CODE\_TEST-3CA82E

Inferred from test source artifact.

| Key          | Value                                      |
|--------------|--------------------------------------------|
| Kind         | test                                       |
| Status       | not-run                                    |
| Source       | inferred_code_test.go                      |
| Name         | Inferred Code Test                         |
| Mentioned In | Interaction Flow View , Inferred Code Test |

VER-INF-TEST-INFERRED\_LAYERS\_TEST-6634AF

Inferred from test source artifact.

| Key          | Value                                        |
|--------------|----------------------------------------------|
| Kind         | test                                         |
| Status       | not-run                                      |
| Source       | inferred_layers_test.go                      |
| Name         | Inferred Layers Test                         |
| Mentioned In | Interaction Flow View , Inferred Layers Test |

VER-INF-TEST-INFERRED\_VERIFICATION\_TEST-6715F6

Inferred from test source artifact.

| Key          | Value                                              |
|--------------|----------------------------------------------------|
| Kind         | test                                               |
| Status       | not-run                                            |
| Source       | inferred_verification_test.go                      |
| Name         | Inferred Verification Test                         |
| Mentioned In | Interaction Flow View , Inferred Verification Test |

VER-INF-TEST-LOAD\_TEST-A58EB3

Inferred from test source artifact.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | test                              |
| Status       | not-run                           |
| Source       | model/load_test.go                |
| Name         | Load Test                         |
| Mentioned In | Interaction Flow View , Load Test |

VER-INF-TEST-LOBSTER\_EXPORT\_TEST-1C1C7E

Inferred from test source artifact.

| Key          | Value                                       |
|--------------|---------------------------------------------|
| Kind         | test                                        |
| Status       | not-run                                     |
| Source       | lobster_export_test.go                      |
| Name         | Lobster Export Test                         |
| Mentioned In | Interaction Flow View , Lobster Export Test |

VER-INF-TEST-MAIN\_TEST-6B61AE

Inferred from test source artifact.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | test                              |
| Status       | not-run                           |
| Source       | cmd/engmcp/main_test.go           |
| Name         | Main Test                         |
| Mentioned In | Interaction Flow View , Main Test |

VER-INF-TEST-OSCAL\_CHAIN\_TEST-C12EF9

Inferred from test source artifact.

| Key  | Value |
|------|-------|
| Kind | test  |

| Key          | Value                                    |
|--------------|------------------------------------------|
| Status       | not-run                                  |
| Source       | oscal_chain_test.go                      |
| Name         | Oscal Chain Test                         |
| Mentioned In | Interaction Flow View , Oscal Chain Test |

VER-INF-TEST-OSCAL\_SSP\_TEST-4B08EF

Inferred from test source artifact.

| Key          | Value                                  |
|--------------|----------------------------------------|
| Kind         | test                                   |
| Status       | not-run                                |
| Source       | oscal_ssp_test.go                      |
| Name         | Oscal Ssp Test                         |
| Mentioned In | Interaction Flow View , Oscal Ssp Test |

VER-INF-TEST-PROJECT\_TEST-2ECF72

Inferred from test source artifact.

| Key          | Value                                |
|--------------|--------------------------------------|
| Kind         | test                                 |
| Status       | not-run                              |
| Source       | view/project_test.go                 |
| Name         | Project Test                         |
| Mentioned In | Interaction Flow View , Project Test |

VER-INF-TEST-RENDER\_TEST-EECC23

Inferred from test source artifact.

| Key    | Value   |
|--------|---------|
| Kind   | test    |
| Status | not-run |

| Key          | Value                               |
|--------------|-------------------------------------|
| Source       | render/mermaid/render_test.go       |
| Name         | Render Test                         |
| Mentioned In | Interaction Flow View , Render Test |

VER-INF-TEST-SCAN\_TEST-B04F81

Inferred from test source artifact.

| Key          | Value                             |
|--------------|-----------------------------------|
| Kind         | test                              |
| Status       | not-run                           |
| Source       | codemap/scan_test.go              |
| Name         | Scan Test                         |
| Mentioned In | Interaction Flow View , Scan Test |

VER-INF-TEST-SERVER\_TEST-389957

Inferred from test source artifact.

| Key          | Value                               |
|--------------|-------------------------------------|
| Kind         | test                                |
| Status       | not-run                             |
| Source       | mcp/server_test.go                  |
| Name         | Server Test                         |
| Mentioned In | Interaction Flow View , Server Test |

VER-INF-TEST-STRUCTURIZR\_EXPORT\_TEST-886374

Inferred from test source artifact.

| Key    | Value                      |
|--------|----------------------------|
| Kind   | test                       |
| Status | not-run                    |
| Source | structurizr_export_test.go |



| Key          | Value                                           |
|--------------|-------------------------------------------------|
| Name         | Structurizr Export Test                         |
| Mentioned In | Interaction Flow View , Structurizr Export Test |

VER-INF-TEST-TELEMETRY\_INGEST\_CONTRACT\_TEST-5811E2

validates telemetry ingest contract fields and accepted request semantics

| Key          | Value                                                                                   |
|--------------|-----------------------------------------------------------------------------------------|
| Kind         | test                                                                                    |
| Status       | not-run                                                                                 |
| Source       | examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go |
| Name         | Telemetry Ingest Contract Test                                                          |
| Mentioned In | Interaction Flow View , Telemetry Ingest Contract Test                                  |

VER-INF-TEST-THREAT\_MODEL\_EXPORT\_TEST-721A8D

Inferred from test source artifact.

| Key          | Value                                            |
|--------------|--------------------------------------------------|
| Kind         | test                                             |
| Status       | not-run                                          |
| Source       | threat_model_export_test.go                      |
| Name         | Threat Model Export Test                         |
| Mentioned In | Interaction Flow View , Threat Model Export Test |

VER-INF-TEST-TRACE\_MATRIX\_TEST-9FB80C

Inferred from test source artifact.

| Key    | Value                |
|--------|----------------------|
| Kind   | test                 |
| Status | not-run              |
| Source | trace_matrix_test.go |
| Name   | Trace Matrix Test    |

| Key          | Value                                     |
|--------------|-------------------------------------------|
| Mentioned In | Interaction Flow View , Trace Matrix Test |

VER-INF-TEST-TRLC\_EXPORT\_TEST-DD1345

Inferred from test source artifact.

| Key          | Value                                    |
|--------------|------------------------------------------|
| Kind         | test                                     |
| Status       | not-run                                  |
| Source       | trlc_export_test.go                      |
| Name         | Trlc Export Test                         |
| Mentioned In | Interaction Flow View , Trlc Export Test |

VER-INF-TEST-VALIDATE\_TEST-498C59

Inferred from test source artifact.

| Key          | Value                                 |
|--------------|---------------------------------------|
| Kind         | test                                  |
| Status       | not-run                               |
| Source       | validate/validate_test.go             |
| Name         | Validate Test                         |
| Mentioned In | Interaction Flow View , Validate Test |

VER-INF-UNIT-CHECKOUT\_DECLINE\_REASON\_TEST-B8BCCD

validates checkout decline reason normalization and retry policy messaging

| Key          | Value                                                                           |
|--------------|---------------------------------------------------------------------------------|
| Kind         | unit                                                                            |
| Status       | not-run                                                                         |
| Source       | examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go |
| Name         | Checkout Decline Reason Test                                                    |
| Mentioned In | Interaction Flow View , Checkout Decline Reason Test                            |

VER-INF-UNIT-OTA\_CAMPAIGN\_TEST-9000FB

verifies OTA campaign signature validation and rollout trigger behavior

| Key          | Value                                                                  |
|--------------|------------------------------------------------------------------------|
| Kind         | unit                                                                   |
| Status       | not-run                                                                |
| Source       | examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go |
| Name         | Ota Campaign Test                                                      |
| Mentioned In | Interaction Flow View , Ota Campaign Test                              |

VER-INF-UNIT-RATE\_LIMIT\_DEFERRAL\_TEST-4C4AEE

verifies rate-limit deferral behavior for pull request review processing

| Key          | Value                                                                               |
|--------------|-------------------------------------------------------------------------------------|
| Kind         | unit                                                                                |
| Status       | not-run                                                                             |
| Source       | examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go |
| Name         | Rate Limit Deferral Test                                                            |
| Mentioned In | Interaction Flow View , Rate Limit Deferral Test                                    |